Religion and Conflict: Evidence from Ramadan[*]

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Abstract

How does a global religious practice affect armed conflict and civilian political behavior? I study Ramadan, exploiting the lunar calendar's rotation through Gregorian months alongside day-to-day and within-day variation in observance. The effects pull in opposite directions across actors. Ramadan intensifies insurgent violence in predominantly Muslim countries, with LLM-based aggressor classification attributing the escalation to offensives from religious-ideological insurgent groups. Hourly military records from Afghanistan corroborate this pattern and reveal that insurgents concentrate strenuous direct fire in post-fast night hours when targeting Afghan state forces, with no such reallocation against better-equipped international coalition units. On the civilian side, demonstrations decline significantly, and Muslims interviewed during the holy month report heightened religiosity alongside reduced political engagement and protest willingness. Non-Muslim respondents in the same countries show null effects.

Keywords: Religion, Conflict, Protests, Ramadan, Time Allocation

JEL classification: Z10; Z12; D74; N30; O57; C55

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1 Introduction

Armed conflicts are at historic peak since 1946 (Rustad, 2025). While their origins range from economic grievances to territorial disputes, religion often plays an important role. It shapes economic, political, and organizational behavior (McCleary and Barro, 2006), including the cohesion and lethality of armed groups (Berman, 2011). Faith can offer combatants solace under existential threat, supply a moral framework that dampens empathy toward outgroups, and bind fighters to a transcendent cause. Despite the importance of this link, we lack granular evidence on whether and how religious practice systematically alters patterns of conflict and peaceful political mobilization (Iyer, 2016).

Ramadan, the holiest month of the Islamic lunar calendar and observed by nearly two billion Muslims worldwide (Pew Research Center, 2017), provides a uniquely tractable case for studying this question. It is the most religiously dense period of the year. Believers fast from dawn to sunset, but the month's significance extends well beyond the daily fast: communal post-fast (*iftar*) meals, intensified daily prayer, nightly congregational prayers (*tarawih*) lasting one to two hours, expanded Qur'anic recitation, and heightened symbolic and doctrinal salience tied to the month in which the Qur'an was first revealed. The final ten nights commemorate the night of revelation (*Laylat al-Qadr*) and concentrate the most intense devotional activity of the year. Observance rates are exceptionally high even among otherwise non-practicing Muslims, making it one of the most widely observed religious obligations across the globe (Lugo et al., 2012).

Theoretically, however, the predicted effect of Ramadan on insurgent violence and civilian protest is ambiguous, with competing channels pointing in different directions. For insurgents, the religious salience of the month could intensify violence, while public-backlash and reputational-cost arguments predict the opposite (Reese et al., 2017; Hodler et al., 2024). For civilians, intensive ritual demands could crowd out political activity, but the same communal gatherings could equally facilitate coordination for collective action. Whether Ramadan amplifies, suppresses, or reshapes patterns of political violence and mobilization is therefore an empirical question, with distinct policy implications across the channels, from scheduling ceasefires and elections to deploying security resources.

To exploit this setting empirically, I leverage the rotation of Ramadan through the Gregorian calendar. Because the Islamic lunar (Hijri) calendar is roughly 11 days shorter than the Gregorian calendar, Ramadan rotates through all solar months over a 33-year cycle, breaking the alignment between the holy month and the solar calendar. This rotation allows me to construct a balanced panel with country-by-Hijri-year and Gregorian month fixed effects, identifying the Ramadan effect from within-country, within-Hijri-year comparisons of conflict outcomes net of solar seasonality. Crucially, the identification strategy isolates the causal effect of the Ramadan *period* rather than religious observance *per se*, yielding an intention-to-treat estimate.

The results reveal a sharp divergence in how Ramadan affects armed actors and ordinary citizens in predominantly Muslim countries where observance is near universal. State-based armed conflict increases by roughly 4% per country during Ramadan.[1] LLM-based aggressor attribution shows that the escalation originates mainly from rebel offensives rather than government-initiated operations. The increase is entirely concentrated among religious-ideological rebel groups and absent among ethno-religious or secular-nationalist organizations operating in the same countries.[2]

Hourly military records from Afghanistan provide additional support for these results and further reveal that insurgents concentrate labor-intensive direct fire attacks into post-fast night hours in Ramadan compared to the same times in non-Ramadan months. This reallocation is sharply asymmetric across targets: insurgents shift attacks against Afghan state forces toward post-iftar night hours but do not reallocate attacks against international coalition forces, whose night-vision optics, drone surveillance, and rapid air support raise the marginal cost of nighttime engagement. The within-day pattern therefore reveals two mechanisms operating jointly, with fasting physiology determining when insurgents attack and technological asymmetry determining whom.

On the civilian side, Ramadan observance systematically reduces protest activity in predominantly Muslim countries, with demonstrations declining by roughly 7%. Within-country evidence from Afghan provinces corroborates this pattern in declassified military records of protests that bypass media reporting, with a decline of nearly 4% during Ramadan concentrated in pre-iftar hours. Individual survey evidence from Muslim-majority countries across the globe shows that respondents interviewed during Ramadan report roughly 7% higher daily prayer rates, 13% lower political interest, and 12% lower willingness to demonstrate than those interviewed in immediately adjacent months within the same country and survey wave. Non-Muslim respondents living in the same countries and interviewed during the same windows show precisely estimated null effects across all three individual outcomes. This periodic lull in civilian mobilization in predominantly Muslim countries may carry significant implications for the strategic timing of authoritative government reforms and the responsiveness of civil society.

I further exploit the precise rate of the drift: the final 10 days of Ramadan in year t overlap with the same Gregorian calendar dates that fall just after Ramadan year t+1, yielding a novel exact-date matching design that nets out any seasonal confound tied to a specific calendar day. This allows me to compare treated and untreated days sharing the same solar position, absorbing any seasonal confound tied to a specific calendar day, whether weather, agricultural cycles, or recurring holidays. The last 10 days are uniquely intense, both religiously and physically. They contain extened nighttime worship, while marking the apex of cumulative fasting fatigue after three weeks of dawn-to-sunset abstention from food and water. The exact-date design therefore identifies off the slice of Ramadan when both religious observance and physi-

[1]For context, the climate-conflict meta-analysis in Burke et al. (2024) reports an average intergroup conflict effect of 2.5% per 1 SD temperature shock, providing a rough order-of-magnitude benchmark; treatment doses are not directly comparable, but the Ramadan estimates lie in at least the same ballpark as effects from this closely studied literature. Replicating the analysis with ACLED political violence data yields effect sizes nearly twice as large and province-level military records from Afghanistan show a 7.3% monthly increase in rebel attacks.

[2]I classify rebel groups by ideology based on their founding charters, affiliations, and the reported role of religious doctrine (Crenshaw and Robinson, 2025).

cal depletion are at their peak. The match confirms the above mentioned results both for armed conflict and civilian protests.

The conceptual framework operates across two levels of aggregation. At the monthly level, the doctrinal channel explains why violence rises during Ramadan relative to other months: for religious-ideological insurgents, the holy month provides a framework that frames offensive warfare as uniquely meritorious. Rebel organizations explicitly invoke battles fought during Ramadan in the early history of Islam, characterizing combat during the fast as obligatory worship whose spiritual rewards are multiplied (Bin Ali, 2017; Amarasingam and Winter, 2017).[3] This generates an incentive for escalation specific to groups with religious governance objectives and absent among ethno-religious or secular-nationalist organizations. At the within-day level, the physiological channel governs the timing of those Ramadan attacks. The dawn-to-sunset fast imposes significant costs on physically demanding operations, so fighters defer high-exertion combat to post-iftar hours when food and water restore physical capacity, while ideological outbidding ensures the underlying intent to attack remains elevated throughout the month. For ordinary civilians, the intensive ritual demands of Ramadan crowd out the time and attention required for political mobilization.

The empirical analysis draws on four data sources: the UCDP Georeferenced Event Dataset for state-based armed conflict across 107 countries (1989–2022), declassified hourly U.S. military records (SIGACTS) from Afghanistan at province level (2002-2015) (Condra et al., 2018; Shaver and Wright, 2017), ACLED demonstration data for 48 African countries (1997-2023), and World Values Survey responses from approximately 40,000 Muslim respondents. To identify the effect, the baseline specification restricts the sample to countries with Muslim population shares above 75%; a complementary specification estimates the model on the full sample, using countries with negligible Muslim populations as a built-in placebo group.

Using the source article descriptions in the UCDP data, I attribute the aggressor in over 100,000 conflict events to either the state or rebel groups via LLM-based text analysis (Asirvatham et al., 2026). I validate this attribution against the SIGACTS military records, which separately log friendly and enemy actions and record the target of each incident. These records also enable within-day analysis: I exploit hourly variation to test whether combatants shift high-exertion operations (direct fire) relative to low-exertion tactics (improvised explosive devices) around the fasting window. Finally, for civilian behavior, I leverage the quasi-random timing of survey interviews in months adjacent to Ramadan to identify individual-level shifts in religiosity and political engagement.

Both the conflict escalation and relative protest suppression results survive a battery of robustness checks. Placebo tests on non-Muslim countries and non-fundamentalist rebel groups in predominantly Muslim regions yield precisely estimated null effects. Both armed conflict and demonstration results from the cross-country samples replicate in declassified military records that bypass media entirely. An event study across all Hijri months shows that Ramadan uniquely drives both the conflict increase and the protest decline, with no comparable

[3] The foundational precedents are the Battle of Badr (624 AD) and the Conquest of Mecca (630 AD), both of which occurred during Ramadan.

effects in any other month. Finally, fasting duration has no independent association with either outcome, indicating that it is the overall observance of Ramadan, not the physical intensity of the fast, that generates the observed effects.

These findings advance research on how religion shapes conflict and political behavior (Clingingsmith et al., 2009; Mitra and Ray, 2014; Becker and Pascali, 2019; Bazzi et al., 2020; Lowes et al., 2025), particularly how quasi-exogenous timing of religious events alters socio-economic behavior (Campante and Yanagizawa-Drott, 2015; Montero and Yang, 2022). More broadly, they speak to ongoing research on the determinants of armed conflict (Miguel et al., 2004; Blattman and Miguel, 2010; Besley and Persson, 2011; Dube and Vargas, 2013; Bazzi and Blattman, 2014; McGuirk and Nunn, 2025), where religion remains underexplored despite its prominence (Iyer, 2016).[4] They also speak to recent evidence that cultural-normative inheritance generates conflict patterns concentrated in narrow windows around specific normative triggers rather than as baseline-level shifts (Cao et al., 2021), a behavioral signature that motivates the exact-date matching design.

Within these two strands of literature, my contributions are fourfold. First, I focus on state-based armed conflict, the most prevalent form of political violence, and decompose effects by group ideology.[5] Second, hourly military records from Afghanistan allow me to study the within-day tactical effects of religious observance at a resolution unavailable in prior work. Third, I introduce a novel exact-date matching design that compares the same Gregorian calendar days across adjacent years, providing the cleanest possible control for seasonal confounds.[6] Finally, by combining cross-country protest data with subnational Afghan records and individual survey responses, I extend the analysis to civilian political behavior and triangulate the underlying mechanisms across three levels of aggregation.[7]

I also build on existing studies of terrorist incidents during Ramadan. Reese et al. (2017) document that insurgents commit fewer terrorist attacks on specific Islamic holidays to avoid public backlash. Hodler et al. (2024) find that longer daylight fasts reduce terrorism in the following year via cumulative attitudinal change. Limodio (2022) exploits Pakistan's government-run Zakat levy to show that mandatory charitable transfers during Ramadan fund capital-intensive attacks.

[4] A growing literature documents the reverse channel: conflict and adversity can themselves intensify religious behavior. Henrich et al. (2019) show that war exposure in Uganda, Sierra Leone, and Tajikistan increased religious participation years later. Zussman (2014) and Blattman (2009) document similar patterns among Israeli civilians and Ugandan ex-combatants, respectively. Dube et al. (2022) use high-frequency telecommunications data to show that religious commitment, measured by mobile phone usage dips during Islamic prayer times, deepens in response to local conflict shocks.

[5] To my knowledge, this project is also the first to systematically attribute the aggressor in a large-scale conflict dataset with an LLM-based text analysis.

[6] Whereas prior work identifies effects either from cross-sectional variation in fasting hours driven by latitude and the Gregorian month in which Ramadan falls (Campante and Yanagizawa-Drott, 2015; Hodler et al., 2024) or from the plausibly random alignment of Ramadan with the Gregorian calendar, I identify effects from within-country, within-Hijri-year variation, further tightened by exact-date matching across consecutive Hijri years.

[7] The paper also contributes to the literature on the determinants of protest and collective action (Madestam et al., 2013; Girod et al., 2018; Cantoni et al., 2019; Enikolopov et al., 2020). The finding that Ramadan crowds out political engagement at the individual level, confirmed by survey data showing heightened religiosity and reduced political interest, provides a mechanism through which religious institutions can shape the political economy of authoritarian governance.

My results diverge from these accounts on outcome, time horizon, methodology, and results. The first two papers document declines in terrorist activity, while I find that state-based armed conflict increases during Ramadan. The backlash channel applies most directly to civilian targeting, where reputational costs are high; in state-based conflict against state forces, these costs operate more weakly. The attitudinal channel in Hodler et al. (2024) measures effects in the year following Ramadan, while my estimates identify the immediate within-Ramadan response from combatants in the field. Limodio's financing channel is institutionally specific to Pakistan. The absence of any Ramadan effect among ethno-religious and secular-nationalist groups in predominantly Muslim countries rules out financing alone and points to a doctrinal mechanism specific to religious-ideological organizations.

While previous studies leverage the quasi-exogenous timing of the lunar month to identify effects on health, educational performance, and other socioeconomic outcomes (Almond and Mazumder, 2011; Majid, 2015; Campante and Yanagizawa-Drott, 2015; Oosterbeek and van der Klaauw, 2013; Hornung et al., 2023; Gulek, 2024), my analysis extends this work to political violence and collective action, examining how different actors respond to the fasting period. Given the well-documented costs of armed conflict for economic growth (e.g., Blattman and Miguel, 2010; Federle et al., 2026), these findings carry significant implications for economic prosperity.

2 Background

Ramadan is the ninth month of the Hijri calendar.[8] It holds profound significance for Muslims as the month in which the Qur'an was first revealed. Observing Ramadan through fasting from early dawn to sunset is a key practice, known as one of the five pillars of Islam. Fasting during Ramadan was made obligatory during the second year of Hijrah (624 AD) after the Muslims migrated from Mecca to Medina in the month immediately preceding Ramadan. Fasting is obligatory for all Muslims who have reached puberty, with exceptions for those who are acutely or chronically ill, traveling, elderly, breastfeeding, diabetic, or pregnant. Fasting during Ramadan involves abstaining from food, drink, tobacco products, and sexual relations.

Ramadan significantly transforms the social and individual lives of those who observe it. Adherents wake up before dawn to eat and drink (*suhur*) and carry on with their daily activities, before breaking their fast at sunset, often in the company of family, friends, and other co-religionists. Mosques also host iftar meals, attracting large gatherings throughout the month.

Among various religions, predominantly Muslim countries have the highest daily and weekly rates of worship (Pew Research Center, 2018). Muslims pray five times a day, either in the mosque or individually, and gather for weekly Friday congregational prayers around noon throughout the year, with heightened emphasis on these practices during Ramadan. Both men and women frequently attend additional congregation prayers (*tarawih*) after iftar, which typically last 1-2 hours. For instance, Pope (2024) uses evidence from US cellphone data to report large spikes in Muslim worship attendance during Ramadan.

[8] It is a purely lunar calendar that consists of twelve months that are either 29 or 30 days long. This causes Ramadan to occur 10 to 12 days earlier each year in relation to the Gregorian calendar.

The theological foundations for this intensified engagement are explicit in the prophetic traditions.[9] One hadith (Sahih Bukhari, Book 2, Hadith 38) on Ramadan states:

> Narrated Abu Huraira: Allah's Apostle said, "Whoever observes fasts during the month of Ramadan out of sincere faith, and hoping to attain Allah's rewards, then all his past sins will be forgiven."[10]

This hadith illustrates the special spiritual status accorded to Ramadan, establishing the month as a period of uniquely heightened religious receptivity for believers. Another hadith (Sahih Muslim, Book 13, Hadith 212) explicitly discourages interpersonal disputes during fasting:[11]

> Abu Huraira reported Allah's Messenger as saying: Allah the Exalted and Majestic said: Every act of the son of Adam is for him, except fasting. It is exclusively meant for Me and I alone will reward it. Fasting is a shield. When any one of you is fasting on a day, he should neither indulge in obscene language, nor raise the voice; or if anyone reviles him or tries to quarrel with him he should say: I am a person fasting.

Together, these traditions create a powerful set of incentives for ordinary Muslims to redirect time and energy from secular activities toward worship, prayer, and self-restraint during the holy month.[12]

While the Hijri calendar governs the timing of religious observances, virtually all Muslim-majority countries use the Gregorian calendar for civil, commercial, and governmental functions (Aydin, 2017). Work schedules, school terms, agricultural cycles, and military operations follow Gregorian months and seasons. Even Saudi Arabia, which historically relied on the Hijri calendar for official purposes, adopted the Gregorian system for administrative functions in 2016. As a result, Ramadan's annual rotation through the Gregorian calendar means that fasting overlaps with different seasonal and institutional conditions each year.

Fighting in Ramadan

Islam designates four sacred months during which warfare is prohibited except in self-defense.[13] Ramadan is not among them. Crucially, foundational military victories in early Islam occurred during Ramadan. These include the Battle of Badr, the first major engagement between early Muslims and the Meccan Quraysh, as well as the Conquest of Mecca, widely considered one of the most pivotal events in Islamic history.

[9] All hadith references follow the standard numbering of the Sahih al-Bukhari collection and the Sahih Muslim collection, which serve as major sources of religious law and moral guidance, accessed via https://sunnah.com.

[10] Translation available at https://sunnah.com/bukhari:38.

[11] Translation available at https://sunnah.com/muslim:1151d.

[12] A natural question is why this injunction against confrontation does not similarly restrain insurgent groups. As discussed in the following subsection, religious-ideological organizations selectively invoke a separate strand of the prophetic tradition that frames offensive warfare during Ramadan as uniquely meritorious, drawing on historical precedents such as the Battle of Badr and the Conquest of Mecca. The behavioral norm against quarreling applies to interpersonal conduct among civilians; insurgent groups reinterpret the martial traditions as superseding it for combatants engaged in what they define as obligatory armed struggle.

[13] These months are Muharram (the first month), Rajab (the seventh month), and the last two months, Dhul Qa'dah and Dhu al-Hijjah (Al-Dawoody, 2011).

Throughout Islamic history, several other major battles have occurred during Ramadan, including the Muslim conquest of the Iberian Peninsula in 711, the Siege of Jerusalem in 1187, and the decisive Mamluk victories over the Mongols at Ain Jalut in 1260 and Marj al-Saffar in 1303. These historical precedents have carried into the modern era, heavily influencing the timing of the 1973 Arab-Israeli War, often referred to in the Arab world as the Ramadan War, and Iran's massive 1982 offensive during the Iran-Iraq War. The latter, involving over 180,000 troops in one of the largest land battles since World War II, was named Operation Ramadan to invoke religious legitimacy and the historical memory of early Islamic victories to motivate forces.

Both state forces and insurgent groups continue to leverage this historical narrative to provide religious and strategic legitimacy for military engagements during the holy month. While classical Islamic teachings often emphasize the "Greater Jihad" (the internal, spiritual struggle for self-restraint), modern radical Islamist organizations selectively prioritize the "Lesser Jihad" (offensive warfare). These groups engage in a sophisticated form of ideological appropriation, framing Ramadan as a unique temporal window where the spiritual rewards for martyrdom and military success are exponentially multiplied. For example, the Afghan Taliban have formally rejected calls for seasonal ceasefires as an "ignorance of religion," instead characterizing combat during the fast as an "obligatory worship" (*ibadat*) for which the spiritual merit is multiplied seventyfold (Taylor and Harooni, 2013).

Similarly, spokesmen for the Islamic State have frequently characterized Ramadan as the "month of conquest and jihad," explicitly drawing a parallel between modern insurgent activities and the foundational victories of the Prophet (Bin Ali, 2017; Amarasingam and Winter, 2017). In this framework, the physical hardship of the fast is rebranded as a form of ascetic preparation that increases the spiritual merit (*thawab*) of the fighter, turning physiological deprivation into a psychological asset.

The doctrinal raw material for this appropriation is readily available in the canonical hadith collections (Cook, 2015; Hegghammer, 2017). A concrete example illustrates how this appropriation works in practice. One canonical tradition on guarding the frontier (*ribat*) states:[14]

> It was narrated from Salman Al-Khair that the Messenger of Allah said, "Whoever guards Ribat (the frontier) in the cause of Allah for one day and one night, he will have (a reward) like that of fasting and praying (night prayers) for a month. If he dies he will continue to receive reward for what he did ..."

The hadith is conventionally understood to honor the discipline of frontier service in defensive contexts. Fundamentalist militants invert this interpretive emphasis to argue that armed combat during Ramadan yields spiritual rewards exceeding the most sacred forms of devotional worship. For example, issues of the Islamic State's Arabic-language weekly *al-Naba* published during Ramadan 1438 (June 2017) describe the fasting month as a period of "ribat, jihad, giving, and martyrdom," and explicitly argue that one night of frontline guard duty during Ramadan surpasses the spiritual merit of worship on Laylat al-Qadr, the most sacred night

[14] Sunan an-Nasa'i, Book 25, Hadith 83. Translation available at `https://sunnah.com/nasai:3167`.

in Ramadan (Zelin, 2026). The interpretive moves required to reach this conclusion, equating modern offensive operations with classical defensive ribat, illustrate how canonical doctrinal raw material is reshaped to serve a specifically militant interpretive agenda.

Finally, insurgent fighters maintain strict observance of the fast even while conducting operations. Hegghammer (2017) documents the prevalence insurgent groups fasting during Ramadan, including suicidal fighters observing the fast before attacks. The Taliban's code of conduct (Layeha) similarly requires fighters to uphold core religious obligations (Munir, 2011).

3 Conceptual Framework

This section develops a conceptual framework linking Ramadan to armed conflict and civilian political behavior. I derive testable predictions from three mechanisms and connect each to the literature on the organizational economics of violent groups and the economics of religious participation.

Mechanism 1: Doctrinal Framing and the Cross-Month Increase in Violence. The historical and doctrinal record establishes that Ramadan occupies a privileged place in the Islamic military tradition. Foundational victories occurred during the fasting month, and modern insurgent organizations actively invoke these precedents to frame offensive warfare as uniquely meritorious (Bin Ali, 2017; Amarasingam and Winter, 2017; Taylor and Harooni, 2013). The combat-as-worship framing operates as a sacred value in the sense of Ginges and Atran (2013): a non-fungible moral commitment that resists material trade-offs and intensifies, rather than softens, under symbolic challenge.

Several reinforcing mechanisms sharpen this prediction. The heightened religious environment creates opportunities for ideological outbidding (Bloom, 2005): launching high-profile attacks during the holy month signals superior religious commitment relative to the state or competing factions. Conducting complex operations while adhering to a strict fast serves as a costly signal critical for recruitment and consolidation of support (Berman and Laitin, 2008; Berman, 2011).[15] This connects to the "club model" of religious organizations (Iannaccone, 1992; Berman, 2011), in which the shared ordeal of fasting under combat conditions screens out uncommitted members and tightens group cohesion, with the shared rhythms of *suhur* and *iftar* providing predictable windows for logistical coordination.

Rather than targeting civilians, which carries high political costs (Clark, 2011; Reese et al., 2017; Hodler et al., 2024), insurgents can concentrate attacks on government forces during periods of high symbolic salience. If the mechanism operates through rebel groups invoking doctrinal justifications for offensive warfare, the escalation should manifest predominantly as rebel-initiated violence, though a smaller government response is consistent with reactive or preemptive counter-insurgency operations triggered by anticipated rebel activity. I test this by classifying the aggressor in each conflict event using LLM-based attribution of unstructured event descriptions.

[15] Berman and Laitin (2008) show that religious militants are more lethal per attack than secular counterparts precisely because costly religious requirements filter for high-commitment operatives. Ramadan fasting under combat conditions intensifies this screening mechanism.

Groups possessing religious identity but primarily nationalist or territorial objectives lack the doctrinal infrastructure to credibly cast Ramadan as a season of obligatory combat; they benefit from religion as a cohesion mechanism year-round, but cannot activate the sacred-period multiplier on violence. Secular-nationalist groups have no basis for such mobilization at all. The Ramadan escalation should therefore be concentrated exclusively among religious-ideological vanguards, with precisely estimated null effects for both ethno-religious and secular-nationalist organizations, even though ethno-religious groups may maintain higher baseline violence due to religious cohesion.

Mechanism 2: Physiological Constraints and the Within-Day Reallocation. The dawn-to-sunset fast imposes acute costs: reduced grip strength, slower reaction times, and diminished aerobic capacity, particularly during afternoon hours when dehydration peaks (Cherif et al., 2016; Chaouachi et al., 2009). These constraints are immediately relaxed after iftar, when fighters eat, rehydrate, and recover the physical capacity necessary for sustained close combat. For insurgent organizations that must rationally allocate scarce combatant resources (Shapiro, 2013), this creates a straightforward optimization problem: operations requiring sustained physical exertion become prohibitively costly during daylight hours, while remotely detonated or victim-activated devices, which are pre-positioned and require negligible physical effort at execution, face no comparable constraint.

The implication is that high-exertion attacks should shift to post-iftar hours while low-exertion attacks should not. This prediction sharply distinguishes the physiological mechanism from alternatives. A public-backlash mechanism (Reese et al., 2017) predicts uniform suppression of all violence, not a within-day shift. A generic nocturnal-preference story predicts identical reallocation for all attack types. Only the fasting-constraint channel predicts a divergence between high-exertion and low-exertion tactics concentrated precisely at the iftar boundary. I test these predictions using declassified military records from Afghanistan, which provide timestamps precise to the minute.

Mechanism 3: Ritual Crowding-Out of Civilian Engagement. For ordinary citizens, Ramadan transforms the allocation of time and attention. The daily rhythm of suhur, five obligatory prayers with heightened emphasis, extended tarawih congregations lasting one to two hours each evening, communal iftars, and Qur'anic recitation collectively impose substantial demands on discretionary time (Pope, 2024). Within the household production framework (Becker, 1965; Azzi and Ehrenberg, 1975; Iannaccone, 1990), the returns to religious investment rise sharply during Ramadan as canonical traditions promise multiplied merit for worship.

The key mechanism is the shadow cost of time: even if the intrinsic desire to engage politically is unchanged, the heightened return to devotion raises the opportunity cost of all competing activities, crowding out political mobilization through the binding time constraint rather than diminished political preferences.[16] This aligns with evidence that religious festivals systematically crowd out secular activities while heightening local religious investments (Montero and Yang, 2022). The doctrinal injunctions reinforce this reallocation: the prophetic

[16] Iannaccone (1990) shows that religious "capital" accumulated through repeated participation raises the marginal return to further religious investment. Ramadan's concentrated period of intensified practice is precisely the kind of shock that activates this mechanism.

tradition instructs the fasting person to avoid confrontation, establishing a behavioral norm against public contention that applies broadly to the civilian population.

A complementary channel operates on the supply side: governments may preemptively intensify their presence around mosques and public spaces during Ramadan, raising the expected cost of collective action. The individual-level survey evidence helps adjudicate between these channels. If preemptive state repression were the sole mechanism, one would expect suppressed protest willingness but no corresponding shift in private religiosity or political interest, since attitudes are not directly constrained by police presence.

Collectively, the three mechanisms generate interlocking predictions testable at different levels of aggregation: cross-country panels for the overall Ramadan effect and its variation by group ideology and Muslim population share; hourly military records for within-day tactical reallocation; and individual survey data for the time-allocation mechanism underlying civilian demobilization.

4 Empirical Framework

4.1 Data

1. Armed conflict. The primary source of armed conflict data is the UCDP Georeferenced Event Dataset (GED) covering the period from 1989 to 2022. I focus on state-based armed conflicts, which account for more than 70% of all armed conflict types. State-based events involve a coherent organized opposition whose ideological position on combat during Ramadan is identifiable, and the rebel group identity and aggressor narrative are recorded consistently — both essential to the ideology and aggressor coding I use later. Non-state and one-sided violence lack this organizational structure, making any Ramadan-mediated shift in behavior harder to identify.

An individual "event" in UCDP-GED involves the use of armed force by the government of a state against one or more opposition groups, with at least one direct battle death (Sundberg and Melander, 2013). Each event records the identity of the rebel group involved, which contains more than 100 distinct entities across the sample. The dataset also includes a source article field containing brief narrative descriptions of each event. This panel dataset covers nearly all Muslim-majority countries and spans one complete cycle of Ramadan's progression through the solar year,[17] with a sample of 107 countries (see Figure A.1).

2. Hours of attacks. Data on hourly insurgent violence come from the U.S. Defense Department's Significant Activities (SIGACTS) dataset for Operation Enduring Freedom, covering more than 200,000 insurgent attacks in Afghanistan from 2002 through early 2015 (Condra et al., 2018; Shaver and Wright, 2017). These records were jointly collected by Afghan security forces and NATO's International Security Assistance Force (ISAF) following standardized military reporting protocols. Unlike media-sourced datasets, military-recorded data of this kind offer more systematic coverage of conflict events and are less susceptible to the reporting

[17] Since a lunar month is about 29.5 days long, the Hijri calendar shifts by around 1.5 weeks each year compared to the Gregorian calendar, with 34 Hijri years approximately corresponding to 33 Gregorian years.

biases documented in media-based incident compilation (Weidmann, 2016). This makes the SIGACTS data a valuable complement to the UCDP records used in the cross-country analysis. Each incident is geo-referenced using military grid coordinates and includes a time stamp, typically precise to the minute.[18]

The data capture three main attack types relevant to this analysis, ordered by the level of physical exertion required of insurgents: direct fire (DF), which involves small arms and rocket-propelled grenades carried out in close combat by insurgents operating in groups and demands sustained presence and effort; indirect fire (IDF), which involves mortars and rockets fired from prepared positions and requires less continuous exertion once the launch site is established; and improvised explosive devices (IEDs), which are typically pre-positioned on or near roadways in advance and triggered remotely or by command wire, requiring minimal real-time effort.

For the within-day analyses, I apply two data-cleaning steps that follow Condra et al. (2018) and address known reporting artifacts in SIGACTS time stamps. First, I drop all incidents reported as occurring at exactly midnight (00:00:00). A disproportionately large number of observations are coded at this exact time, which Condra et al. (2018) interpret as evidence that the midnight designation was assigned to events whose actual time was not recorded; observations at any other minute of the midnight hour (e.g., 12:24 AM) are retained. Second, although event times are typically recorded down to the minute, the minute-level distribution is skewed toward natural rounding numbers: events are more frequently listed on the 45th or 30th minute of an hour than on the 43rd or 31st minute, reflecting reporter rounding behavior rather than the precise timing of attacks. To avoid this minute-level reporting bias, I aggregate event times to the hour for the hourly event study and to half-day shifts (pre-iftar versus post-iftar) for the within-day reallocation tests.

3. Demonstrations. For data on civilian protests, I use the Armed Conflict Location & Event Data (ACLED) project in Africa over the period 1997-2023, involving 48 countries (Raleigh et al., 2023). The project started by focusing on Africa, allowing for coverage dating back to 1997, thus making it the only continent with the most extensive data. The rationale behind additionally focusing on Africa in this part is its significant Muslim population, with over half a billion Muslims and at least 17 Muslim-majority countries, comprising nearly a third of the global Muslim population (Kettani, 2010). In a global survey examining the significance of religion within various religious traditions, Pew Research Center (2018) documents that Africa exhibits the highest average percentages of individuals who consider religion to be very important in their lives as well as the highest percentages of individuals who pray daily (around 80%). The countries are shown in Figure A.2.

Within ACLED, I restrict the outcome to *peaceful protests*, defined as gatherings in which demonstrators do not engage in violence or property destruction and are not met with force or intervention. The vast majority of recorded demonstrations are peaceful, and using this sub-

[18] The use of high-frequency records to capture granular, circadian behavioral shifts closely parallels recent methodological advances in the economics of religion, which leverage high-frequency telecommunications metadata to track localized, prayer-time network dips (Dube et al., 2022).

event type isolates non-violent civilian political behavior as the conceptual counterpart to the political violence outcomes examined elsewhere in the paper.

I also use the demonstration variable in the SIGACTS dataset for Afghan provinces as a within-country complement to the ACLED cross-country evidence. Because these events are logged by on-the-ground military personnel rather than media stringers, the SIGACTS data provide consistent province-level coverage including remote districts where media coverage is sparse. The SIGACTS demonstration category is broader than ACLED's peaceful-protest definition, since military reporters did not consistently code participant identity, and I therefore treat the SIGACTS evidence as a within-country corroboration of the ACLED protest result.

4. Share of Muslims. I separately merge the UCDP and ACLED datasets with datasets on the Muslim population share in each country from the World Religion Project (WRP) (Maoz and Henderson, 2013) and the Hackett et al. (2025). The WRP provides data on the number of followers and total population in five-year intervals from 1985 to 2010. I interpolate these figures to create an annual cross-country dataset for the years between the WRP's five-year markers. For the period from 2010 to 2023, I interpolate data from the Pew Research Center, which offers similar data at ten-year intervals. For the armed conflict analysis, the monthly panel includes 31 countries with an average Muslim population share of at least 75%. For the demonstration analysis, 14 African countries meet the 75% Muslim population threshold over the sample period.

5. Individual survey outcomes. To investigate the individual-level mechanisms driving the decline in civilian demonstrations, I use World Values Survey (WVS) data spanning four waves between 1999 and 2022 (Haerpfer et al., 2022). Restricting the sample to approximately 40,000 Muslim respondents across more than 20 Muslim-majority countries with exact interview dates, I construct three binary indicators to capture shifts in religious and political engagement. First, to measure daily religious intensity, I construct a prayer frequency indicator coded as 1 for respondents who report praying at least once or several times a day, and 0 otherwise.[19] Second, to capture political interest, I create an indicator coded as 1 if the respondent is ″very″ or ″somewhat″ interested in politics, and 0 otherwise. Finally, to measure protest willingness, I construct an indicator coded as 1 if the respondent ″has done″ or ″might do″ peaceful demonstrations, and 0 if they assert they ″would never″ participate.

6. Fasting hours. To explore whether the intensity of fasting mediates the relationship between Ramadan and conflict, I construct a measure of daily fasting duration for each country-year, calculated as the difference between civil twilight and sunset times at each capital city using data web-scraped from the U.S. Naval Observatory's Astronomical Applications Department.[20] Fasting hours vary across countries and years due to two factors: Ramadan's rotation through the Gregorian calendar, which shifts it between longer summer days and shorter winter days, and latitude, which amplifies seasonal daylight variation. Equatorial countries experience nearly constant fasting durations, while countries farther from the equator face swings

[19]This high threshold isolates intense daily observance, though baseline religiosity is high; nearly 72% of the sample reports praying at least once a month.

[20]Civil twilight occurs when the sun is 6° below the horizon, closely approximating the 15° angle traditionally used by Muslims to determine the start of fasting.

of up to seven hours across Ramadan cycles. Because this variation is determined entirely by geography and astronomy, it is orthogonal to conflict conditions. I include this variable in selected specifications to test whether longer fasting hours independently affect armed conflict or protest activity; I find no significant relationship with either outcome.

In all balanced panels, hours, days or months with no reported incidents are coded as zero. All Gregorian dates are converted to the Hijri calendar using a popular Python package and cross-checked against independent sources.[21] Summary statistics for all the key variables presented in this section are available in Table A.1 and Table A.2.

4.2 Estimating Equations

To examine the effect of Ramadan on conflict frequency, I use a balanced panel at the country-month level, organized by the Hijri calendar. The baseline model, estimated on countries with a Muslim population share above 75%, is:

$$\log(y_{cmt} + 1) = \beta_1 \text{Ramadan}_{mt} + \gamma_g + \theta_{ct} + \epsilon_{cmt} \tag{1}$$

where y_{cmt} is the number of state-based armed conflicts or peaceful protests in country c, Hijri month m, and Hijri year t. Ramadan_{mt} is an indicator equal to 1 if month m in year t corresponds to Ramadan, and 0 otherwise. Since Ramadan falls in the same Hijri month across all countries, this variable does not vary by country. γ_g denotes Gregorian month fixed effects, which absorb seasonal patterns tied to the solar calendar such as weather, agricultural cycles, and fighting seasons. θ_{ct} denotes country-by-Hijri-year fixed effects, which absorb all time-varying shocks specific to a given country and year, including economic conditions, political transitions, and national holidays.

Identification comes from the rotation of Ramadan through the Gregorian calendar. Because the Hijri calendar shifts approximately 10–12 days earlier each year, Ramadan falls in different Gregorian months across years, cycling through all seasons over a 33-year period. After netting fixed effects out, β_1 is identified from the year-to-year shift in which Gregorian month coincides with the fasting period. Because the rotation of Ramadan through the Gregorian calendar is governed by lunar astronomy, its timing is not determined by or responsive to conflict conditions in any given country or year.

The log transformation reduces skewness in the count data and allows interpretation of coefficients as approximate percentage changes. An alternative is Poisson pseudo-maximum likelihood (PPML), which directly models count data and accommodates zeros without a log transformation (Silva and Tenreyro, 2006; Chen and Roth, 2024). However, the identification strategy in this paper requires both Gregorian month fixed effects, which break the collinearity between Ramadan and solar-calendar seasonality that would otherwise confound identification, and country-by-Hijri-year fixed effects, which ensure the estimate is not driven by country-specific annual shocks.

[21] I use the HijriDate Python package (available at `https://pypi.org/project/hijridate`) and cross-check dates using Islamic Philosophy Online (`https://www.muslimphilosophy.com/ip/hijri.htm`) from the Institute of Oriental Studies, Zürich University.

This high-dimensional structure creates severe estimation problems for PPML on sparse conflict data: when an entire fixed-effect group contains only zeros, the likelihood has no finite maximum and the algorithm drops that group entirely, a well-documented consequence of perfect statistical separation in nonlinear models (Correia et al., 2020; Fernández-Val and Weidner, 2016). Separation is particularly costly in this setting, as it systematically excludes peaceful country-years that form the counterfactual against which the Ramadan effect is estimated. I therefore retain the log-linear OLS specification as the baseline throughout.[22].

To examine how the effect of Ramadan varies with the relative size of the Muslim population, I estimate the following specification on the full sample of countries:

$$\log(y_{cmt}+1) = \beta_1 \text{Ramadan}_{mt} + \beta_2 \text{Ramadan}_{mt} \times \%\text{Muslim}_{ct} + \gamma_g + \theta_{ct} + \epsilon_{cmt} \tag{2}$$

where β_2 captures how the effect of Ramadan on conflict changes as the Muslim population share increases. Since the specification includes country-by-year fixed effects, the level of $\%\text{Muslim}_{ct}$ is absorbed and does not enter separately.

The linear interaction in Equation 2 imposes a constant marginal effect of Muslim population share on the Ramadan coefficient. To allow for nonlinearities, I also estimate a more flexible specification that captures the effect of Ramadan within discrete population-share categories:

$$\log(y_{cmt}+1) = \beta_1 \text{Ramadan}_{mt} + \sum_{k=1}^{4} \theta_k \cdot (D_{k,ct} \times \text{Ramadan}_{mt}) + \gamma_g + \theta_{ct} + \epsilon_{cmt} \tag{3}$$

where $D_{k,ct}$ represents indicator variables for different Muslim population share categories.[23] The coefficient θ_1, for instance, captures the differential effect of Ramadan on conflict in countries with more than 75% Muslim population relative to the baseline category.

Subnational Intraday Specifications

To examine the intraday tactical substitution driven by the physiological constraints of fasting, I utilize high-frequency military data from Afghanistan. I structure the data as a panel at the province, Hijri month, and shift level. I first divide each day into two periods: a daylight fasting window from 05:00 to 17:59 and a nighttime post-iftar and pre-suhoor windows from 18:00 to 04:59.[24] After balancing the daily panel to include periods with zero incidents, I aggregate the data to the monthly level by shift to estimate the following:

[22]Columns 3 and 6 of Table 7 implement PPML within the exact-date matching design for armed conflict and protests, respectively.

[23]$D_{1,ct}$, $D_{2,ct}$, $D_{3,ct}$, and $D_{4,ct}$ are indicators for countries where the Muslim population share exceeds 75% (31 countries for UCDP and 14 for ACLED), is between 50–75% (7 and 3), between 25–50% (8 and 6), and between 1–25% (39 and 19), respectively. The baseline category consists of countries with less than 1% Muslim population (22 and 6).

[24]These cutoffs approximate the average fasting window in Afghanistan over the 2002–2015 sample period.

$$\log(y_{pmst} + 1) = \beta_1 \,\text{Ramadan}_{mt} + \beta_2 \,\text{PostIftar}_s + \beta_3 \,(\text{Ramadan}_{mt} \times \text{PostIftar}_s) + \theta_{pt} + \gamma_{g(mt)} + \epsilon_{pmst} \tag{4}$$

The dependent variable y_{pmst} is the count of attacks of a given type in province p during Hijri month m, Hijri year t, and shift s, where $s \in \{\text{day}, \text{night}\}$ denotes whether the attack occurred during the daytime fasting window (05:00–17:59) or the post-iftar and pre-suhoor window (18:00–04:59). These cutoffs correspond closely to the sample-averaged sunset (18:05) and civil twilight (05:13) times in Afghanistan over the 2002–2015 period, computed from US Naval Observatory data.

Ramadan_{mt} is an indicator equal to one when m is the ninth Hijri month, and PostIftar_s is an indicator equal to one for the nighttime shift. The coefficient of interest is β_3, the difference-in-differences estimate of the within-day Ramadan reallocation: the change in attack frequency during the post-iftar window in Ramadan months relative to non-Ramadan months, net of the average post-iftar level and the average Ramadan level. The total Ramadan effect during the post-iftar window, relative to the same window in non-Ramadan months, is given by $\beta_1 + \beta_3$; the corresponding effect during the daytime window is given by β_1 alone. The contrast between these two quantities indicates whether Ramadan reallocates combat across the day (large β_3) or shifts attack frequency uniformly (small β_3).

To map this circadian reallocation precisely, I expand the temporal resolution to the hour level and estimate a flexible event study specification:

$$\log(y_{pdht} + 1) = \beta_1 \text{Ramadan}_{dt} + \sum_{j \neq 2} \alpha_j \text{Hour}_j + \sum_{j \neq 2} \delta_j (\text{Hour}_j \times \text{Ramadan}_{dt}) + \gamma_g + \theta_{pt} + \epsilon_{pdht} \tag{5}$$

where Hour_j are indicator variables for each hour of the day local time. The base category is 02:00 AM, which serves as a neutral nighttime reference point falling comfortably between post-sunset mobilization and the pre-dawn meal. The coefficients δ_j trace out the intraday trajectory of violence during Ramadan relative to this baseline hour. Plotting these coefficients provides a direct visual test of whether labor-intensive attacks surge exclusively after sunset during Ramadan.

Fasting Hours

Following Campante and Yanagizawa-Drott (2015), I also test whether the duration of the daily fast influences the frequency of violence. I estimate the baseline equations by interacting the Ramadan indicator with the logarithm of daily fasting hours. I rely on national variation for this analysis rather than adopting a subnational grid approach (as in Hodler et al., 2024). Global subnational conflict panels are exceedingly sparse, georeferencing in standard datasets is often imprecise (Eck, 2012), and local demographic data on religious affiliation are systematically unavailable across the sample.

5 Basic Results

5.1 Effects on Armed Conflicts

Table 1 presents the baseline results on the impact of Ramadan on state-based armed conflict. Columns (1) through (5) restrict the sample to 31 countries with a Muslim population share exceeding 75%. The Ramadan coefficient is stable at 0.037 across all five specifications, significant at the 1% level, and unaffected by the progressive inclusion of Gregorian month fixed effects, country fixed effects, year fixed effects, and the most demanding country-by-year fixed effects. This stability indicates that the result is not driven by seasonality, cross-country differences in conflict levels, or country-specific annual shocks. In levels, the coefficient implies approximately 0.5 additional conflict events per predominantly Muslim country during Ramadan relative to other months, or 15 additional episodes across the sample in a single Ramadan. Given an average of about 7 fatalities per conflict incident per month in these countries, this translates into a conservative estimate of approximately 100 extra conflict-related deaths during each Ramadan.

Columns (6) and (7) expand the sample to all 107 countries. In column (6), the Ramadan indicator and its linear interaction with Muslim population share are both statistically insignificant, suggesting that the relationship does not follow a simple linear pattern. Column (7) addresses this by interacting Ramadan with discrete population-share categories. The coefficient on the interaction with 31 countries exceeding 75% Muslim population indicates that the Ramadan period raises armed conflict frequency in these countries by approximately 3.3% relative to countries with less than 1% Muslim population ($p < 0.05$). All other population-share interactions are statistically insignificant and close to zero, confirming that the Ramadan effect is concentrated in countries where Islamic observance is most widespread.

The results withstand several sensitivity checks. Restricting the control group to months immediately adjacent to Ramadan yields similar estimates (Table A.3 reports the specification using three months before and after). A placebo test on 22 countries with less than 1% Muslim population produces precisely estimated null effects, confirming that the results are not driven by non-religious seasonal factors (Table A.4, columns 1–6). Replacing the Muslim population interactions with analogous Christian population variables in the same table (columns 7–8) shows that Ramadan's effect on conflict declines as the Christian share rises, consistent with the reference group of low-Christian countries being predominantly Muslim.

As a cross-dataset validation, I replicate the analysis using politically violent events from ACLED, restricted to incidents involving state and rebel groups to approximate the UCDP definition of state-based conflict. These results, presented in Table A.5, are consistent with the main findings, with effect sizes nearly doubling in magnitude and significant at the 1% level. Finally, the estimates are generally robust to excluding any single country from the sample (Figure A.3).

Group Ideology

The baseline results establish that Ramadan is associated with an increase in state-based armed conflict in Muslim-majority countries. A natural follow-up question is *which* groups drive this increase. If the mechanism operates through religious doctrine and historical precedent, as the conceptual framework suggests, then the Ramadan effect should be concentrated among groups that explicitly invoke religious justification for armed struggle. Groups with religious identity but primarily nationalist goals should not respond as much to Ramadan, since their strategic calculus is not anchored in Islamic jurisprudence on warfare. Secular groups, whose objectives are entirely divorced from religious framing, should show no response at all, serving as a falsification benchmark. Furthermore, null or negative effects among these groups serve as a falsification of a generic routine-disruption story. If Ramadan simply reshuffled daily rhythms in a way that mechanically altered all conflict, non-religious actors operating within the same cultural geography would exhibit an identical operational response.

To test these predictions, I manually classify all 139 rebel groups in the sample into three ideological categories based on each group's founding charter, stated objectives, and primary recruitment ideology, drawing on the UCDP Actor Dataset (Brosché and Sundberg, 2024) and the Mapping Militants Project (Crenshaw and Robinson, 2025). The complete list and detailed coding logic for all groups are provided in Appendix B.[25]

The manual classification proceeds as follows:

(i) Religious-ideological (n=34) groups treat Islamic identity as constitutive of group membership and frame the establishment of religious governance, whether sharia rule, an Islamic state, or a caliphate, as the principal objective of armed struggle. Combat itself is characterized as obligatory religious worship rather than as an instrumental means to a political end. Examples include the Afghan Taliban, the Islamic State, al-Qaeda and its regional affiliates (AQAP, AQIM, JNIM), al-Shabaab, and Palestinian Islamic Jihad. These groups draw most directly on Islamic precedent and doctrine, making them the most theoretically likely to alter conflict behavior during Ramadan.

(ii) Ethno-religious (n=21) groups carry a strong religious identity but pursue primarily territorially bounded objectives such as statehood, autonomy, or localized regime change. Religion serves as a powerful mechanism for cohesion, recruitment, and external solidarity rather than as the doctrinal justification for armed struggle. Examples include Hamas, Hezbollah, and broad umbrella coalitions such as Syrian Insurgents and Kashmir Insurgents that encompass component factions with diverse political and territorial objectives. These groups allow testing whether religious identity alone is sufficient to produce a Ramadan effect, or whether an explicitly religious-ideological framework is required.

(iii) Secular-nationalist (n=84) groups operate entirely outside any religious-ideological framework. Their stated objectives are strictly ethno-nationalist, Marxist-Leninist, or factional, and religion plays no constitutive role in the armed struggle. Examples include the Kurdistan

[25] The classification covers all 139 organized armed groups identified as engaging in state-based conflict against government forces in 30 Muslim-majority countries in the sample period (1989–2022). State actors appearing as side B in the UCDP data (e.g., Government of Pakistan in interstate episodes) are dropped.

Workers' Party (PKK) and its affiliates, the Baloch separatist groups (BLA, BRA, BRAS), and Fatah. These groups serve as the falsification benchmark: if the theory holds, Ramadan should have no systematic effect on their conflict behavior.

The boundary between religious-ideological and ethno-religious is occasionally contested, particularly for groups with evolving ideological trajectories or umbrella coalitions encompassing heterogeneous factions. To verify that the empirical results are not driven by classification choices on contested groups, I re-estimate all main specifications excluding these groups sequentially and as a bloc. The estimates are statistically and substantively unchanged (available upon request).

Table 2 examines the impact of Ramadan on conflict intensity and fatalities in Columns (1)–(3) and Columns (4)–6 by group ideology, respectively. The results reveal a sharp gradient. Religious-ideological groups exhibit a nearly 3.3% increase in conflict events during Ramadan (column 1) and a 4.6% increase in fatalities (column 4), the results are statistically different below the 5% and 1% thresholds, respectively. The fatality response exceeds the event response, consistent with Berman and Laitin (2008)'s evidence that religious militant organizations achieve higher lethality per attack than secular counterparts because costly ritual requirements screen for high-commitment operatives. The coefficients for both ethno-religious and secular-nationalist groups show a precisely estimated null effect on both outcomes.

Two features of these results stand out. First, the null effects for ethno-religious and secular nationalist groups provide a sharp test of the mechanism. Despite operating in the same countries and sharing the same religious calendar as religious-ideological factions, groups like the Syrian insurgent coalition exhibit no comparable escalation. This suggests the mechanism is strictly doctrinal: mobilizing for violence requires an ideological framework justifying warfare, not simply a religious label. Second, these results weigh heavily against a generic time-reallocation channel that drives all conflict seasonality. If mechanical disruptions to daily routines were the primary driver, they would similarly alter the operational tempo of all armed actors in these environments, regardless of ideology.

The finding that this escalation is strictly isolated to religious-ideological groups also aligns cleanly with the "club model" of religious terrorism (Berman and Laitin, 2008; Berman, 2011). While fighters across various groups may observe the fast, religious-ideological organizations uniquely couple this physical ordeal with a strict doctrinal mandate for offensive warfare. In this framework, the extreme sacrifice required to execute combat operations while fasting acts as a severe screening mechanism. It weeds out free-riders and tightens group cohesion, enhancing operational lethality specifically for organizations that demand both ritual purity and martial sacrifice.

Aggression Attribution

The preceding analysis establishes *which* groups are involved in the Ramadan conflict increase, but not *who initiates* the violence. The UCDP GED records both parties to each conflict event but does not systematically code which side was the aggressor. If religious-ideological rebels are driving the Ramadan increase by invoking Islamic precedents for offensive warfare, the effect should concentrate in rebel-initiated events. If instead governments intensify (preemptive) strikes during Ramadan, the coefficient would appear stronger on government-initiated events.

To distinguish aggressor from target, I classify each conflict event using a large language model.[26] Each event's source description, together with the identities of Side A (government) and Side B (rebel group), is passed to GPT-5-mini through the GABRIEL framework (Asirvatham et al., 2026), which returns one of three labels: Side A, Side B, or Neutral. Events whose descriptions do not unambiguously identify an aggressor are coded Neutral, a conservative default that trades sample size for attribution accuracy.[27] Neutral events constitute 57% of the sample and are excluded from the aggressor analysis. The remaining 43% split into 26% government-initiated and 17% rebel-initiated (see Table A.6 for examples).

Table 3 presents the results, disaggregating the data by both ideological category and aggressor type to unpack the mechanisms driving the Ramadan escalation. The findings sharpen the baseline results considerably. Among religious-ideological groups, rebel-initiated conflict events surge by 3.9% during Ramadan, a result that is statistically significant at the 1% level (column 1). Concurrently, government-initiated events against these same groups also rise by 1.2%, significant at the 5% level (column 2). Because both sides exhibit a statistically significant escalation, the data points to a broader, interactive intensification of the conflict environment rather than a strictly one-sided offensive. However, the magnitude of the rebel-driven escalation is more than three times larger. This dynamic is highly consistent with the hypothesis that religious-ideological groups actively operationalize Islamic historical precedents to justify offensive campaigns during the holy month, which in turn draws a corresponding (though smaller) military response from state forces.

The subsequent columns provide a rigorous falsification test. For ethno-religious groups, both rebel-initiated (column 3) and government-initiated (column 4) events return precisely estimated null effects. This confirms a crucial theoretical boundary: a shared Islamic identity, absent a strict religious-ideological mandate for governance, is insufficient to trigger a Ramadan escalation from either side of the conflict.

For secular-nationalist groups, rebel-initiated events yield a precise zero (column 5), offering the cleanest possible rejection of the mechanical across-the-board time-reallocation alternative. Notably, there is a very small increase in government-initiated events against secular targets that is only marginally significant (column 6). Interpreted cautiously, this may suggest a strategic substitution by the state. Governments might opportunistically redirect certain

[26] An alternative approach cross-references UCDP events against the Global Terrorism Database and codes events appearing in both as rebel-initiated (see Gassebner et al., 2023). This method is restricted to incidents meeting the GTD's narrower terrorism definition and miscodes conventional rebel-initiated military engagements. The LLM-based approach classifies each event on its own description, accommodating the full range of conflict types in UCDP.

[27] GABRIEL's `classify` function uses log-probability extraction at temperature zero, yielding deterministic and reproducible output. Events with missing source text or containing only the UCDP aggregation tag (approximately 1,390 of 156,685 observations) are automatically coded as Neutral. Full implementation, validation, and benchmarking of the framework are documented in Asirvatham et al. (2026); all prompts and code used here are in the replication package.

military operations toward secular insurgents during Ramadan, capitalizing on the lower domestic political costs of striking unobservant communities during a sacred period.[28]

While the baseline classification relies strictly on the stated objectives and founding charters of armed groups, insurgent ideological boundaries are frequently contested in qualitative conflict literature. To ensure the core findings are not artifacts of subjective coding decisions, I re-estimate the primary specifications using a ”strict” classification sample that excludes prominent borderline cases. From the religious-ideological category, I remove the Afghan Taliban, Palestinian Islamic Jihad, and the Macina Liberation Front due to their respective overlaps with Pashtun tribal codes, territorial nationalism, and localized Fulani grievances. From the ethno-religious category, I exclude broad umbrella coalitions—namely Syrian and Kashmir insurgents—as well as Hamas and Hezbollah, whose foundational religious charters complicate their current nationalist operational realities. Finally, because the secular-nationalist category is defined by the explicit absence of a religious governance mandate, its empirical boundaries are inherently less ambiguous and require fewer exclusions. Nevertheless, the result is robust to dropping any one group. The baseline findings remain unchanged when these contested groups are excluded, confirming that the divergent behavioral responses to Ramadan are deeply structural rather than driven by idiosyncratic conflicts.[29]

The results point to a nuanced causal chain. Ramadan systematically intensifies armed conflict in predominantly Muslim countries, but this intensification is exclusively concentrated in conflicts involving insurgencies whose foundational ideologies demand strict religious governance. Within these specific conflicts, while both sides significantly escalate their operations, the surge is disproportionately driven by rebel-initiated violence. In contrast, neither broad religious identity nor secular objectives produce a comparable insurgent behavioral shift.

5.2 Timing Attacks: Evidence from Afghanistan

How do insurgents operationally manage the Ramadan escalation given the physical constraints of fasting? Strict adherence to the dawn-to-sunset fast makes sustained close combat physiologically hard during peak fasting hours. If insurgent organizations rationally allocate scarce resources to survive (Shapiro, 2013), high-exertion attacks should shift toward post-iftar hours when fighters have eaten and rehydrated, while attacks requiring minimal real-time effort should show no such circadian reallocation.

The three principal attack types in the SIGACTS data sit at different points on this exertion spectrum and therefore provide a natural test of the mechanism. Direct fire is the most demanding: small-arms ambushes and rocket-propelled grenade engagements require fighters to maneuver into firing position, sustain accurate fire under return contact, and exfiltrate, all while bearing weapons and ammunition over uneven terrain. Indirect fire is intermediate. Mortars and rockets are launched from prepared positions at distance from the target, so once

[28] As an additional robustness check, I also classified the unstructured event descriptions using a standard zero-shot prompting approach and manual classification of 500 incidents for training via the OpenAI API (`gpt-4o-mini`) rather than GABRIEL's log-probability extraction. The regression estimates remain highly robust to this alternative text classification strategy and are available upon request.

[29] These results are available upon request.

the firing point is established and the rounds are pre-positioned, the engagement itself is brief and the team can disperse before counter-battery fire arrives; the physical bottleneck is the setup, not the strike. IED explosions sit at the bottom of the gradient: the device is emplaced hours or days in advance and detonated either by command wire, pressure plate, or radio trigger, so the effort at the moment of attack is essentially zero.[30] If physiological constraints are the binding factor during Ramadan, the within-day reallocation should track exactly this ordering: largest for DF, smaller for IDF, and smallest for IED explosions.

Table 4 has two panels. Panel A reports the monthly Ramadan effect on insurgent violence in Afghanistan; Panel B decomposes each province-Hijri-month into pre-iftar (dawn-to-sunset, fasting) and post-iftar (sunset-to-dawn) shifts. The first three columns disaggregate insurgent attacks by physical exertion: direct fire requires sustained close combat, indirect fire requires brief positioning effort, and IED explosions require none. If fasting drives the violence patterns, post-iftar reallocation should follow this ordering. Column 4 reports state-initiated direct fire against insurgents. Columns 5 and 6 disaggregate the column 1 direct-fire effect by target group (Coalition forces and Afghan state forces) to probe whether the Ramadan response varies with the technological capability of the opponent.

Column 1 shows that the monthly insurgent DF rises by 8.7% during Ramadan months (Panel A, significant at the 1% level), a sizeable aggregate effect consistent with the cross-country UCDP findings at the province level. The within-day decomposition in Panel B reveals that this aggregate is entirely a circadian reallocation rather than a uniform escalation. The Ramadan day effect is null, while the Ramadan$\times$Post-Iftar interaction term is 0.127, significant at the 1% level, implying a 13.5% shift toward post-iftar hours. The entire Ramadan-period DF elevation concentrates after sunset, exactly when fasting fatigue lifts and physical capacity returns. Insurgents do not attack more in total during Ramadan daytime hours; they shift the operational tempo from a daytime window into a more effective nighttime window.

Column 2 reports indirect fire attacks such as mortars, rockets, and recoilless rifle fire launched from prepared positions some distance from the target. The monthly Ramadan coefficient is statistically zero (1.0% in Panel A), with no aggregate escalation in IDF activity during Ramadan months. Within day, the interaction term is 0.047 (Panel B, significant at the 10% level), a modest 4.8% post-iftar shift. The magnitude is smaller than the direct-fire reallocation in column 1, exactly as the exertion logic predicts. Launcher teams still need to reach the firing position and operate the launcher, but mortar and rocket engagements are brief enough that the constraint binds less tightly than for sustained direct-fire firefights.

Column 3 reports IED explosions. Monthly IEDs rise by 10.9% during Ramadan (Panel A, significant at the 1% level), the largest aggregate effect in the table. Within day, the picture flips relative to direct fire: the interaction collapses to 0.025 and is not significant, while the Ramadan main effect is 0.094 (Panel B, significant at the 1% level), a 9.9% pre-iftar rise distributed

[30]Throughout, I focus on IED *explosions* rather than total IED events recorded in SIGACTS, which would also include cache discoveries and pre-detonation finds by patrols. Pre-detonation discoveries reflect coalition counter-IED operations rather than insurgent attack decisions, so including them would conflate the supply of insurgent attacks with the demand from coalition search activity. Restricting to explosions isolates the insurgent decision margin.

evenly across both shifts. Detonating a pre-positioned device requires no real-time physical exertion: the device is emplaced in advance, often days or weeks earlier, and triggered remotely or by command wire. Fasting does not constrain when devices are triggered, so the Ramadan escalation in IEDs scales up activity uniformly across the fasting and non-fasting hours alike. Across columns 1 through 3, the three insurgent attack types map onto the exertion ordering as cleanly as the data allow: direct fire reallocates sharply, indirect fire reallocates modestly, and IED explosions do not reallocate at all.

Column 4 shows that state DF rises by 7.9% during Ramadan (Panel A, significant at the 1% level), but the within-day decomposition fails the placebo test in the relevant direction. The interaction term in Panel B is 0.015 and not significant, against 0.127 for insurgent DF in column 1. State forces do not redistribute their attacks across the day during Ramadan in the way insurgents do, which rules out generic conflict-intensity confounds and the alternative reading that coalition patrol movements drive the column 1 and column 2 patterns. State DF does rise by 3.5% pre-iftar (significant at the 10% level), most parsimoniously reflecting reactive engagement with the elevated insurgent IED activity in the same window documented in column 3. The comparison is already informative at the pooled state level: insurgents reallocate sharply toward post-iftar, state forces do not, and the modest state daytime response tracks insurgent IED timing rather than the religious calendar.[31]

Column 5 reports insurgent direct fire against Coalition forces. Monthly DF on Coalition targets rises by 7.3% during Ramadan (Panel A, significant at the 1% level), in line with the aggregate column 1 effect. Within day, however, the reallocation pattern is conspicuously absent: the interaction is 0.02 and not significant (Panel B). The Coalition-target Ramadan effect is uniform daytime escalation rather than a circadian shift toward post-iftar hours. Insurgents attack Coalition forces more during Ramadan, but they do not move those attacks into the nighttime window.

Column 6 reports insurgent direct fire against Afghan state forces. Monthly DF on Afghan targets is statistically zero. The within-day decomposition shows that Ramadan day effect in Panel B is a 4.4% pre-iftar decline significant at the 10% level, while the interaction shows a 12.6% post-iftar shift significant at the 1% level. The monthly null masks a pure circadian redistribution: daytime attacks on Afghan forces fall, nighttime attacks rise correspondingly, and the totals across the day cancel almost exactly to zero. The aggregate column 1 effect, which pools across Coalition and Afghan targets, therefore captures only the Coalition-side increase that has no reallocation component. The contrast between columns 5 and 6 is the central asymmetry in the table. Insurgents do not reallocate attacks against Coalition forces, but they reallocate sharply against Afghan forces. Pure fasting fatigue cannot generate this pattern, since fasting affects insurgents identically regardless of whom they attack. The asymmetry follows naturally once technological capability enters. Coalition forces possess night-vision optics, drone surveillance, and rapid air support, all of which raise the marginal cost of attacking them after dark; Afghan units lack equivalent night-fighting capability. Insur-

[31] A finer decomposition of state DF into Coalition-led versus Afghan-led operations would add useful texture but lies outside what SIGACTS captures for Friendly Action events, where the target rather than the actor is recorded in the data.

gents therefore optimize jointly: physiology pushes them toward post-iftar hours, and target vulnerability pushes them toward Afghan rather than Coalition forces in those hours. The within-day decomposition reveals not one mechanism but two: fasting physiology determines when insurgents attack, and technological asymmetry determines whom.

Figure 1 plots the hour-by-hour coefficients from Equation 5. The direct fire estimates (dark teal triangles) exhibit a sharp circadian profile: a brief morning increase around hours 7–8, presumably reflecting residual energy from the pre-dawn suhoor meal; a steady decline through midday as caloric deficit and dehydration mount, with point estimates falling below zero around hours 14–15 when sports-medicine evidence places the trough of fasted physical performance (Chaouachi et al., 2009); and a pronounced upward break immediately after iftar, with elevated coefficients sustained through the night.

The indirect fire coefficients (black squares) trace a similar but flatter trajectory, consistent with their intermediate exertion intensity — the morning bump and afternoon dip are visible but muted, and the post-iftar rebound is smaller. The IED-explosion coefficients (midnight blue circles) lack the post-iftar inflection entirely: they rise in the morning and remain elevated and roughly flat through the rest of the day, including during peak fasting hours when DF activity collapses. The hour-by-hour figure visualizes the same pattern that Table 4 captures in the half-day specification — the exertion gradient maps directly onto the within-day reallocation profile, and the reallocation is sharpest precisely where physical capacity matters most.

5.3 Effects on Civilian Political Behavior

Table 5 examines Ramadan's effect on civilian protest activity, beginning with cross-country evidence from ACLED. Column 1, restricted to the 14 African countries with Muslim population shares above 75%, shows a marginally significant 6.3% decline in peaceful protests during Ramadan (significant at the 10% level). The small sample limits statistical power, but the underlying pattern emerges clearly once the full 48-country sample is introduced. Column 2 interacts Ramadan with the continuous Muslim population share: the interaction coefficient of -0.117 (p=0.01) indicates that Ramadan's suppressive effect strengthens substantially as religious composition rises. The positive Ramadan main effect of 0.052 (significant at the 1% level) captures the baseline trend in non-Muslim countries, confirming that the calendar timing carries no inherent depressive effect where Ramadan lacks social salience. Column 3 replaces the linear interaction with discrete population-share categories. Countries with Muslim populations above 75% experience a 13.2% relative decline in demonstrations during Ramadan compared to the baseline group of countries below 1% Muslim (significant at the 5% level), and no other share category reaches conventional significance. The protest suppression is structurally concentrated where Muslim observance is widespread.

Columns 4 and 5 turn to within-country evidence using the SIGACTS demonstration variable for Afghan provinces. Column 4 shows that monthly demonstrations fall by 3.7% during Ramadan at the province level (significant at the 1% level), corroborating the cross-country result within a predominanly Muslim setting using a non-media data source. Column 5 decomposes the monthly effect into pre-iftar and post-iftar half-day shifts. The Ramadan suppression is concentrated in the daytime: pre-iftar demonstrations fall by 2.8% (significant at the 1% level), while the combined post-iftar effect (β_1 + β_3 = -0.006) is statistically indistinguishable from zero. The post-iftar coefficient should not be read as a behavioral response. The Post-Iftar effect of -0.051 confirms that demonstrations cluster in daylight hours and are rare after sunset. The post-iftar baseline is already near a floor, leaving little for Ramadan to suppress in that window. The pattern is consistent with the time-allocation channel of Mechanism 3: Ramadan intensifies religious observance and modified work routines precisely during the daytime hours when civilian protests typically occur, crowding out the conventional mobilization window.

Individual Survey Evidence

Having documented that demonstrations decline at the country and subnational levels during Ramadan, I now ask whether the same pattern appears at the individual level using World Values Survey data from 23 Muslim-majority countries, where respondents' exact interview dates allow me to identify Ramadan exposure through matching to the Hijri calendar.

A natural concern in matching WVS respondents to the Hijri calendar is that the Ramadan and non-Ramadan interview groups may differ on observable characteristics in ways that confound the comparison. Appendix Table A.7 reports balance tests on eight demographic covariates. Five are statistically balanced: gender, marital status, number of children, education, and employment show no significant differences across the two groups. Three imbalances are detectable, all small in magnitude. Respondents interviewed during Ramadan are on average 3.1 years older, an offset of roughly 8% of the age standard deviation; their reported income is 0.13 points lower on the 10-point scale, about 6% of the income standard deviation; and they are 1.8 percentage points more likely to report religion as "very important" rather than "rather important." The age and income imbalances are absorbed by the demographic controls in the main regressions. The religion-importance imbalance is small in magnitude and the regressions control directly for self-reported religious importance, which absorbs this residual compositional difference. Fieldwork intensity is well balanced: the average number of interviews per country-Hijri-year-Hijri-month cell is statistically indistinguishable across Ramadan and non-Ramadan periods (342 vs. 399), as is the average number of distinct interview days per cell (8.6 vs. 9.3), consistent with standardized fieldwork schedules driven by logistical and budgetary constraints rather than the religious calendar.

The analytical sample is restricted to Muslim respondents from 23 Muslim-majority countries who report religion as either "very important" or "rather important", a criterion met by approximately 95% of Muslim respondents in the sample. This individual-level restriction is conceptually analogous to the country-level restriction to Muslim-majority contexts: the Ramadan mechanism operates through religious observance during the holy month, and the relevant population is the one for whom the treatment is plausibly active. Respondents who report low or no religious importance are unlikely to alter their practice during Ramadan and would contribute attenuating noise to the estimates without adding identification leverage.

Within this sample, country-by-Hijri-year fixed effects absorb differences in fieldwork intensity and respondent composition across survey waves, while Gregorian month fixed effects absorb seasonal scheduling patterns and any climatic or institutional rhythms that systematically vary across calendar months. What remains is variation in whether a given respondent happened to be interviewed during Ramadan as opposed to an adjacent Hijri month within the same country-wave-season cell.

Table 6 presents the estimates for three individual-level outcomes, with each outcome reported in two columns: a specification restricted to a seven-month Hijri window centered on Ramadan (Jamada-Al-Thani through Dhul-Hijjah) and a specification using the full twelve-month sample. The motivation for the narrower window is to compare Ramadan-period respondents to interviews conducted in Hijri-adjacent months, where seasonality, fieldwork composition, and respondent demographics are closest to those during Ramadan; the full sample provides the broader robustness check.

Columns (1)–(2) show that high-intensity prayer increases by 5.1–5.2 percentage points during Ramadan, significant at the 10% level, off a baseline of roughly 76%. Daily or near-daily prayer is a habitual and largely rigid behavior, and the elevated baseline reflects the religious-importance restriction; that a 5 percentage-point shift is detectable even in this already-observant population confirms that Ramadan measurably heightens religious engagement at the individual level. Columns (3)–(4) show that interest in politics declines by 5.4–5.5 percentage points, significant at the 5% and 10% levels respectively, about a 12% reduction from the baseline mean. Columns (5)–(6) show that willingness to attend lawful peaceful demonstrations declines by 3.9–4.2 percentage points, significant at the 5% level, about an 11% reduction off baseline. The coefficients are nearly identical across the seven-month-window and full-sample specifications, indicating that the results do not depend on the specific Hijri-month comparison group and that the Gregorian month fixed effects already absorb the seasonality component the narrower window further isolates.

The pattern across the three outcomes is internally consistent and difficult to attribute to selection alone. For compositional shifts to explain the results, respondents interviewed during Ramadan would need to simultaneously exhibit higher religiosity, lower political interest, and lower protest willingness relative to those interviewed in adjacent weeks within the same country-wave-season cell, conditional on all observables. The demographic controls in the main regressions absorb the small observable imbalances documented in the balance test, including the residual religious-importance difference that operates in the same direction as the predicted effects. The more parsimonious interpretation is that Ramadan redirects individual attention from political engagement toward religious observance.

To assess whether the effects in Table 6 reflect individual religious observance rather than country-level fieldwork artifacts or seasonal patterns the fixed effects fail to absorb, I estimate the same six specifications on non-Muslim respondents living in the same 23 Muslim-majority countries (Appendix Table A.8). Non-Muslims share the same fieldwork timing, country-wave composition, and broader societal context as Muslim respondents but do not observe Ramadan, so the Ramadan indicator should have no effect on their religiosity or political engagement if the main results operate through the religious-observance mechanism. None of the six placebo coefficients reaches statistical significance. The prayer placebo coefficients are slightly negative (-0.033 and -0.039), pointing in the opposite direction from the positive Muslim coefficients and confirming that Muslim prayer responds to the religious calendar while non-Muslim prayer in the same societal context does not. The rest of the results are not statistically different from zero, supporting the interpretation of the Ramadan effects as operating through individual religious observance rather than through country-level confounds.

6 Discussion

The central results are clear. Religious-ideological insurgent groups escalate attacks during Ramadan, reallocating labor-intensive operations to the post-iftar window when fasting fatigue lifts, while ethno-religious and secular-nationalist insurgents operating in the same conflict environments show precisely estimated null effects. By contrast, Ramadan suppresses civilian protest at the country level, and individual survey evidence shows that respondents shift toward intensified prayer and away from political interest and demonstration willingness, with parallel placebo estimates among non-Muslim respondents in the same Muslim-majority countries returning statistically null effects. Both the conflict escalation and the civilian demobilization are concentrated in countries where Muslim observance is socially prevalent and disappear where Ramadan lacks social salience. Before concluding, I address the following alternative accounts that could in principle generate or contribute to the observed patterns: seasonal confounds tied to the Gregorian calendar, anticipatory timing, differential media coverage during Ramadan, and the physiological intensity of the fast itself.

Seasonal Confounds

A natural concern is that the observed Ramadan effect simply captures seasonal patterns, such as weather, agricultural cycles, or fighting seasons. Several features of the empirical design rule this out. First, no solar season can explain the intraday shift of high-exertion attacks to post-iftar hours in Afghanistan, nor the quasi-random survey evidence linking interview timing to immediate behavioral shifts. Second, the panel data span nearly a full 33-year lunar cycle (34 years for UCDP, 27 for ACLED). Because Ramadan rotates through every solar month over this period, its timing breaks mechanically from any fixed Gregorian confounder. Finally, the models explicitly absorb seasonality via the consistent inclusion of Gregorian month fixed effects.

As a further robustness check, I implement a novel Gregorian calendar matching design that exploits the precise rate at which Ramadan drifts through the solar calendar. Because the Islamic lunar year is about 10 days shorter than the Gregorian year, the last 10 days of Ramadan in year t fall on nearly the same Gregorian dates as the early post-Ramadan period in year $t+1$, by which point the lunar calendar has shifted forward and Ramadan has already concluded. This creates a natural within-date comparison: for each treated day during Ramadan in year

t, the same calendar date in year $t+1$ serves as an untreated control. Figure 3 illustrates this construction using the 2017–2018 cycle as an example.

I restrict the sample to these narrow windows, collapse the data to the country-day level, and estimate a daily analogue of Equation 1, retaining Gregorian month fixed effects but replacing country-by-Hijri-year fixed effects with country-by-cycle-year fixed effects that group each treated window with its matched control. This specification is deliberately demanding: the sample shrinks to roughly 20 days per country-cycle, standard errors are clustered at the country level, and identification relies entirely on within-pair variation across adjacent years sharing the same calendar dates. To my knowledge, this exact-date matching approach has not previously been applied in the Ramadan literature.

The treated window is based on the annual drift and deliberately covers the last 10 days of Ramadan, the religiously most intense period of the fasting month. Islamic tradition designates these days as the time when Laylat al-Qadr, the night on which the Qur'an was first revealed, occurs on one of the odd-numbered nights, and Muslims observe heightened prayer, congregation, and devotional intensity throughout this window. The physiological cost of fasting also accumulates over the month, so by the final ten days the cumulative caloric deficit, dehydration, and altered sleep schedules from suhur and tarawih reach their peak. Both the ideological-mobilization and physiological mechanisms therefore predict the largest Ramadan effects in precisely this window, making the last ten days the most powerful test of the religious-observance hypothesis as well as the most natural treated period to match against the same calendar dates in the following year.

Table 7 reports the results. Columns (1) and (2) give the cross-country day-level estimates for state-based armed conflict. The Ramadan coefficient is positive and significant in both, with the OLS estimate implying roughly a 1% daily increase per country during the last 10 days of Ramadan and the PPML estimate corresponding to an approximately 10% increase ($p < 0.01$).[32] Columns (3) and (4) probe the mechanism using within-day variation in Afghan SIGACTS direct-fire attacks, splitting each province-day into a fasting (pre-iftar) and a non-fasting (post-iftar) shift. The *Ramadan* $\times$ *Post-iftar* interaction is positive and significant in both. Direct fire therefore shifts from fasting hours toward the post-iftar window during the last ten days, with the PPML differential corresponding to roughly a 14% relative increase in post-iftar attacks. The pattern matches both the doctrinal salience of the final ten days, anchored by *Laylat al-Qadr* (the Night of Power), which creates a peak incentive structure for religious-ideological insurgents to mobilize offensive operations, and the physiological reality of combat under fasting conditions: rebels concentrate the most exertion-intensive operations in hours when they are fed and hydrated.

For demonstrations, columns (5) and (6) show that the day-level effect is negative and significant at the 1% level, with the PPML estimate implying a roughly 28% daily decline per country. The strengthening of the demonstration result in this narrower window is consistent

[32] To illustrate the severity of the separation problem discussed in Section 4.1, applying PPML with the same fixed-effects structure to the main country-month panel reduces the Muslim-majority sample from 22,196 to 7,214 observations, a loss of two-thirds, and drops 9 of 31 countries entirely. The Ramadan coefficient nonetheless remains positive and significant at the 1% level.

with the time-allocation mechanism: civilian observance peaks in the final ten days, and the crowd-out of secular political activity should be sharpest precisely there. Columns (7) and (8) carry the within-day decomposition over to Afghan SIGACTS demonstrations. The Ramadan main effect is negative and significant in both OLS and PPML, indicating that fasting hours see fewer demonstrations in the daytime during the last ten days; the *Ramadan* $\times$ *Post-iftar* interaction is positive in sign but statistically insignificant, likely reflecting the low base rate of demonstrations in the Afghan panel. Taken together, both sides of the Ramadan effect, the increase in insurgent attacks and the decline in demonstrations, survive an identification strategy that perfectly differences out the seasonal component of any specific calendar date.[33]

Anticipation Story

A more pointed alternative to the seasonal-confound concern is that the apparent Ramadan effect reflects strategic intertemporal substitution rather than within-Ramadan religious mobilization. Under this story, insurgents would deliberately suppress operations in the months leading up to Ramadan in order to conserve resources, stockpile materiel, or redirect violence into the holy month itself, artificially depressing the pre-Ramadan baseline and inflating the estimated Ramadan jump. The same concern applies to demonstrations: civilians might front-load political mobilization into pre-Ramadan months in anticipation of the upcoming religious restraint, again inflating the apparent Ramadan-to-non-Ramadan gap. Both versions of the anticipation alternative would undermine the religious-mobilization interpretation by attributing the Ramadan coefficient to behavioral planning around the calendar rather than to contemporaneous religious effects.

Neither version is plausible on substantive grounds. The conflict literature has consistently shown that the timing of insurgent operations is shaped primarily by logistics, terrain, weather, the operational tempo of opposing forces, and economic shocks (Kalyvas, 2006; Berman et al., 2011; Sexton, 2016), none of which respond meaningfully to a religious calendar more than a few weeks out. Insurgents do not have the operational flexibility to substantially compress or delay attacks based on anticipated religious dates months in advance; the constraints of materiel, intelligence cycles, and counterinsurgent activity dominate the timing decision. Demonstration timing is similarly driven by the salience of underlying political grievances and the immediacy of the issue at hand, with mobilization triggered by contemporaneous shocks such as electoral fraud, economic crises, or specific repressive acts, rather than by anticipatory calendar planning.

Despite the substantive implausibility of the anticipation story (at least more than a month ahead), the event-study design provides a direct empirical test. I estimate an event study that jointly includes indicators for each Hijri month, with country-by-Hijri-year and stricter Gregorian year-by-quarter fixed effects. Identification comes from within-country-Hijri-year, within-year-quarter variation across Hijri months, exploiting the fact that the lunar drift causes

[33] The results are robust to excluding the Eid al-Fitr holiday from the control window. Both the armed conflict and demonstration coefficients are similar.

each Hijri month to rotate through all four Gregorian quarters over the 33-year cycle. Figure 2 presents the results with Dhul-Hijjah (last month) as the reference month.

In panel (a), Ramadan is the only Hijri month to drive a positive and marginally significant increase in armed conflict (β = 0.040, p=0.052). The pre-fasting months of Rajab and Sha'ban display mildly positive and statistically insignificant coefficients (0.024 and 0.025), directly cutting against the alternative that insurgents systematically hold back operations before Ramadan; under that story the pre-Ramadan coefficients would be negative. The early Hijri-calendar months are all modestly negative and insignificant. The isolated Ramadan spike confirms that the conflict escalation is uniquely tied to the holy month rather than to slow-moving seasonal trends or strategic anticipation effects.

In panel (b), Ramadan yields the largest negative point estimate across the Islamic calendar (β=-0.1, p=0.038). All Hijri months prior to Sha'ban are insignificant and slightly positive, ruling out the symmetric concern that civilians front-load demonstrations ahead of the fasting period. The marginally significant partial suppression in Sha'ban (β = -0.056, p = 0.072) is consistent with the religiously significant character of that month in Islamic tradition, during which many Muslims observe voluntary fasting and heightened spiritual preparation for Ramadan, and reflects extension of the religious-observance mechanism rather than strategic anticipation. Shawwal remains negative and significant (β = -0.073, p = 0.029), consistent with the continued social rhythm of Eid al-Fitr celebration. The calendar-specific cluster of demobilization centered on Ramadan and extending through the month after supports the interpretation that the demonstration decline reflects contemporaneous religious observance rather than intertemporal calendar planning.[34]

Fasting Hours

One might ask whether variation in fasting intensity, rather than Ramadan observance per se, drives the results. Hodler et al. (2024) argue that longer daylight fasts mechanically suppress violence in the following year through erosion of public support for religious extremism. The theoretical link between fasting duration and behavioral change is tenuous: metabolic adaptation stabilizes the body after roughly ten hours (Anton et al., 2018; de Cabo and Mattson, 2019; Lessan and Ali, 2019), countries with longer fasts compensate with shortened work schedules and other accommodations.

Table A.10 tests the fasting-intensity channel directly by restricting the sample to Ramadan months and regressing outcomes on log fasting hours, following the panel structure in Campante and Yanagizawa-Drott (2015) and Hodler et al. (2024). Fasting hours show no association with armed conflict in Muslim-majority countries, with the precisely-estimated null ruling out the Hodler suppression channel for the violence outcome. The non-Muslim placebo sample yields a similar null (column 2), and the pooled interaction specification shows no significant

[34] The main monthly Ramadan results are also robust to dropping the two Hijri months immediately before (Rajab and Sha'ban) or immediately after (Shawwal and Dhul-Qadah) Ramadan from the comparison period. This specification mechanically eliminates any anticipation or post-Ramadan-spillover concern by removing the adjacent months entirely. The Ramadan coefficients retain their sign, statistically significant, and quantitatively similar to the baseline estimate (available upon request).

coefficient for the most-Muslim category (column 3), the group where the mechanism should operate most strongly under the Hodler hypothesis.

For demonstrations, the coefficient is significant but in an unexpected direction. Column 4 yields a positive coefficient, the opposite of the prediction that longer fasts suppress mobilization through exhaustion.[35] The non-Muslim placebo sample yields a precisely estimated null (column 5), and the interaction specification confirms that the most-Muslim category does not show the negative fasting-hours effect predicted by the Hodler mechanism (column 6). Taken together with the conflict results, the table indicates that the occurrence of Ramadan, not the duration of the daily fast, generates the observed effects on both armed conflict and civilian protest.

News Coverage

News coverage and media attention may fluctuate during religious holidays (Durante and Zhuravskaya, 2018; Eisensee and Strömberg, 2007), raising the concern that differential reporting during Ramadan could mechanically inflate or deflate recorded conflict events and demonstrations. Three features of the results argue against this. First, the armed conflict increase replicates in declassified military records from Afghanistan (SIGACTS) that bypass media reporting entirely, with within-day reallocation patterns that no plausible reporting story can generate. Second, the demonstration decline is corroborated within the same SIGACTS data at the province level and reinforced by individual survey responses on prayer frequency, political interest, and protest willingness from the WVS, which are immune to news cycles. Third, the opposite-signed effects on armed conflict and demonstrations within the same media-sourced datasets are difficult to attribute to a single reporting shift: a Ramadan-induced decline in coverage would most likely suppress both outcomes uniformly, not raise one while lowering the other.

7 Conclusion

This paper estimates the causal effect of religious observance on armed conflict and civilian political behavior, exploiting the quasi-exogenous rotation of the Hijri lunar calendar through the Gregorian solar calendar to identify the Ramadan effect under stringent within-country fixed-effect structures. The central contribution is that a single global religious shock can move armed actors and ordinary citizens in opposite directions along the political-economy spectrum. The two responses align with distinct microfoundations. The insurgent escalation is consistent with the club-good and costly-signaling logic of religious (rebel) organizations (Iannaccone, 1992; Berman and Laitin, 2008), in which sacred time activates a multiplier on

[35] This result should be interpreted cautiously. The ACLED Muslim-majority sample contains only 14 countries concentrated in North Africa above the equator (see Figure A.2), making the cross-country variation in fasting hours extremely limited. The positive sign likely reflects summer Ramadans coinciding with warmer weather and longer daylight, which naturally facilitate outdoor civic activity. The pooled interaction specification in column 6 shows no significant effect for the most-Muslim category, consistent with the column 4 estimate being driven by a narrow sample with little variation rather than by a structural fasting-hours channel.

doctrinally justified violence specific to groups with religious governance objectives. The civilian demobilization reflects a binding time-allocation constraint in the spirit of Becker (1965) and Iannaccone (1990), in which the rising shadow cost of devotional time crowds out political participation along the intensive margin. The within-day, individual-level, and exact-date matching evidence collectively reject the leading alternative mechanisms.

One scope condition is worth flagging. The design identifies an intention-to-treat parameter, isolating the effect of the Ramadan *period* rather than compliance with fasting itself; recovering the average treatment effect on the treated would require individual-level fasting compliance data unavailable across the sample. This caveat does not challenge the internal or external validity of the estimates: the empirical strategy combines cross-country panel evidence across 107 countries with subnational corroboration from Afghanistan, individual-level evidence from thousands of respondents across 23 Muslim-majority countries, and converging results across multiple datasets, levels of analysis, and identification strategies.

The findings carry implications for both empirical research and policy design. The asymmetric response across rebel ideologies indicates that static cross-sectional measures of religiosity are insufficient to characterize the equilibrium relationship between faith and violence; temporal interaction between religious observance and organizational ideology is important. For policy, international mediators have repeatedly attempted to leverage Ramadan as a de-escalation window in Afghanistan, Yemen, and Syria, on the assumption that the holy month's spiritual significance creates an opening for peace. The three-way ideology gradient suggests that sacred-time ceasefires may hold promise where insurgent actors lack a doctrinal mandate for offensive warfare, but are least likely to succeed where they are most frequently attempted. Experimental work by Ginges and Atran (2013) documents that material side-payments offered in exchange for sacralized commitments backfire by signaling that the commitment is for sale, intensifying moral outrage among those who hold the value; the symmetric prediction here is that conventional Ramadan ceasefire packages extended to religious-ideological groups may harden rather than soften their operational posture.

The civilian-side result raises a complementary question for the political economy of authoritarian governance: if the observance of Ramadan demobilizes the street, autocratic regimes may treat the period as a window to advance unpopular reforms or tighten constraints on civil society with reduced risk of public backlash.

The Ramadan-induced conflict escalation also carries downstream consequences for economic development, since armed violence disrupts labor markets, destroys infrastructure, and displaces productive activity (Federle et al., 2026; Novta and Pugacheva, 2021; Blattman and Miguel, 2010; Miguel et al., 2004). The conflict spikes documented here constitutes an unmeasured micro-foundation for the broader macroeconomic costs of Ramadan identified in prior work (Campante and Yanagizawa-Drott, 2015). The physiological constraints and doctrinal mandates that bifurcate mobilization during Ramadan offer a template for examining sacred time in other major faiths.

References

Al-Dawoody, Ahmed, *The Islamic law of war: Justifications and regulations*, Springer, 2011.

Ali, Mohamed Bin, "IS Ramadan Attacks: Perverted Interpretation of Fasting Month," 2017. RSIS Commentary No.115.

Almond, Douglas and Bhashkar Mazumder, "Health Capital and the Prenatal Environment: The Effect of Ramadan Observance During Pregnancy," *American Economic Journal: Applied Economics*, 2011, *3* (4), 56–85.

Amarasingam, Amarnath and Charlie Winter, "ISIS's Bloody Interpretation of Ramadan," *The Atlantic*, 2017.

Anton, Stephen D, Keelin Moehl, William T Donahoo, Krisztina Marosi, Stephanie A Lee, Arch G Mainous, Christiaan Leeuwenburgh, and Mark P Mattson, "Flipping the Metabolic Switch: Understanding and Applying the Health Benefits of Fasting," *Obesity*, 2018, *26* (2), 254–268.

Asirvatham, Hemanth, Elliott Mokski, and Andrei Shleifer, "GPT as a Measurement Tool," Technical Report, National Bureau of Economic Research 2026.

Aydin, Cemil, *The idea of the Muslim world: A global intellectual history*, Harvard University Press, 2017.

Azzi, Corry and Ronald Ehrenberg, "Household allocation of time and church attendance," *Journal of political Economy*, 1975, *83* (1), 27–56.

Bazzi, Samuel and Christopher Blattman, "Economic shocks and conflict: Evidence from commodity prices," *American Economic Journal: Macroeconomics*, 2014, *6* (4), 1–38.

___, Gabriel Koehler-Derrick, and Benjamin Marx, "The institutional foundations of religious politics: Evidence from indonesia," *The Quarterly Journal of Economics*, 2020, *135* (2), 845–911.

Becker, Gary S, "A Theory of the Allocation of Time," *The economic journal*, 1965, *75* (299), 493–517.

Becker, Sascha O. and Luigi Pascali, "Religion, Division of Labor, and Conflict: Anti-Semitism in Germany Over 600 Years," *American Economic Review*, 2019, *109* (5), 1764–1804.

Berman, Eli, *Radical, religious, and violent: The new economics of terrorism*, MIT press, 2011.

___ and David D Laitin, "Religion, terrorism and public goods: Testing the club model," *Journal of public Economics*, 2008, *92* (10-11), 1942–1967.

___, Jacob N Shapiro, and Joseph H Felter, "Can hearts and minds be bought? The economics of counterinsurgency in Iraq," *Journal of political Economy*, 2011, *119* (4), 766–819.

Besley, Timothy and Torsten Persson, "The logic of political violence," *The quarterly journal of economics*, 2011, *126* (3), 1411–1445.

Blattman, Christopher, "From violence to voting: War and political participation in Uganda," *American political Science review*, 2009, *103* (2), 231–247.

___ and Edward Miguel, "Civil War," *Journal of Economic Literature*, 2010, *48* (1), 3–57.

Bloom, Mia, *Dying to kill: The allure of suicide terror*, Columbia University Press, 2005.

Brosché, Johan and Ralph Sundberg, "What they are fighting for–Introducing the UCDP Conflict Issues Dataset," *Journal of Conflict Resolution*, 2024, *68* (10), 2128–2157.

Burke, Marshall, Joel Ferguson, Solomon Hsiang, and Edward Miguel, "New evidence on the economics of climate and conflict," *Handbook of the Economics of Conflict*, 2024, *1*, 249–305.

Campante, Filipe and David Yanagizawa-Drott, "Does Religion Affect Economic Growth and Happiness? Evidence from Ramadan," *Quarterly Journal of Economics*, 2015, *130* (2), 615–658.

Cantoni, Davide, David Y Yang, Noam Yuchtman, and Y Jane Zhang, "Protests as strategic games: experimental evidence from Hong Kong's antiauthoritarian movement," *The Quarterly Journal of Economics*, 2019, *134* (2), 1021–1077.

Cao, Yiming, Benjamin Enke, Armin Falk, Paola Giuliano, and Nathan Nunn, "Herding, Armed Conflict, and a Culture of Honor: Global Evidence," Technical Report, National Bureau of Economic Research 2021.

Chaouachi, Anis, Aaron J Coutts, Karim Chamari, del P Wong, Mustapha Chaouachi, Moktar Chtara, Rachida Roky, and Mohamed Amri, "Effect of Ramadan intermittent fasting on aerobic and anaerobic performance and perception of fatigue in male elite judo athletes," *The Journal of Strength & Conditioning Research*, 2009, *23* (9), 2702–2709.

Chen, Jiafeng and Jonathan Roth, "Logs with Zeros? Some Problems and Solutions," *The Quarterly Journal of Economics*, 2024, *139* (2), 891–936.

Cherif, Anissa, Bart Roelands, Romain Meeusen, and Karim Chamari, "Effects of intermittent fasting, caloric restriction, and Ramadan intermittent fasting on cognitive performance at rest and during exercise in adults," *Sports medicine*, 2016, *46* (1), 35–47.

Clark, Kate, "The Layha," *AAN Thematic Report*, 2011, pp. 1–32.

Clingingsmith, David, Asim Ijaz Khwaja, and Michael Kremer, "Estimating the Impact of the Hajj: Religion and Tolerance in Islam's Global Gathering," *Quarterly Journal of Economics*, 2009, *124* (3), 1133–1170.

Condra, Luke N., James D. Long, Andrew C. Shaver, and Austin L. Wright, "The Logic of Insurgent Electoral Violence," *American Economic Review*, 2018, *108* (11), 3199–3231.

Cook, David, *Understanding jihad*, Univ of California Press, 2015.

Correia, Sergio, Paulo Guimarães, and Tom Zylkin, "Fast Poisson estimation with high-dimensional fixed effects," *The Stata Journal*, 2020, *20* (1), 95–115.

Crenshaw, Martha and Kris Robinson, "Mapping Militants Project," 2025.

de Cabo, Rafael and Mark P Mattson, "Effects of Intermittent Fasting on Health, Aging, and Disease," *New England Journal of Medicine*, 2019, *381* (26), 2541–2551.

Dube, Oeindrila and Juan F. Vargas, "Commodity Price Shocks and Civil Conflict: Evidence from Colombia," *Review of Economic Studies*, 2013, *80* (4), 1384–1421.

——, **Joshua Blumenstock, and Michael Callen**, "Measuring religion from behavior: Climate shocks and religious adherence in Afghanistan," Technical Report, National Bureau of Economic Research 2022.

Durante, Ruben and Ekaterina Zhuravskaya, "Attack when the world is not watching? US news and the Israeli-Palestinian conflict," *Journal of Political Economy*, 2018, *126* (3), 1085–1133.

Eck, Kristine, "In data we trust? A comparison of UCDP GED and ACLED conflict events datasets," *Cooperation and Conflict*, 2012, *47* (1), 124–141.

Eisensee, Thomas and David Strömberg, "News droughts, news floods, and US disaster relief," *The Quarterly Journal of Economics*, 2007, *122* (2), 693–728.

Enikolopov, Ruben, Alexey Makarin, and Maria Petrova, "Social media and protest participation: Evidence from Russia," *Econometrica*, 2020, *88* (4), 1479–1514.

Federle, Jonathan, André Meier, Gernot J Müller, Willi Mutschler, and Moritz Schularick, "The price of war," *American Economic Review*, 2026, *116* (3), 791–827.

Fernández-Val, Iván and Martin Weidner, "Individual and time effects in nonlinear panel models with large N, T," *Journal of Econometrics*, 2016, *192* (1), 291–312.

Gassebner, Martin, Paul Schaudt, and Melvin H. L. Wong, "Armed Groups: Competition and Political Violence," *Journal of Development Economics*, 2023, *162*, 103052.

Ginges, Jeremy and Scott Atran, "Sacred values and cultural conflict," *Advances in culture and psychology*, 2013, *4*, 273–301.

Girod, Desha M, Megan A Stewart, and Meir R Walters, "Mass protests and the resource curse: The politics of demobilization in rentier autocracies," *Conflict Management and Peace Science*, 2018, *35* (5), 503–522.

Gulek, Ahmet, "Driving while hungry: The effect of fasting on traffic accidents," *Journal of Development Economics*, 2024, *168*, 103260.

Hackett, Conrad, Marcin Stonawski, Yunping Tong, Stephanie Kramer, and Anne Fengyan Shi, "Dataset of Global Religious Composition Estimates for 2010 and 2020," 2025.

Haerpfer, Christian, Ronald Inglehart, Alejandro Moreno, Christian Welzel, Kseniya Kizilova, Jaime Diez-Medrano, Marta Lagos, Pippa Norris, Eduard Ponarin, Bjorn Puranen et al., "World values survey: Round seven-country-pooled datafile version 5.0," 2022.

Hegghammer, Thomas, *Jihadi culture: The art and social practices of militant Islamists*, Cambridge University Press, 2017.

Henrich, Joseph, Michal Bauer, Alessandra Cassar, Julie Chytilová, and Benjamin Grant Purzycki, "War Increases Religiosity," *Nature Human Behaviour*, 2019, *3* (2), 129–135.

Hodler, Roland, Paul A. Raschky, and Anthony Strittmatter, "Religion and Terrorism: Evidence from Ramadan Fasting," *Journal of Peace Research*, 2024, *61* (3), 351–365.

Hornung, Erik, Guido Schwerdt, and Maurizio Strazzeri, "Religious Practice and Student Performance: Evidence from Ramadan Fasting," *Journal of Economic Behavior & Organization*, 2023, *205*, 100–119.

Iannaccone, Laurence R, "Religious practice: A human capital approach," *Journal for the scientific study of religion*, 1990, pp. 297–314.

——, "Sacrifice and stigma: Reducing free-riding in cults, communes, and other collectives," *Journal of political economy*, 1992, *100* (2), 271–291.

Iyer, Sriya, "The New Economics of Religion," *Journal of Economic Literature*, 2016, *54* (2), 395–441.

Kalyvas, Stathis N, *The logic of violence in civil war*, Cambridge University Press, 2006.

Kettani, Houssain, "Muslim Population in Africa: 1950–2020," *International Journal of Environmental Science and Development*, 2010, *1* (2), 136.

Lessan, Nader and Tomader Ali, "Energy Metabolism and Intermittent Fasting: The Ramadan Perspective," *Nutrients*, 2019, *11* (5), 1192.

Limodio, Nicola, "Terrorism financing, recruitment, and attacks," *Econometrica*, 2022, *90* (4), 1711–1742.

Lowes, Sara, Benjamin Marx, and Eduardo Montero, "Religion in emerging and developing regions," *Annual Review of Economics*, 2025, *18.*

Lugo, Luis, Alan Cooperman, James Bell, Erin O'Connell, and Sandra Stencel, "The World's Muslims: Unity and Diversity," *World*, 2012.

Madestam, Andreas, Daniel Shoag, Stan Veuger, and David Yanagizawa-Drott, "Do political protests matter? evidence from the tea party movement," *The Quarterly Journal of Economics*, 2013, *128* (4), 1633–1685.

Majid, Muhammad Farhan, "The Persistent Effects of In Utero Nutrition Shocks over the Life Cycle: Evidence from Ramadan Fasting," *Journal of Development Economics*, 2015, *117*, 48–57.

Maoz, Zeev and Errol A. Henderson, "The World Religion Dataset, 1945–2010: Logic, Estimates, and Trends," *International Interactions*, 2013, *39* (3), 265–291.

McCleary, Rachel M. and Robert J. Barro, "Religion and Economy," *Journal of Economic Perspectives*, 2006, *20* (2), 49–72.

McGuirk, Eoin F and Nathan Nunn, "Transhumant pastoralism, climate change, and conflict in Africa," *Review of Economic Studies*, 2025, *92* (1), 404–441.

Miguel, Edward, Shanker Satyanath, and Ernest Sergenti, "Economic Shocks and Civil Conflict: An Instrumental Variables Approach," *Journal of Political Economy*, 2004, *112* (4), 725–753.

Mitra, Anirban and Debraj Ray, "Implications of an economic theory of conflict: Hindu-Muslim violence in India," *Journal of political economy*, 2014, *122* (4), 719–765.

Montero, Eduardo and Dean Yang, "Religious festivals and economic development: Evidence from the timing of mexican saint day festivals," *American Economic Review*, 2022, *112* (10), 3176–3214.

Munir, Muhammad, "The Layha for the Mujahideen: an analysis of the code of conduct for the Taliban fighters under Islamic law," *International Review of the Red Cross*, 2011, *93* (881), 81–102.

Novta, Natalija and Evgenia Pugacheva, "The macroeconomic costs of conflict," *Journal of Macroeconomics*, 2021, *68*, 103286.

Oosterbeek, Hessel and Bas van der Klaauw, "Ramadan, Fasting and Educational Outcomes," *Economics of Education Review*, 2013, *34*, 219–226.

Pew Research Center, "The Changing Global Religious Landscape," April 5 2017. Report.

——, "The Changing Global Religious Landscape," June 13 2018. Report.

Pope, Devin G., "Religious Worship Attendance in America: Evidence from Cellphone Data," Technical Report, National Bureau of Economic Research 2024.

Raleigh, Clionadh, Roudabeh Kishi, and Andrew Linke, "Political Instability Patterns Are Obscured by Conflict Dataset Scope Conditions, Sources, and Coding Choices," *Humanities and Social Sciences Communications*, 2023, *10* (1), 1–17.

Reese, Michael J., Keven G. Ruby, and Robert A. Pape, "Days of Action or Restraint? How the Islamic Calendar Impacts Violence," *American Political Science Review*, 2017, *111* (3), 439–459.

Rustad, Siri Aas, "Conflict trends: a global overview, 1946–2024," *Oslo: Peace Research Institute Oslo (PRIO)*, 2025.

Sexton, Renard, "Aid as a tool against insurgency: Evidence from contested and controlled territory in Afghanistan," *American Political Science Review*, 2016, *110* (4), 731–749.

Shapiro, Jacob N, "The terrorist's dilemma: Managing violent covert organizations," 2013.

Shaver, Andrew and Austin Wright, "Data on Combatant Activity during Afghanistan War Advance Scientific Investigation of Insurgency," Technical Report, Empirical Studies of Conflict Project (ESOC) 2017.

Silva, JMC Santos and Silvana Tenreyro, "The Log of Gravity," *The Review of Economics and Statistics*, 2006, *88* (4), 641–658.

Sundberg, Ralph and Erik Melander, "Introducing the UCDP Georeferenced Event Dataset," *Journal of Peace Research*, 2013, *50* (4), 523–532.

Taylor, Rob and Mirwais Harooni, "Afghan Taliban Vow Attacks over Ramadan after Being Hacked. Reuters.," 2013.

Weidmann, Nils B, "A closer look at reporting bias in conflict event data," *American Journal of Political Science*, 2016, *60* (1), 206–218.

Zelin, Aaron Y., "Jihadology," `https://jihadology.net` 2026. Online archive of militant Islamist primary source materials.

Zussman, Asaf, "The Effect of Political Violence on Religiosity," *Journal of Economic Behavior & Organization*, 2014, *104*, 64–83.

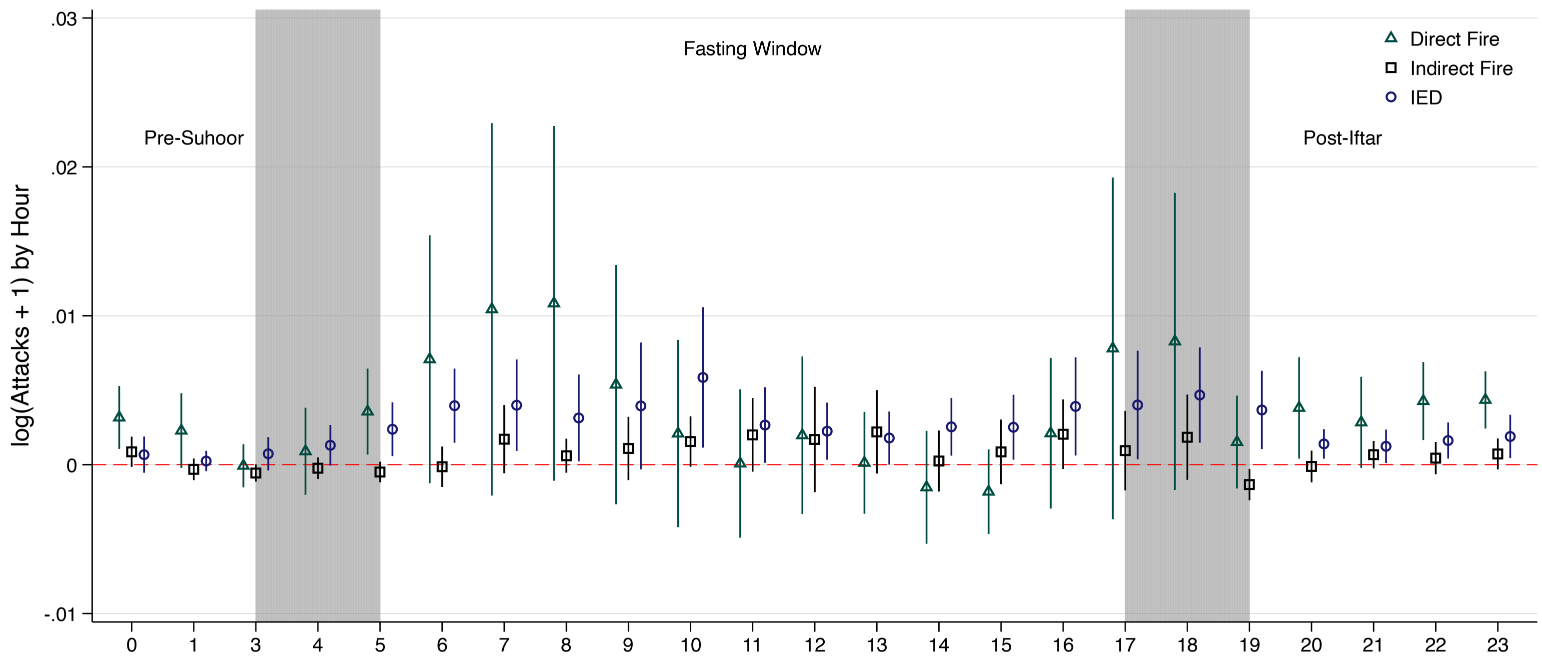

Figure 1: Within-Day Tactical Substitution During Ramadan by Attack Type: Direct Fire, Indirect Fire, and IEDs

This figure plots point estimates and 95% confidence intervals from the interaction between a Ramadan indicator and hour-of-day indicators, estimated separately for three attack-type outcomes at the province-day-hour level. The dependent variable in each model is the log-transformed hourly number of attacks plus one ($\ln(y+1)$)*. The omitted base category is 2:00 AM (local time), representing a neutral nighttime resting state. Dark teal hollow triangles represent direct fire (DF) attacks involving small arms and rocket-propelled grenades, which require sustained physical exertion in close combat. Black hollow squares represent indirect fire (IDF) attacks (mortars and rockets fired from prepared positions), which involve less continuous exertion. Midnight blue hollow circles represent IED explosions, which are pre-positioned and triggered remotely or by command wire, requiring minimal real-time effort. The middle region denotes the approximate daylight fasting window, bordered on the left by the sample-averaged pre-dawn meal (Suhoor) and on the right by sunset (Iftar). All models include province, Hijri-year, and Gregorian month fixed effects to account for cross-province differences, annual conflict trajectories, and solar seasonality. Robust standard errors are clustered at the province level (34 provinces).*

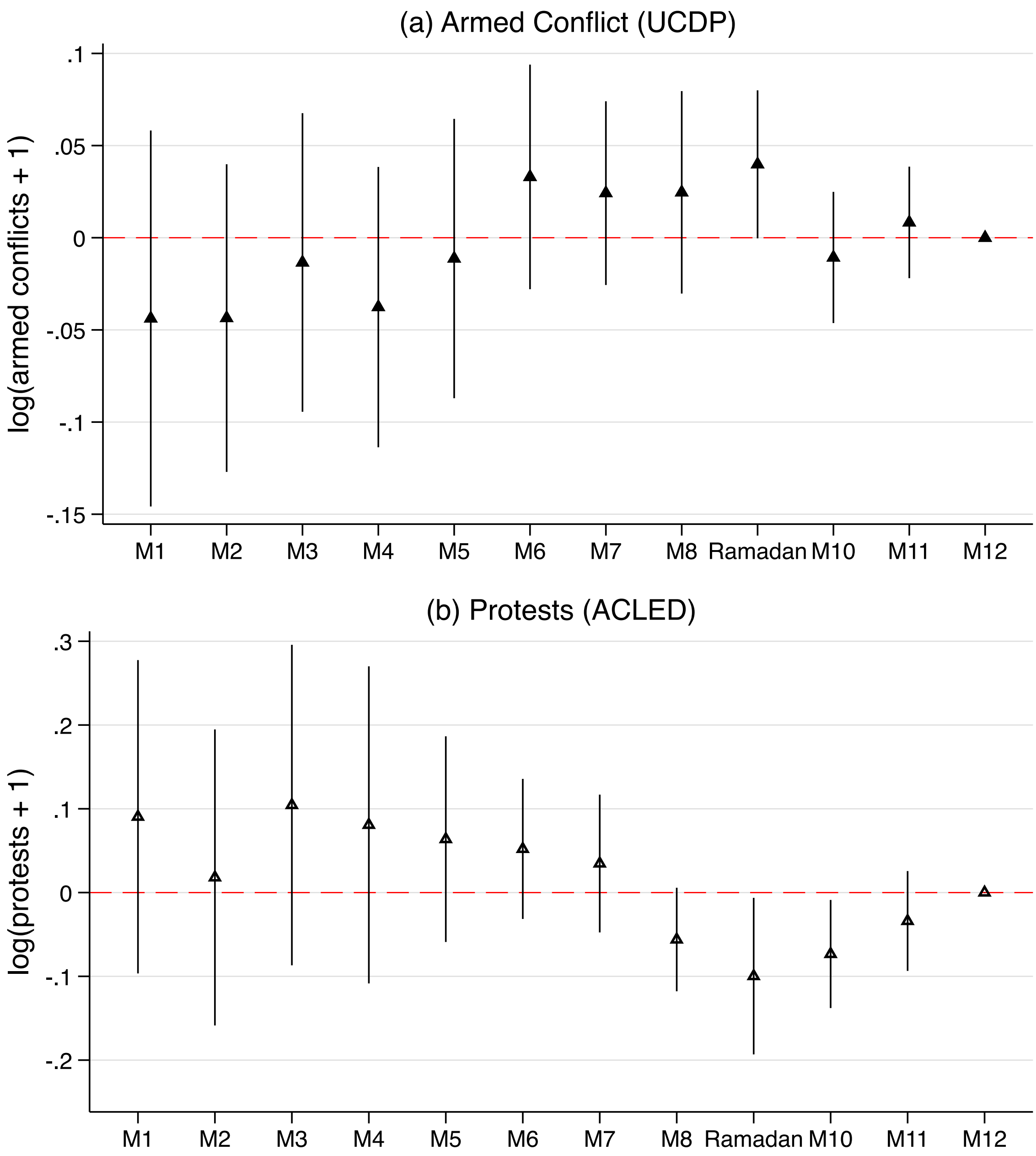

Figure 2: The Ramadan Effect Across All Hijri Months

Notes: This figure plots coefficients on all twelve Hijri month indicators from a single regression for each outcome, with Dhu al-Hijjah (M12) as the omitted base category. Panel (a) estimates the effect on log(state-based armed conflicts + 1) using UCDP data for 31 Muslim-majority countries (1989–2022). Panel (b) estimates the effect on log(demonstrations + 1) using ACLED data for 14 African Muslim-majority countries (1997–2023). Both specifications include country-by-Hijri-year and year-by-quarter fixed effects; the latter absorb year-specific seasonal shocks rather than just average seasonality, providing a more demanding test than the Gregorian month fixed effects used in the baseline regressions. Bars represent 95% confidence intervals based on robust standard errors clustered at the country level (31 clusters in panel a, 14 in panel b). Wider confidence intervals in panel (b) reflect the smaller sample.

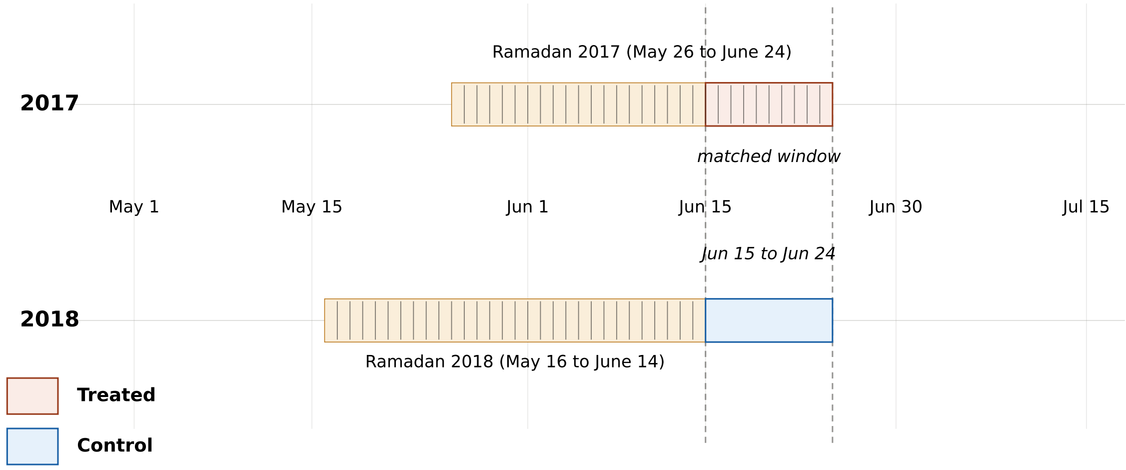

Figure 3: Exact-Date Matching Design

Notes: The Hijri calendar drifts approximately 10 days per year relative to the Gregorian calendar, so the last 10 days of Ramadan in year t fall on the same Gregorian calendar dates as the days immediately after Ramadan ends in year $t+1$. The figure illustrates the 2017–2018 cycle. In 2017 (top row), Ramadan ran from May 26 to June 24; the last 10 days, June 15 to June 24, constitute the treated window. In 2018 (bottom row), Ramadan ran from May 16 to June 14, and the same Gregorian dates, June 15 to June 24, constitute the control window. The two windows share the same solar position, absorbing any seasonal confound tied to a specific calendar day, such as weather, agricultural cycles, or recurring holidays, that month-level fixed effects cannot capture. The design pools all such consecutive-year pairs across the panel. Daily tick marks within the Ramadan blocks indicate individual days; the matched windows are highlighted with dashed vertical guides.

Table 1: The Relative Effects of Ramadan on State-Based Armed Conflict, 1989–2022

	Dependent Variable: log(armed conflict+1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	0.037***	0.037***	0.037***	0.037***	0.037***	0.008	0.002
	(0.012)	(0.013)	(0.013)	(0.012)	(0.012)	(0.008)	(0.009)
Ramadan × % Muslim						0.021	
						(0.016)	
Ramadan× >75% Muslim							0.033**
							(0.015)
Ramadan× 50–75% Muslim							-0.017
							(0.025)
Ramadan× 25–50% Muslim							-0.023
							(0.025)
Ramadan× 1–25% Muslim							0.020
							(0.014)
Conflict mean	11.84	11.84	11.84	11.84	11.84	5.046	5.046
Sample	Muslim	Muslim	Muslim	Muslim	Muslim	All	All
Observations	12648	12648	12648	12648	12648	43656	43656
Gregorian month FE		✓	✓	✓	✓	✓	✓
Country FE			✓	✓			
Year FE				✓			
Country-by-year FE					✓	✓	✓

Notes: Armed conflict is the count of state-based armed conflict events per country-month, drawn from the UCDP Georeferenced Event Dataset (GED), 1989–2022. Ramadan equals one if the calendar month corresponds to Ramadan in the Hijri calendar. In columns (1)–(5), the monthly panel sample consists of the 31 countries with at least 75% Muslims on average for the given period (World Religion Project and Pew). In columns (6)–(7), the sample consists of 107 countries. In column (7), the reference group consists of 22 countries with an average Muslim population of 1% or less. All specifications except column (1) include Gregorian month fixed effects to absorb seasonal patterns. Conflict mean is the average of the outcome in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: The Ramadan Effect on Armed Conflict and Fatalities by Group Ideology

	log(Conflict Events + 1)			log(Deaths + 1)		
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.033**	0.003	0.013	0.046***	-0.006	0.008
	(0.012)	(0.005)	(0.009)	(0.017)	(0.012)	(0.014)
Group Ideology	Relig-Ideo	Ethno-Relig	Secular	Relig-Ideo	Ethno-Relig	Secular
Outcome mean (levels)	5.325	5.554	1.210	36.77	26.97	14.49
Observations	12240	12240	12240	12240	12240	12240
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-year FE	✓	✓	✓	✓	✓	✓

Notes: This table examines the effect of Ramadan on armed conflict events and fatalities by rebel group ideology in Muslim-majority countries. Rebel groups are classified into three categories: *religious-ideological* (primary stated objective is establishing religious governance), *ethno-religious* (religious identity but primarily nationalist or territorial objectives), and *secular* (nationalist, separatist, or political objectives with no significant religious framing). Columns (1)–(3) use the log-transformed number of initiated conflict events plus one as the dependent variable. Columns (4)–(6) use the log-transformed number of fatalities (best estimate) plus one. All specifications include country-by-Hijri-year and Gregorian month fixed effects. Outcome mean reports the average of the dependent variable in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 3: The Ramadan Effect on Conflict Initiation by Group Ideology and Aggressor

	log(Initiated Conflict Events + 1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.038***	0.012**	-0.002	-0.002	-0.000	0.009*
	(0.013)	(0.005)	(0.004)	(0.003)	(0.007)	(0.005)
Group Ideology	Relig-Ideo	Relig-Ideo	Ethno-Relig	Ethno-Relig	Secular	Secular
Aggressor	Rebel	Government	Rebel	Government	Rebel	Government
Outcome mean (levels)	1.352	1.973	0.526	0.757	0.226	0.302
Observations	12240	12240	12240	12240	12240	12240
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-year FE	✓	✓	✓	✓	✓	✓

Notes: This table isolates the effect of Ramadan on conflict initiation, disaggregated by rebel group ideology and the initiating actor. The dependent variable across all columns is the log-transformed number of initiated armed conflict events plus one. Aggressor status (Rebel vs. Government) was determined by parsing unstructured UCDP event descriptions using generative AI (LLM log-probability extraction). Rebel groups are classified into *religious-ideological*, *ethno-religious*, and *secular* categories based on their primary stated objectives. All specifications include country-by-Hijri-year and Gregorian month fixed effects. The sample is restricted to Muslim-majority countries. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 4: Effect of Ramadan on Insurgent Violence in Afghanistan: Monthly Aggregates and Within-Day Reallocation

	Insurgent Attacks			State Attacks	Insurgent DF by Target	
	log(DF+1)	log(IDF+1)	log(IED+1)	log(State DF+1)	log(DF+1)	log(DF+1)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Monthly Aggregates						
Ramadan	0.083***	0.010	0.103***	0.076***	0.070***	-0.025
	(0.027)	(0.028)	(0.023)	(0.022)	(0.024)	(0.021)
Outcome mean (levels)	20.37	4.546	6.955	1.551	8.076	10.63
Observations	5,406	5,406	5,406	5,406	5,406	5,406
Panel B: Within-Day Decomposition (pre-iftar vs. post-iftar)						
Ramadan	-0.018	-0.031	0.094***	0.035*	0.028	-0.044*
	(0.024)	(0.028)	(0.021)	(0.018)	(0.021)	(0.023)
Post-Iftar	-0.367***	-0.195***	-0.509***	-0.084***	-0.258***	-0.281***
	(0.055)	(0.055)	(0.057)	(0.017)	(0.049)	(0.040)
Ramadan × Post-Iftar	0.127***	0.047*	0.025	0.015	0.020	0.119***
	(0.025)	(0.027)	(0.025)	(0.020)	(0.017)	(0.025)
Outcome mean (levels)	9.646	2.199	3.325	0.669	3.976	5.012
Observations	10,880	10,880	10,880	10,880	10,880	10,880
Target	Govt	Govt	Govt	Rebels	State Govt	Afghan Govt
Gregorian month FE	✓	✓	✓	✓	✓	✓
Province-by-Hijri-year FE	✓	✓	✓	✓	✓	✓

Notes: This table estimates the effect of Ramadan on insurgent violence in Afghanistan using the SIGACTS dataset (2002–2015). Panel A reports the monthly aggregate effect using a province-by-Hijri-month-by-Hijri-year panel. Panel B decomposes the monthly effect using a half-day shift panel that splits each province-Hijri-month into pre-iftar (dawn to sunset, encompassing the daytime fast) and post-iftar (sunset to dawn) windows. Columns 1–3 disaggregate rebel-initiated attacks by exertion type: direct fire (DF, small-arms and rocket-propelled grenade attacks requiring sustained close combat), indirect fire (IDF, mortars and rockets fired from prepared positions with less ongoing exertion), and improvised explosive device explosions (IED, pre-positioned and triggered remotely or by command wire, requiring minimal real-time effort). Column 4 reports friendly direct fire (state-initiated DF against insurgents). Columns 5 and 6 disaggregate the Column 1 DF result by target: Column 5 restricts to enemy DF events with at least one Coalition target affiliation (ISAF members and OEF allies), and Column 6 restricts to enemy DF events with at least one Afghan State Forces target affiliation (ANSF, including Afghan National Army and Afghan National Police). Target categories are not mutually exclusive when a single attack records multiple affiliations. In Panel B, the Ramadan main effect captures the pre-iftar (fasting-period) differential; the Ramadan × Post-Iftar interaction measures the additional reallocation toward post-iftar hours during Ramadan relative to non-Ramadan months. The total post-iftar Ramadan effect is the sum of the main effect and the interaction. Post-iftar windows use sample-averaged sunset (18:05) and dawn (05:13) cutoffs. All specifications include province-by-Hijri-year and Gregorian month fixed effects. Outcome mean reports the average level of the dependent variable in the estimation sample. Robust standard errors are in parentheses, clustered at the province level (34 provinces). Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 5: Effect of Ramadan on Civilian Protest Activity

	Cross-Country (ACLED)			Afghanistan (SIGACTS)	
	log(Peaceful Protests + 1)			log(Demonstrations + 1)	
	(1)	(2)	(3)	(4)	(5)
Ramadan	-0.065*	0.052***	0.062	-0.037***	-0.028***
	(0.031)	(0.019)	(0.045)	(0.013)	(0.009)
Ramadan × % Muslim		-0.117***			
		(0.039)			
Ramadan × >75% Muslim			-0.142**		
			(0.053)		
Ramadan × 50–75% Muslim			-0.039		
			(0.072)		
Ramadan × 25–50% Muslim			0.032		
			(0.054)		
Ramadan × 1–25% Muslim			-0.050		
			(0.047)		
Post-Iftar					-0.051***
					(0.015)
Ramadan × Post-Iftar					0.022**
					(0.009)
Outcome mean	7.569	4.215	4.215	0.304	0.136
Sample	Muslim	All	All	Afghanistan	Afghanistan
Observations	4,536	15,552	15,552	5,406	10,880
Gregorian month FE	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓		
Province-by-Hijri-year FE				✓	✓

Notes: This table estimates the effect of Ramadan on civilian protest activity. Columns 1–3 use the ACLED dataset for 48 African countries (1997–2023). The outcome is the logarithm of peaceful protests plus one at the country-Hijri-month level, where ACLED defines peaceful protests as demonstrations in which participants do not engage in violence or property destruction and are not met with force or intervention. Column 1 restricts the sample to 14 countries with a Muslim population share above 75%. Columns 2 and 3 use the full sample, interacting the Ramadan indicator with the continuous Muslim population share (column 2) and with discrete share categories (column 3); the baseline category in column 3 is countries with less than 1% Muslim population. Columns 4 and 5 use the SIGACTS dataset for Afghan provinces (2002–2015). The outcome is the logarithm of demonstration events plus one at the province-Hijri-month level (column 4) and at the province-Hijri-month-shift level (column 5), where each province-month is split into pre-iftar (dawn to sunset) and post-iftar (sunset to dawn) windows. In column 5, the Ramadan main effect captures the pre-iftar (fasting-period) differential and the Ramadan × Post-Iftar interaction measures the additional reallocation toward post-iftar hours during Ramadan; the total post-iftar Ramadan effect is the sum of the two coefficients. All specifications include Gregorian month fixed effects. Columns 1–3 include country-by-Hijri-year fixed effects and columns 4–5 include province-by-Hijri-year fixed effects. Outcome mean reports the average level of the dependent variable in the estimation sample. Robust standard errors are in parentheses, clustered at the country level in columns 1–3 and at the province level in columns 4–5 (34 provinces). Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 6: The Ramadan Effect on Spirituality and Political Engagement

	Spirituality/Prayer		Interest in Politics		Attending Demonstrations	
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.052*	0.051*	-0.055**	-0.054*	-0.042**	-0.039**
	(0.025)	(0.026)	(0.026)	(0.026)	(0.018)	(0.017)
Outcome mean	0.770	0.751	0.433	0.480	0.342	0.368
Sample	$\pm$3 mo.	All	$\pm$3 mo.	All	$\pm$3 mo.	All
Observations	16,744	33,149	19,326	41,183	16,718	36,813
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓

Notes: This table examines the effect of Ramadan on individual-level measures of spirituality and political engagement using data from 23 Muslim-majority countries from the World Values Survey (1999–2023). The analytical sample is restricted to Muslim respondents from Muslim-majority countries who report religion as either "very important" or "rather important" on the WVS importance-of-religion scale; approximately 95% of Muslim respondents in the WVS sample meet this criterion. The restriction is motivated by the mechanism under study: the Ramadan effect operates through religious observance during the holy month and is unlikely to affect respondents whose self-reported religious commitment is low. This individual-level restriction is conceptually analogous to the country-level restriction to Muslim-majority contexts. Observation counts vary across specifications because certain survey questions were not administered to respondents in all countries and waves. Columns (1)–(2) use a binary indicator for high-intensity prayer (praying daily or more than once a week). Columns (3)–(4) use a binary indicator for political interest (reporting being "very interested" or "somewhat interested" in politics). Columns (5)–(6) use a binary indicator for demonstration willingness (reporting having attended or willingness to attend lawful peaceful demonstrations). Odd-numbered columns restrict the sample to interviews conducted in the seven Hijri months centered on Ramadan (Jamada-Al-Thani through Dhul-Hijjah); even-numbered columns use the full twelve-month sample. All specifications include country-by-Hijri-year and Gregorian month fixed effects and the full demographic control set: gender, age, age-squared, marital status, number of children, education, employment status, income, and self-reported importance of religion. Outcome mean reports the baseline average of the dependent variable. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 7: Last 10 Days of Ramadan vs. Same Dates in Year $t+1$

	Y=Armed Conflict				Y=Demonstrations			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ramadan	0.009*	0.099***	-0.025***	-0.068	-0.020***	-0.327***	-0.004**	-0.938**
	(0.005)	(0.026)	(0.009)	(0.094)	(0.006)	(0.060)	(0.002)	(0.458)
Post-iftar			-0.126***	-1.240***			-0.005**	-1.431***
			(0.033)	(0.104)			(0.002)	(0.397)
Ramadan×Post-iftar			0.023**	0.129**			0.003	0.620
			(0.009)	(0.056)			(0.002)	(0.598)
Outcome mean (levels)	0.423	0.423	0.414	0.414	0.282	0.282	0.005	0.005
Estimator	OLS	PPML	OLS	PPML	OLS	PPML	OLS	PPML
Dep. variable	Log(Y+1)	# of Y	Log(Y+1)	# of Y	Log(Y+1)	# of Y	Log(Y+1)	# of Y
Observations	22,196	7,214	17,408	11,260	8,218	2,924	17,408	1,824
Country/Province × Cycle-year FE	✓	✓	✓	✓	✓	✓	✓	✓
Greg. month FE	✓	✓	✓	✓	✓	✓	✓	✓
Within-day			✓	✓			✓	✓

Notes: This table reports estimates from an exact Gregorian calendar matching design. Because the Islamic lunar year is approximately 10 days shorter than the Gregorian year, the last 10 days of Ramadan in year t fall on nearly the same calendar dates as the early post-Ramadan period in year $t+1$, which encompasses the 10 days of the month after Ramadan. Country (or province) by (cycle-)year fixed effects group each treated window with its matched control, ensuring that identification comes entirely from within-pair variation; Gregorian month fixed effects absorb residual cross-cycle seasonality. Columns (1)–(2) and (5)–(6) report the cross-country day-level results: state-based armed conflicts from UCDP and demonstrations from ACLED, respectively, restricted to countries where the average Muslim population share exceeds 75%. Columns (3)–(4) and (7)–(8) zoom in on Afghanistan using the SIGACTS incident database (direct-fire attacks and demonstrations), with the panel constructed at the province × Gregorian date × shift level. The shift indicator *Post-iftar* equals one for local hours 18:00–04:59 (the non-fasting window during Ramadan) and zero for the 05:00–17:59 fasting window. In within-day columns the *Ramadan* main effect captures the fasting-period differential, the *Post-iftar* coefficient captures the average evening lull, and the interaction measures the additional reallocation toward post-iftar hours during Ramadan; the total post-iftar Ramadan effect is the sum of the main effect and the interaction. OLS specifications (odd columns) use $\log(Y+1)$ as the dependent variable; PPML specifications (even columns) model the raw event count directly and are robust to the prevalence of zeros in daily data (Silva and Tenreyro, 2006; Chen and Roth, 2024). PPML drops observations separated by fixed effects, reducing the effective sample. The outcome mean reports the unconditional average of the level variable across the estimation sample (per country-day in columns 1–2 and 5–6; per province-day-shift in columns 3–4 and 7–8). Robust standard errors are in parentheses, clustered at the country (columns 1–2, 5–6) or province (columns 3–4, 7–8) level. Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Appendix

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A Figures & Tables

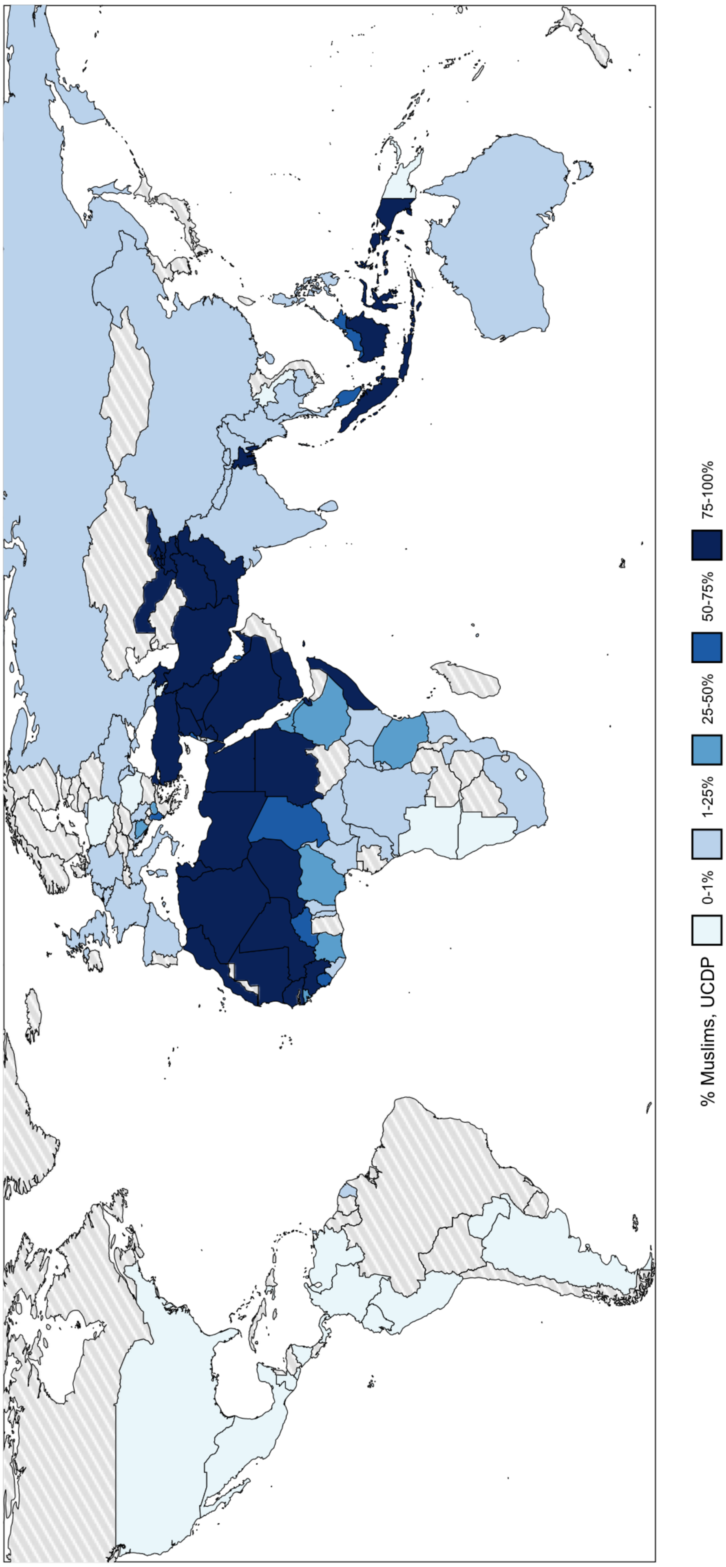

Figure A.1: Countries by Share of Muslim Population

Notes: There are 107 in-sample countries in the UCDP dataset. Hatched countries in gray do not appear in the dataset. Countries are grouped according to their average share of Muslim population from 1989 to 2022.

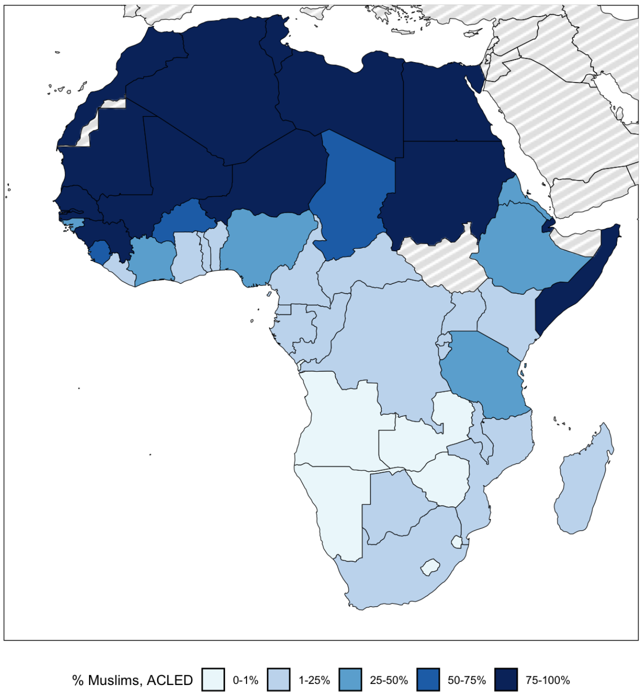

Figure A.2: African Countries by Share of Muslim Population

Notes: There are 48 in-sample countries in the ACLED dataset. Hatched countries in gray do not appear in the dataset. Countries are grouped according to their average share of Muslim population from 1997 to 2023.

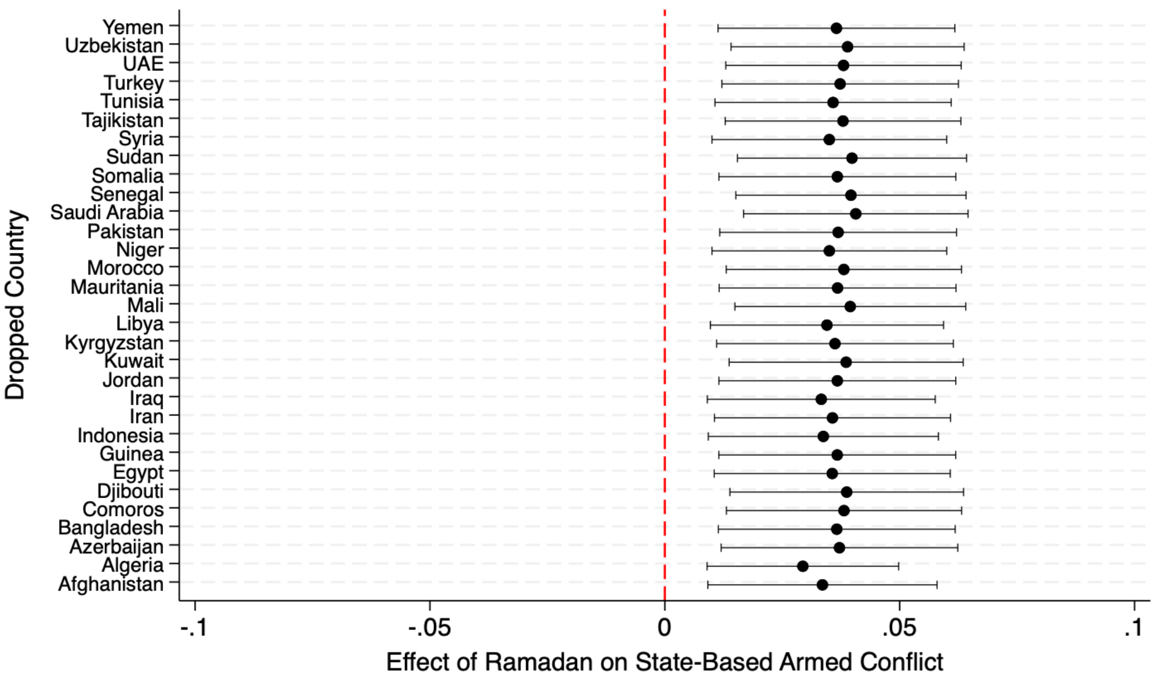

Figure A.3: Relative Effects of Armed Conflict in Ramadan in Muslim Countries Excluding 1 Country

Notes: This figure shows a leave-one-country-out sensitivity analysis of the effect of Ramadan on the intensity of state-based armed conflict, based on UCDP data from 1989–2022. The outcome is the natural logarithm of the number of conflict events plus one, measured monthly. Each point reports the Ramadan coefficient from a separate regression excluding one of the 31 Muslim-majority countries (defined as having a Muslim population share above 75%). All regressions include Gregorian month and country-by-year fixed effects. Standard errors are clustered at the country level. The red vertical dashed line denotes the zero-effect benchmark. This robustness check assesses the influence of individual countries on the overall result.

Table A.1: Summary Statistics: Armed Conflict

Variable	(1) N	(2) Mean	(3) S.D.	(4) Min	(5) Max
Panel A — UCDP Cross-Country Hijri Month (Muslim Share $>$ 75%, 31 countries)					
Log(State Conflicts+1)$_{cmt}$	12,648	0.729	1.298	0	7.602
Panel B — UCDP by Rebel Ideology (Muslim Share $>$ 75%, 30 countries)					
Log(Religious-Ideological Events+1)$_{cmt}$	12,240	0.436	1.089	0	6.787
Log(Ethno-Religious Events+1)$_{cmt}$	12,240	0.106	0.649	0	7.602
Log(Secular-Nationalist Events+1)$_{cmt}$	12,240	0.262	0.701	0	4.595
Panel C — UCDP by Aggression Attribution (Muslim Share $>$ 75%, 30 countries)					
Rebel-Initiated Events					
Log(Religious-Ideological Events+1)$_{cmt}$	12,240	0.253	0.734	0	4.745
Log(Ethno-Religious Events+1)$_{cmt}$	12,240	0.051	0.374	0	6.174
Log(Secular-Nationalist Events+1)$_{cmt}$	12,240	0.089	0.353	0	3.466
Government-Initiated Events					
Log(Religious-Ideological Events+1)$_{cmt}$	12,240	0.216	0.724	0	6.479
Log(Ethno-Religious Events+1)$_{cmt}$	12,240	0.048	0.400	0	5.481
Log(Secular-Nationalist Events+1)$_{cmt}$	12,240	0.097	0.393	0	3.892
Panel D — SIGACTS Afghanistan					
D.1 Province-Month Panel					
Log(Rebel Attacks+1)$_{pmt}$	5,542	2.252	1.803	0	7.581
Log(Direct Fire+1)$_{pmt}$	5,542	1.833	1.638	0	7.268
Log(IED Attacks+1)$_{pmt}$	5,542	1.608	1.556	0	6.645
D.2 Province-Month-Shift Panel (Pre/Post Iftar)					
Log(Rebel Attacks+1)$_{pmst}$	11,084	1.717	1.583	0	7.401
Log(Direct Fire+1)$_{pmst}$	11,084	1.356	1.416	0	7.090
Log(IED Attacks+1)$_{pmst}$	11,084	1.139	1.313	0	6.506
D.3 Province-Day-Hour Panel					
Log(Rebel Attacks+1)$_{pdht}$	3,804,192	0.038	0.182	0	3.434
Log(Direct Fire+1)$_{pdht}$	3,804,192	0.023	0.138	0	3.367
Log(IED Attacks+1)$_{pdht}$	3,804,192	0.017	0.118	0	2.944

Notes: This table presents summary statistics for all armed conflict variables. Subscripts follow the regression specifications: c denotes country, m Hijri month, t Hijri year, p province, s shift, d Gregorian day, and h hour. Panel A reports descriptives for state-based armed conflict from the UCDP GED (1989–2022) for 31 countries with a Muslim population share above 75%. The full sample covers 106 countries (43,248 observations). Panel B disaggregates log conflict events by rebel group ideology. Panel C further decomposes events by LLM-based aggression attribution using the GABRIEL framework. Panel D reports statistics from the declassified SIGACTS dataset covering insurgent attacks in Afghanistan (2002–2015) at three levels of temporal aggregation. Column (1) reports the number of observations, (2) and (3) the mean and standard deviation, and (4) and (5) the minimum and maximum values.

Table A.2: Summary Statistics: Demonstrations, Survey, Fasting Hours, and Raw Counts

Variable	(1) N	(2) Mean	(3) S.D.	(4) Min	(5) Max
Panel A — ACLED Cross-Country Hijri Month (Muslim Share > 75%, 14 countries)					
Log(Demonstrations+1)$_{cmt}$	4,536	1.082	1.361	0	5.730
Log(Political Violence+1)$_{cmt}$	4,536	0.742	1.166	0	5.652
Panel B — World Values Survey (Individual-Level, Muslim Respondents)					
High-Frequency Prayer (Dummy)$_i$	35,449	0.732	0.443	0	1
Interviewed during Ramadan (Prayer Sample)$_i$	35,449	0.121	0.326	0	1
Participation in Demonstrations (Dummy)$_i$	39,049	0.364	0.481	0	1
Interviewed during Ramadan (Demo Sample)$_i$	39,049	0.106	0.308	0	1
Interest in Politics (Dummy)$_i$	43,582	0.483	0.500	0	1
Interviewed during Ramadan (Politics Sample)$_i$	43,582	0.099	0.299	0	1
Panel C — Fasting Hours (Ramadan-Only Sample)					
C.1 ACLED Sample (Muslim Share > 75%, 14 countries)					
Log(Demonstrations+1)$_{ct}$	1,296	0.855	1.159	0	5.537
Log(Fasting Hours)$_{ct}$	1,296	2.526	0.063	2.318	2.719
C.2 UCDP Sample (Muslim Share > 75%, 31 countries)					
Log(State Conflicts+1)$_{ct}$	3,710	0.494	1.058	0	7.298
Log(Fasting Hours)$_{ct}$	3,710	2.527	0.113	2.062	2.917
Panel D — Raw Event Counts (Levels)					
D.1 UCDP Armed Conflict (Muslim Share > 75%)					
State Conflicts$_{cmt}$	12,648	11.904	80.440	0	2,001
Fatalities$_{cmt}$	12,648	79.768	507.713	0	20,214
Religious-Ideological Events$_{cmt}$	12,240	5.427	29.696	0	885
Ethno-Religious Events$_{cmt}$	12,240	5.461	72.670	0	2,001
Secular-Nationalist Events$_{cmt}$	12,240	1.220	5.494	0	98
D.2 SIGACTS Afghanistan					
Rebel Attacks$_{pt}$	5,542	45.931	130.828	0	1,960
Direct Fire$_{pt}$	5,542	26.487	76.880	0	1,433
IED Attacks$_{pt}$	5,542	19.444	59.471	0	768
D.3 ACLED Demonstrations (Muslim Share > 75%)					
Demonstrations$_{cmt}$	4,536	9.789	28.352	0	307

Notes: This table presents summary statistics for civilian demonstrations, individual-level survey outcomes, fasting intensity, and raw count levels for all conflict variables. Subscripts indicate the level of observation: c denotes country, m Hijri month, t Hijri year, p province, and i individual respondent. Panel A reports descriptive statistics from the ACLED dataset. Panel B reports individual-level responses from Muslim participants across four waves of the World Values Survey (1999–2022). Panel C reports cross-sectional fasting hours calculated as the time between civil twilight and sunset at each capital city, restricted strictly to Ramadan months to test for physiological intensity mechanisms. Panel D provides the unlogged raw counts (levels) corresponding to the core armed conflict and demonstration samples. Column (1) reports the number of observations, (2) and (3) the mean and standard deviation, and (4) and (5) the minimum and maximum values.

Table A.3: The Relative Effects of Ramadan on State-Based Armed Conflicts, 1989–2022 (Months 6–12)

	Dependent Variable: log(armed conflict+1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	0.030**	0.034***	0.034***	0.030**	0.030**	0.008	0.003
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.008)	(0.009)
Ramadan × % Muslim						0.017	
						(0.016)	
Ramadan× >75% Muslim							0.025*
							(0.015)
Ramadan× 50–75% Muslim							-0.020
							(0.024)
Ramadan× 25–50% Muslim							-0.011
							(0.025)
Ramadan× 1–25% Muslim							0.015
							(0.014)
Conflict mean	12.19	12.19	12.19	12.19	12.19	5.192	5.192
Sample	Muslim	Muslim	Muslim	Muslim	Muslim	All	All
Observations	7378	7378	7378	7378	7378	25466	25466
Gregorian month FE		✓	✓	✓	✓	✓	✓
Country FE			✓	✓			
Year FE				✓			
Country-by-year FE					✓	✓	✓

Notes: The control group of months includes the three months preceding and following Ramadan (Hijri months 6–12). Armed conflict is the count of state-based armed conflict events per country-month, drawn from the UCDP Georeferenced Event Dataset (GED), 1989–2022. Ramadan equals one if the calendar month corresponds to Ramadan in the Hijri calendar. In columns (1)–(5), the monthly panel sample consists of the 31 countries with at least 75% Muslims on average for the given period (World Religion Project and Pew). In columns (6)–(7), the sample consists of 107 countries. In column (7), the reference group consists of 22 countries with an average Muslim population of 1% or less. All specifications except column (1) include Gregorian month fixed effects to absorb seasonal patterns. Conflict mean is the average of the outcome in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: The Relative Effects of Ramadan on Conflict in Non-Muslim Countries, 1989–2022

	Dependent Variable: log(armed conflict+1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	0.002	0.002	0.002	0.002	0.002	0.029**	0.058***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.011)	(0.020)
Ramadan $\times$ % Christian						-0.028*	
						(0.016)	
Ramadan$\times$ >75% Christian							-0.044**
							(0.022)
Ramadan$\times$ 50–75% Christian							-0.069***
							(0.024)
Ramadan$\times$ 25–50% Christian							-0.070**
							(0.035)
Ramadan$\times$ 1–25% Christian							-0.034
							(0.024)
Conflict mean	0.885	0.885	0.885	0.885	0.885	5.046	5.046
Sample	Non-Muslim	Non-Muslim	Non-Muslim	Non-Muslim	Non-Muslim	All	All
Observations	8976	8976	8976	8976	8976	43656	43656
Gregorian month FE		✓	✓	✓	✓	✓	✓
Country FE			✓	✓			
Year FE				✓			
Country-by-year FE					✓	✓	✓

Notes: This table serves as a placebo test, estimating the effect of Ramadan on state-based armed conflict in countries where Ramadan observance is minimal or nonexistent. Armed conflict is the count of state-based armed conflict events per country-month, drawn from the UCDP Georeferenced Event Dataset (GED), 1989–2022. Ramadan equals one if the calendar month corresponds to Ramadan in the Hijri calendar. In columns (1)–(5), the monthly panel sample consists of 22 countries with at most 1% Muslims on average for the given period (World Religion Project and Pew). In columns (6)–(7), the sample consists of 107 countries. In column (7), the reference group consists of 12 countries with an average Christian population of 1% or less. All specifications except column (1) include Gregorian month fixed effects to absorb seasonal patterns. Conflict mean is the average of the outcome in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: The Relative Effects of Ramadan on Political Violence, 1997–2023 (ACLED)

	Dependent Variable: log(Political Violence +1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	0.065***	0.079***	0.079***	0.076***	0.076***	-0.001	0.006
	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.016)	(0.017)
Ramadan × % Muslim						0.070***	
						(0.025)	
Ramadan× >75% Muslim							0.065**
							(0.024)
Ramadan× 50–75% Muslim							-0.029
							(0.056)
Ramadan× 25–50% Muslim							0.020
							(0.055)
Ramadan× 1–25% Muslim							0.005
							(0.023)
Political Violence mean	5.260	5.260	5.260	5.260	5.260	3.437	3.437
Sample	Muslim	Muslim	Muslim	Muslim	Muslim	All	All
Observations	4536	4536	4536	4536	4536	15552	15552
Gregorian month FE		✓	✓	✓	✓	✓	✓
Country FE			✓	✓			
Year FE				✓			
Country-by-year FE					✓	✓	✓

Notes: Political violence is the count of state-based armed conflict events per country-month, drawn from the ACLED dataset, 1997–20223. In columns (1)–(5), the sample consists of 14 African countries with at least 75% Muslims on average for the given period in the World Religion Project (WRP) and PEW database. In columns (6)–(7), the sample and the control group consist of 48 and 6 African countries, respectively. Political violence mean is the average of the outcome in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Examples of LLM Aggressor Classification from UCDP Source Text

Side A (State)	Side B (Rebel)	UCDP Source Article Text	Class.
Gov. of Somalia	Al-Shabaab	"Xinhua News Agency, 2022-09-23, Somali forces kill 15 al-Shabab militants"	1
Gov. of Pakistan	TTP	"Agence France Presse, 2014-01-22, Bomb blast kills five in northwest Pakistan"	2
Gov. of Syria	Syrian insurgents	"SOHR, 2014-04-19, 203 were killed yesterday"	0
Gov. of Syria	Syrian insurgents	"VDC, 2013-09-20, Unidentified"	0
Gov. of Afghanistan	Hizb-i Islami (Khalis)	"The Globe and Mail, 1989-03-20, Rebel toll 215 in siege at Jalalabad, Kabul says"	1
Gov. of Afghanistan	Taleban	"Reuters, 1995-05-08, Afghan forces claim gains against Taleban militia"	1
Gov. of Syria	Syrian insurgents	"SOHR, 2014-01-05, Nasra Front executes soldiers from regular forces"	2
Gov. of Tunisia	JAK-T	"Agence France Presse, 2016-11-06, IS claims killing of Tunisian soldier"	2
Gov. of Turkey	PKK	"Xinhua News Agency, 2007-06-22, Two PKK members killed in operations in eastern Turkey"	1
Gov. of Uzbekistan	IMU	"AP, 2000-08-07, Militants invade Uzbekistan villages, clash with troops"	2
Gov. of Yemen	AQAP	"Agence France Presse, 2014-08-06, 'Qaeda ambush' kills 5 Yemen soldiers: security"	2

Notes: This table provides illustrative examples of how the GABRIEL LLM framework classified unstructured text from the UCDP GED dataset. Classifications are defined as: 1 = Side A (Government) initiated, 2 = Side B (Rebel) initiated, and 0 = Neutral/Indeterminate. The model conservatively defaults to 0 when the text describes casualties without clear attribution.

Table A.7: Balance Test on Demographic Covariates: Ramadan vs. Non-Ramadan Interviews

	Non-Ramadan	Ramadan	Difference	p-value
Female	0.511	0.517	-0.006	0.513
	(0.500)	(0.500)		
Age	38.135	41.262	-3.127***	0.000
	(13.652)	(14.713)		
Married	0.683	0.678	0.005	0.560
	(0.465)	(0.467)		
Number of children	1.943	1.949	-0.006	0.851
	(1.724)	(1.919)		
Education	1.759	1.770	-0.011	0.448
	(0.821)	(0.807)		
Employed	0.531	0.523	0.008	0.398
	(0.499)	(0.500)		
Income (1–10 scale)	4.756	4.629	0.127***	0.001
	(2.068)	(2.110)		
Religion very important	0.862	0.879	-0.018***	0.003
	(0.345)	(0.326)		
N	15,567	3,759	19,326	

Notes: This table reports balance tests on demographic covariates for the analytical sample used in Table 6. The sample consists of Muslim respondents in 23 Muslim-majority countries from the World Values Survey (1999–2023) who report religion as either "very important" or "rather important" on the WVS importance-of-religion scale (approximately 95% of Muslim respondents in the sample) and who were interviewed in the seven Hijri months centered on Ramadan (Jamada-Al-Thani through Dhul-Hijjah). The Difference column reports the mean difference (Non-Ramadan − Ramadan) from a two-sample t-test with unequal variances; the p-value column reports the corresponding two-sided p-value. Standard deviations are in parentheses below the means. The Employed indicator equals 1 for respondents reporting full-time, part-time, or self-employed work and 0 for retirees, housewives, students, unemployed respondents, and other categories. The Religion very important indicator equals 1 for respondents reporting religion as "very important" and 0 for those reporting "rather important." Ramadan-period interviews account for 19.5% of the seven-month window, above the 14.3% share expected under uniform fieldwork timing within the window. The average number of interviews per country-Hijri-year-Hijri-month cell is statistically indistinguishable across Ramadan and non-Ramadan periods (342 vs. 399, p=0.69), as is the average number of distinct interview days per cell (8.6 vs. 9.3, p=0.76). Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.8: Placebo Test: Non-Muslims in Muslim-Majority Countries

	Spirituality/Prayer		Interest in Politics		Attending Demonstrations	
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	-0.033	-0.039	-0.045	-0.072	-0.154	-0.161
	(0.059)	(0.058)	(0.050)	(0.061)	(0.108)	(0.103)
Outcome mean	0.544	0.595	0.526	0.497	0.370	0.412
Sample	$\pm$3 mo.	All	$\pm$3 mo.	All	$\pm$3 mo.	All
Observations	3,330	6,115	3,460	6,321	3,308	6,109
Adj. R^2	0.219	0.331	0.097	0.084	0.089	0.118
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓

Notes: This table reports placebo estimates of the Ramadan effect on non-Muslim respondents living in the same 23 Muslim-majority countries used in the main analysis. The sample consists of non-Muslim respondents from the World Values Survey (1999–2023). Because non-Muslims living in Muslim-majority countries do not observe Ramadan but share the same fieldwork timing, country-wave composition, and broader societal context as the Muslim sample, the Ramadan indicator should have no effect on individual-level religiosity or political engagement if the main results documented in Table 6 operate through individual religious observance rather than country-level fieldwork artifacts. Columns (1)–(2) use a binary indicator for high-intensity prayer (praying daily or more than once a week). Columns (3)–(4) use a binary indicator for political interest (reporting being "very interested" or "somewhat interested"). Columns (5)–(6) use a binary indicator for demonstration willingness (reporting having attended or willingness to attend lawful peaceful demonstrations). Odd-numbered columns restrict the sample to interviews conducted in the seven Hijri months centered on Ramadan; even-numbered columns use the full twelve-month sample. All specifications include country-by-Hijri-year and Gregorian month fixed effects and the full demographic control set: gender, age, age-squared, marital status, number of children, education, employment status, income, and self-reported importance of religion. Outcome mean reports the baseline average of the dependent variable. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.9: The Ramadan Effect and News Reporting

	Dependent Variable: log(armed conflict + 1)			Dependent Variable: log(demonstrations + 1)		
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.020*	0.029**	0.033*	0.115*	0.105*	0.107*
	(0.010)	(0.012)	(0.017)	(0.065)	(0.058)	(0.057)
Ramadan× >75% Muslim				-0.187**	-0.149**	-0.155**
				(0.078)	(0.068)	(0.067)
Ramadan× 50–75% Muslim				-0.004	-0.050	-0.036
				(0.093)	(0.070)	(0.063)
Ramadan× 25–50% Muslim				-0.025	-0.065	-0.064
				(0.073)	(0.061)	(0.062)
Ramadan× 1–25% Muslim				-0.083	-0.096	-0.099
				(0.067)	(0.060)	(0.059)
Dataset	UCDP	UCDP	ACLED	ACLED	ACLED	ACLED
Fatality threshold	≥5	None	≥5	None	None	None
News sources per event	≥1	≥2	≥1	≥2	≥1	≥1
Source type	Any	Any	Any	Any	International	Non-International
Outcome mean (levels)	10.05	8.696	0.797	8.866	8.209	8.441
Observations	12240	11016	4536	15552	15552	15552
Adj. R^2	0.880	0.895	0.716	0.796	0.859	0.858
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-year FE	✓	✓	✓	✓	✓	✓

Notes: This table tests whether differential news reporting during Ramadan can account for the baseline results on armed conflict and demonstrations. A concern is that media outlets may reduce coverage during Ramadan, mechanically lowering recorded events, or that international attention shifts in ways correlated with the holy month. Columns (1)–(3) estimate the effect on state-based armed conflict, restricting the sample to countries with a Muslim population share exceeding 75% (31 countries for UCDP, 14 for ACLED). Column (1) uses UCDP data restricted to events with at least five fatalities, under the assumption that high-casualty incidents receive consistent media coverage regardless of timing. Column (2) uses UCDP data restricted to events corroborated by at least two independent sources. Column (3) uses ACLED political violence data (events involving at least one state actor) restricted to events with at least five fatalities. Columns (4)–(6) estimate the effect on civilian demonstrations using the full sample of 48 African countries and report the coefficient on the interaction between the Ramadan indicator and an indicator for countries with a Muslim population share exceeding 75%, with countries below 1% Muslim population as the omitted baseline. This interaction specification exploits low-Muslim countries as a built-in control group for reporting trends. Column (4) restricts to demonstration events corroborated by at least two independent sources. Columns (5) and (6) decompose demonstrations by source type: column (5) retains only events covered by at least one international media outlet, while column (6) retains events reported exclusively by local, national, or subnational sources. Columns (1)–(3) include country-by-Hijri-year and Gregorian month fixed effects. Columns (4)–(6) additionally include Muslim population share category-by-Hijri-year fixed effects to absorb group-specific trends. Outcome mean reports the sample average of the dependent variable in levels for Muslim-majority countries. Robust standard errors are in parentheses, clustered at the country level (31 countries in columns 1–2, 44 in column 3, and 48 in columns 4–6). Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: The Effect of Fasting Intensity on Armed Conflicts and Demonstrations

	Dependent Variable: log(armed conflict+1)			Dependent Variable: log(demonstrations+1)		
	(1)	(2)	(3)	(4)	(5)	(6)
RFH	-0.753	0.301	0.262	6.430***	1.008	0.591
	(3.224)	(0.183)	(0.181)	(2.052)	(6.820)	(3.203)
RFH × >75% Muslim			0.477			4.585
			(2.061)			(3.895)
RFH × 50–75% Muslim			-0.682			-13.941***
			(1.309)			(4.791)
RFH × 25–50% Muslim			-1.958*			1.729
			(1.180)			(3.547)
RFH × 1–25% Muslim			-0.077			-2.097
			(0.555)			(4.118)
Outcome Mean (levels)	12.52	0.891	5.636	8.566	2.469	5.730
Sample	Muslim	Non-Muslim	All	Muslim	Non-Muslim	All
Observations	945	770	3,710	378	162	1,296
Country FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Share-by-year FE			✓			✓

Notes: This table tests whether fasting intensity, measured as the logarithm of daily Ramadan fasting hours (RFH), predicts armed conflict or demonstrations. The sample is restricted to the month of Ramadan only, so each country contributes one observation per Hijri year. Fasting hours vary across country-years because Ramadan rotates through Gregorian seasons: the same country experiences longer fasts when Ramadan falls in summer and shorter fasts when it falls in winter. All regressions include controls for annual population levels. Columns (1) and (4) restrict to Muslim-majority countries (Muslim population share above 75%), columns (2) and (5) to non-Muslim countries (below 1%) as a placebo. Columns (3) and (6) pool all countries and interact RFH with indicators for Muslim population share categories; the omitted baseline is countries with less than 1% Muslim population. Country fixed effects absorb time-invariant latitude differences, and Hijri year fixed effects absorb which Gregorian season Ramadan falls in globally. Share-by-year FE controls for the muslim population share bin dummies and their interactions with year dummies. The RFH coefficient is identified from the interaction of latitude and seasonal timing: within a given year, countries at different latitudes experience different fasting durations, and within a given country, fasting duration varies as Ramadan rotates across seasons. Robust standard errors are in parentheses, clustered at the country level (31 countries in column 1, 22 in column 2, 106 in column 3, 14 in column 4, 6 in column 5, and 48 in column 6). Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B: Rebel Group Classifications

This appendix lists the 139 organized rebel groups in the sample with their assigned ideological category and a brief justification for each, and documents the procedure used to assign them to one of three categories: religious-ideological, ethno-religious, or secular-nationalist.

Decision criteria. For each group, I assess two dimensions. *Religious identity* is present when the founding charter, recruitment base, or organizational rhetoric explicitly treats Islamic identity as constitutive of group membership, beyond the incidental fact that the population from which fighters are drawn is Muslim. *Doctrinal mandate for offensive warfare* is present only when the group's principal stated objective is the establishment of religious governance, whether sharia rule, an Islamic state, or a caliphate, and the group frames armed struggle itself as religiously obligatory rather than as an instrumental means to a political end. Groups satisfying both dimensions are coded as religious-ideological. Groups satisfying religious identity but not the doctrinal mandate are coded as ethno-religious. Groups satisfying neither are coded as secular-nationalist. The classification rests on stated objectives and foundational documents rather than tactical behavior alone, since organizationally similar operational profiles can arise from very different ideological bases.

The boundary between religious-ideological and ethno-religious is occasionally contested; the boundary with secular-nationalist groups, by contrast, is rarely ambiguous. Where the distinction between religious-ideological and ethno-religious is genuinely contested, two pieces of behavioral evidence enter as tiebreakers: known operational alliances with secular factions, and formal participation in established state governance structures. Both are inconsistent with a strict doctrinal mandate for offensive religious warfare, since either would require subordinating religious-governance objectives to political pragmatism. Cooperation with secular allies or integration into official state institutions therefore weighs against the religious-ideological coding.

Sources. Classification draws on three layers of evidence. The UCDP Actor Dataset (Brosché and Sundberg, 2024) provides the canonical organizational identifier used throughout the conflict data. The Mapping Militants Project at Stanford's Center for International Security and Cooperation (Crenshaw and Robinson, 2025) provides standardized profiles of major armed organizations, including founding ideology, organizational lineage, and primary recruitment networks. Where these sources are silent, ambiguous, or contested, I consult the group's own founding charter or manifesto and contemporaneous reporting.

Coding procedure. Each group was evaluated against the two dimensions in sequence, with the religious identity dimension assessed first and the doctrinal mandate dimension assessed second. The assignment rule is deterministic given the criteria: a group failing the first dimension is automatically classified as secular-nationalist regardless of the second; a group passing the first but failing the second is classified as ethno-religious; a group passing both is classified as religious-ideological.

Robustness to borderline cases. A small number of groups occupy contested ideological space. Within the religious-ideological category, the Afghan Taliban combine Deobandi religious mandate with Pashtun tribal codes; Palestinian Islamic Jihad frames a religiously obligatory struggle within a territorially bounded conflict; the Macina Liberation Front operates under JNIM's al-Qaeda-affiliated umbrella while drawing heavily on localized Fulani communal grievances. Within the ethno-religious category, broad coalition labels such as Syrian Insurgents and Kashmir Insurgents encompass component groups with widely varying ideologies, and Hamas and Hezbollah present founding charters that sit uneasily with their current operational realities.

Religious-Ideological

1. **Afghan Taliban (Taleban)** — Deobandi Islamist movement founded in 1994 in Kandahar; established the Islamic Emirate of Afghanistan in 1996 and again in 2021. Primary stated objective is establishing governance based on sharia under the authority of an Amir al-Mu'minin (Commander of the Faithful). The group's founding charter declares that sovereignty belongs only to God and designates sharia as the sole source of legislation. Ideology rooted in Hanafi jurisprudence and the Deobandi tradition; armed jihad is invoked as an obligatory religious duty. The ideology of the Taliban is considered a shift from traditional Islamist views held by anti-Soviet mujahedeen fighters in the 1980s and early 1990s. It can be characterized as a combination of strict anti-modern Pashtun tribal ideology mixed with certain radicalized Deobandi interpretations of Islam. The UN Security Council grouped the Taliban with Al-Qaeda.

2. **Tehrik-i-Taliban Pakistan (TTP)** — Deobandi jihadist umbrella organization formed in December 2007 from approximately 40 militant factions in Pakistan's Federally Administered Tribal Areas (FATA). Stated objectives include enforcing a strict Deobandi interpretation of sharia throughout Pakistan and ultimately establishing an Islamic caliphate. The group explicitly frames its armed struggle as defensive jihad and seeks to replace Pakistan's secular constitution with religious law. The TTP claims that its jihad against the Pakistani government is an act of self-defense against an apostate and puppet regime of the United States.

3. **TTP – Tariq Afridi faction (TTP - TA)** — Splinter faction of the TTP operating in Pakistan's tribal regions, retaining the parent organization's ideological orientation. Same doctrinal basis as the TTP: enforcement of sharia and armed jihad against the Pakistani state on explicitly religious grounds.

4. **Islamic Movement of Uzbekistan (IMU)** — Islamist militant organization co-founded in 1998 by Tahir Yuldashev and Juma Namangani in Afghanistan. Declared jihad to overthrow the secular government of Uzbekistan and establish an Islamic state governed by sharia. Later expanded its goal to the creation of a caliphate across all of "Turkestan" (Central Asia from the Caspian Sea to western China). Closely affiliated with al-Qaeda; received substantial financial backing from Osama bin Laden and fought alongside the Taliban.

5. **Jamaat-ul-Ahrar (JuA)** — Splinter faction that broke from the TTP in August 2014, merging elements of TTP's Mohmand Agency branch with Ahrar-ul-Hind. Retained the TTP's core objective of imposing sharia in Pakistan. Its founding emir, Omar Khalid Khorasani, explicitly called for overthrowing the Pakistani government, seizing its nuclear weapons, and waging jihad until a global caliphate is established. The group pledged to continue attacks until sharia law was implemented nationwide.

6. **Lashkar-e-Islam (LeI)** — Deobandi militant group founded in 2004 in Pakistan's Khyber Agency (now Khyber District) by Mufti Munir Shakir. Established with the explicit purpose of enforcing strict Deobandi sharia in Khyber Agency. Under the subsequent leadership of Mangal Bagh Afridi, the group set up parallel Islamic courts, mandated mosque attendance, imposed Islamic dress codes, and banned activities it deemed un-Islamic. LeI only accepted sharia as legitimate law and refused to recognize elected Pakistani officials. Before 2008, LeI was primarily a religious reformist organization. After Mangal Bagh assumed leadership, LeI sought to gain extensive control over the Khyber District and the Khyber Pass. During this period, LeI began conducting more suicide and violent attacks against members of the Pakistani state. Since 2014, this goal has largely been com-

promised because of LeI's relocation to Afghanistan. However, LeI still launches attacks and raises funds from Afghanistan with the hope of returning to the Khyber District.

7. **Ansar al-Islam (Jama'at Ansar al-Islam)** — Kurdish Salafist-jihadist group formed in December 2001 in Iraqi Kurdistan from the merger of Jund al-Islam and a splinter faction of the Islamic Movement of Kurdistan. Founded with the stated objective of expelling non-Muslim influences from Kurdistan and governing all territory under strict sharia. Established an "Islamic Emirate of Kurdistan" where it implemented Taliban-style governance: barring women from education and employment, banning music and television, and threatening amputation, flogging, and stoning for religious offenses. Funded by al-Qaeda and closely affiliated with its network.

8. **Ittihad-i Islami (High Council of Afghanistan Islamic Emirate)** — Islamic Union for the Liberation of Afghanistan, a Wahhabist mujahideen faction founded in 1981 by Abdul Rasul Sayyaf. The only major Afghan mujahideen party to embrace Wahhabi theology explicitly and in its entirety. Sayyaf, a Saudi-trained theologian with an MA from Al-Azhar University, maintained close ties to the Saudi religious establishment and to Osama bin Laden. The organization was strongly committed to sharia governance and, in the post-Taliban era, secured control over parts of Afghanistan's judiciary to advance that agenda.

9. **Forces of Mullo Abdullo** — Armed faction led by Abdullo Rahimov (known as Mullo Abdullo), a veteran Islamist field commander of the UTO during Tajikistan's 1992–1997 civil war. Mullo Abdullo rejected the 1997 peace agreement and fled to Afghanistan, where he sheltered with the Taliban and maintained links to the Islamic Movement of Uzbekistan. He returned to Tajikistan in 2009 with a multinational band of insurgents and proclaimed jihad against the Rahmon government. His forces carried out a major ambush in the Rasht Valley in September 2010 that killed at least 28 Tajik military personnel. Tajik authorities identified him as an al-Qaeda-linked commander.

10. **East Turkestan Islamic Movement (ETIM)** — Uyghur jihadist organization founded in 1997 by Hasan Mahsum in Xinjiang, China; also known as the Turkistan Islamic Party (TIP). The group's stated objective is the creation of an independent Islamic state of "East Turkestan" governed by sharia. Since its establishment, ETIM has maintained close organizational ties with al-Qaeda and the Taliban, with its leader Abdul Haq reportedly serving on al-Qaeda's shura council from 2005. The group relocated its headquarters to Taliban-controlled Kabul in 1998, trained fighters in al-Qaeda camps in Afghanistan, and later deployed several thousand fighters to Syria's civil war under the Turkistan Islamic Party banner. Its propaganda calls explicitly for armed jihad against the Chinese state, framing the struggle in religious terms against "atheist" occupation.

11. **Forces of the Caucasus Emirate (Imirat Kavkaz)** — North Caucasus jihadist umbrella organization founded in October 2007 by Chechen militant leader Doku Umarov. Umarov's declaration of the Caucasus Emirate marked a decisive shift from Chechen separatist nationalism toward a transnational jihadist agenda. The group's stated objective was the establishment of an Islamic state governed by sharia across the entire North Caucasus, encompassing Chechnya, Dagestan, Ingushetia, Kabardino-Balkaria, and other republics. Umarov explicitly rejected the secular nationalist goals of the earlier Chechen independence movement and declared armed jihad against Russia as a religious obligation. The organization conducted suicide bombings against civilian targets in Moscow and elsewhere, and was designated a terrorist organization by the United States, the United Nations, and Russia.

12. **Islamic State (IS)** — Salafi-jihadist organization that emerged from al-Qaeda in Iraq (founded 2004 by Abu Musab al-Zarqawi) and declared a global caliphate in June 2014 under Abu Bakr al-Baghdadi. The group's central objective is the establishment and expansion of a caliphate governed by a literalist interpretation of sharia. The declaration of the caliphate explicitly invoked religious authority, claiming sovereignty over all Muslims worldwide and demanding their allegiance. IS implemented comprehensive sharia governance in territories it controlled across Iraq and Syria, including religious courts, enforcement of hudud punishments, and systematic persecution of religious minorities. The group frames its armed struggle as an obligatory religious duty incumbent upon all Muslims.

13. **Al-Qaeda** — Transnational Salafi-jihadist organization founded in 1988 by Osama bin Laden and Abdullah Azzam in Peshawar, Pakistan. Al-Qaeda's stated objectives center on a global armed jihad to expel Western influence from Muslim lands and ultimately establish Islamic governance under sharia across the Muslim world. The group's founding ideology frames armed struggle as an individual religious obligation (fard ayn) for all Muslims. Al-Qaeda's 1998 fatwa explicitly invoked Quranic authority to call for attacks on Americans and their allies worldwide. The organization has served as the ideological and operational nucleus for a network of regional affiliates (AQAP, AQIM, al-Shabaab, JNIM) that share its religious-ideological framework.

14. **Al-Qaeda in the Arabian Peninsula (AQAP)** — Regional al-Qaeda affiliate formed in January 2009 from the merger of the Saudi and Yemeni branches of al-Qaeda, based in Yemen. AQAP's stated objective is the establishment of Islamic governance under sharia across the Arabian Peninsula and the expulsion of Western influence from Muslim lands. The group operates within al-Qaeda's global jihadist framework, framing armed struggle as a religious obligation. AQAP has pursued both local insurgency against the Yemeni government and transnational attacks, including the 2009 attempted bombing of a U.S. airliner and the 2015 attack on the offices of Charlie Hebdo in Paris.

15. **Al-Qaeda in the Islamic Maghreb (AQIM)** — Regional al-Qaeda affiliate that formally adopted its current name in 2007, evolving from Algeria's Salafist Group for Preaching and Combat (GSPC, founded 1998). AQIM's objectives include the establishment of an Islamic caliphate across North and West Africa and the expulsion of Western presence from Muslim lands. The group pledged allegiance to al-Qaeda's central leadership and adopted its transnational jihadist agenda. AQIM has conducted kidnappings, bombings, and armed attacks across Algeria, Mali, Mauritania, Niger, and Burkina Faso, and its Saharan branch later merged into JNIM.

16. **Armed Islamic Group of Algeria (GIA)** — Salafi-jihadist organization founded in 1992 during the Algerian Civil War following the military's cancellation of the 1991 elections that the Islamic Salvation Front (FIS) was poised to win. The GIA's stated objective was the violent overthrow of the Algerian secular state and the establishment of an Islamic state governed by sharia. The group rejected any political compromise and pursued an uncompromising takfiri ideology, declaring all who did not support its armed struggle to be apostates. The GIA conducted mass civilian massacres, assassinations, and bombings throughout the 1990s.

17. **Takfir wa'l Hijra** — Egyptian jihadist group (formally Jama'at al-Muslimin) founded in the 1960s–1970s by Shukri Mustafa, a former Muslim Brotherhood member radicalized during imprisonment. The group's ideology centers on the concept of takfir (excommunication of other Muslims deemed insufficiently pious) and hijra (withdrawal from corrupt society). It declared Egyptian society to be in a state of pre-Islamic ignorance

(jahiliyya) and sought to establish a purified Islamic community that would eventually overthrow the secular state. The group carried out the 1977 kidnapping and murder of a former Egyptian minister of religious endowments.

18. **Al-Gama'a al-Islamiyya (Egyptian Islamic Group)** — Egyptian Islamist militant organization active primarily from the 1970s through the late 1990s. The group's stated objective was the overthrow of the Egyptian secular state and the establishment of an Islamic state governed by sharia, drawing ideological inspiration from Sayyid Qutb's writings. Al-Gama'a al-Islamiyya carried out attacks on government officials, Coptic Christians, secular intellectuals, and foreign tourists, most notably the 1997 Luxor massacre that killed 62 people. Its spiritual leader, Omar Abdel-Rahman, was convicted for his role in the 1993 World Trade Center bombing conspiracy.

19. **Ansar Bayt al-Maqdis** — Sinai-based jihadist group that emerged around 2011 in the security vacuum following the Egyptian revolution. Initially focused on attacks against the Israel-Egypt gas pipeline and Israeli targets, the group expanded to attack Egyptian military and police forces in the Sinai Peninsula. In November 2014, Ansar Bayt al-Maqdis pledged allegiance to the Islamic State and rebranded as "Wilayat Sinai" (Sinai Province). The group seeks to establish Islamic governance in the Sinai through armed jihad against the Egyptian state, which it considers apostate.

20. **Signed-in-Blood Battalion (al-Muwaqi'un bi-l-Dima)** — AQIM splinter group formed in 2012 by veteran jihadist Mokhtar Belmokhtar after his expulsion from AQIM's leadership. The group operated within a Salafi-jihadist framework and is most known for the January 2013 attack on the In Amenas gas facility in Algeria, which killed 39 foreign hostages. Belmokhtar framed the attack as retaliation for France's military intervention in Mali. The group subsequently merged with MUJAO to form al-Murabitun in August 2013.

21. **Movement for Unity and Jihad in West Africa (MUJAO)** — West African jihadist organization that split from AQIM in 2011. MUJAO's stated objective was to spread jihad across West Africa, particularly targeting what it considered apostate governments and Western interests. The group occupied the city of Gao in northern Mali in 2012, where it imposed strict sharia governance including amputations and public floggings. MUJAO merged with Belmokhtar's Signed-in-Blood Battalion to form al-Murabitun in August 2013.

22. **Al-Murabitun** — Jihadist organization formed in August 2013 through the merger of MUJAO and the Signed-in-Blood Battalion, led by Mokhtar Belmokhtar. Al-Murabitun's stated goal was to spread jihad across North and West Africa. The group reaffirmed its allegiance to al-Qaeda in 2015 and conducted joint operations with AQIM in Mali, Burkina Faso, and Ivory Coast, including the November 2015 Radisson Blu hotel attack in Bamako. In March 2017, al-Murabitun merged with AQIM's Saharan branch and other groups to form Jama'at Nusrat al-Islam wal-Muslimin (JNIM).

23. **Al-Shabaab** — Somali Salafi-jihadist organization that emerged from the militant wing of the Islamic Courts Union following the 2006 Ethiopian intervention, formally pledging allegiance to al-Qaeda in 2012. Al-Shabaab's stated objective is the establishment of an Islamic state governed by sharia across Somalia and the broader Horn of Africa. The group has implemented strict sharia governance in territories under its control, including religious courts, hudud punishments, and bans on music, cinema, and secular education. Al-Shabaab frames its armed struggle as jihad against the "apostate" Somali government and its foreign backers, particularly the African Union Mission in Somalia (AMISOM).

24. **Hizbul Islam** — Somali Islamist alliance formed in 2009 from the merger of several factions, led by Sheikh Hassan Dahir Aweys. Hizbul Islam sought to establish Islamic governance in Somalia and opposed the Transitional Federal Government and the Ethiopian military presence. The group imposed sharia in areas under its control and rejected any political accommodation with the Western-backed government. In December 2010, Hizbul Islam formally merged with al-Shabaab after losing most of its territory to the larger organization.

25. **Ansar Dine** — Malian Salafist Islamist group founded in late 2011 by Iyad ag Ghaly, a veteran Tuareg political and military leader. Ansar Dine's stated objective is the implementation of sharia law across all of Mali, distinguishing it from the secular Tuareg separatist movement (MNLA) with which it briefly cooperated. During the 2012 occupation of northern Mali, the group imposed strict sharia governance in Timbuktu and other cities, destroying Sufi shrines designated as UNESCO World Heritage Sites. In March 2017, Ansar Dine merged into JNIM under ag Ghaly's overall leadership.

26. **Jama'at Nusrat al-Islam wal-Muslimin (JNIM)** — Al-Qaeda-affiliated jihadist umbrella organization formed in March 2017 from the merger of Ansar Dine, AQIM's Saharan branch, al-Murabitun, and the Macina Liberation Front, led by Iyad ag Ghaly. JNIM's stated purpose is to unite jihadist efforts across the Sahel under al-Qaeda's global framework. The group pursues the establishment of sharia governance, the expulsion of French and Western military forces, and the overthrow of what it considers apostate governments in Mali, Burkina Faso, and Niger. JNIM has become the dominant jihadist force in the Sahel, conducting attacks across a vast operational area.

27. **Macina Liberation Front (FLM / Katiba Macina)** — Fulani-dominated jihadist organization founded in 2015 by the radical preacher Amadou Koufa, operating primarily in central Mali. The FLM seeks to impose sharia governance in the Macina region (the historic Fulani heartland of the inner Niger Delta) and draws on the legacy of the 19th-century Macina Empire, a Fulani theocratic state. While the group recruits heavily along Fulani ethnic lines and taps into longstanding grievances over land rights and intercommunal marginalization, it operates under JNIM's al-Qaeda-affiliated umbrella and explicitly frames its struggle in religious terms. Koufa's preaching calls for armed jihad and the rejection of secular state authority.

28. **Jama'atu Ahlis Sunna Lidda'awati wal-Jihad (Boko Haram)** — Nigerian Salafi-jihadist organization founded around 2002 by Mohammed Yusuf in Maiduguri, Borno State. The group's name translates roughly to "People Committed to the Propagation of the Prophet's Teachings and Jihad," and it rejects Western secular education and governance as incompatible with Islam. Following Yusuf's extrajudicial killing in 2009, the group launched a full-scale insurgency under Abubakar Shekau, seeking to establish an Islamic state governed by sharia in northeastern Nigeria and the broader Lake Chad Basin.

29. **Al-Itihaad al-Islamiya (AIAI)** — Somali Salafi-jihadist organization active primarily in the 1990s, seeking to establish an Islamic state in Somalia and unite ethnic Somali territories in the Horn of Africa. Founded in the early 1980s and influenced by Wahhabi ideology, AIAI drew support from Saudi and Gulf donors and operated training camps in Somalia. The group imposed sharia governance in areas under its control, notably in the Gedo and Luuq regions. AIAI was linked to al-Qaeda and was accused of involvement in the 1998 U.S. embassy bombings in East Africa. Following Ethiopian military operations and internal fragmentation, many former AIAI members later joined the Islamic Courts Union and subsequently al-Shabaab.

30. **Islamic Front for Armed Jihad (FIAA / FIDA)** — Islamist militant group that split from Algeria's GIA in 1993 during the Algerian Civil War. The group pursued the establishment of an Islamic state in Algeria through armed jihad, sharing the broader Islamist insurgency's goal of replacing the secular government with sharia-based governance. It operated during the most violent phase of Algeria's civil conflict. The Islamic Front for Armed Jihad (FIDA) was notorious during the Algerian Civil War for a very specific tactical choice: they primarily assassinated secular intellectuals, journalists, feminists, and secular politicians. Their violence was not aimed at capturing territory or redressing an ethnic grievance; it was explicitly designed to purge Algerian society of secular and Western ideological influence to pave the way for a strict Sharia state.

31. **Jama'at al-Islamiyya al-Gama'a (JIG)** — Algerian Islamist militant faction active during the Algerian Civil War. Operating as part of the broader constellation of armed jihadist groups that emerged in the 1990s, the group rejected secular state authority and sought to establish strict Islamic governance in Algeria through armed struggle, framing its violent campaign against the state as a religious mandate rather than a conventional political dispute.

32. **Palestinian Islamic Jihad (PIJ)** — Palestinian Islamist militant organization founded in 1981 as an explicit break from the secular nationalist factions of the PLO. PIJ was established precisely to place Islamic religious mandate at the center of the Palestinian armed struggle, and its founding ideology invokes jihad as a religious obligation rather than a strategic instrument. Unlike Hamas, PIJ maintains no formal political wing or social services apparatus, reflecting an organizational structure oriented almost entirely around religiously framed armed resistance. While its operational theater is territorially bounded to the Palestinian context, its primary justification for violence is doctrinal rather than nationalist, placing it in the religious-ideological category.

33. **Abu Sayyaf Group (ASL)** — A violent jihadist militant network founded in 1991 in the southern Philippines. While operating within the ethno-religious geography of Mindanao, the group split from the broader Moro separatist movement specifically to pursue a hardline transnational jihadist agenda. The group rejected political autonomy in favor of establishing an independent Islamic state governed by strict sharia, maintained deep historical ties to al-Qaeda, and later officially pledged allegiance to the Islamic State.

34. **Jama'atu Ansaril Muslimina Fi Biladis Sudan (Ansaru)** — A Salafi-jihadist militant organization based in northeastern Nigeria that splintered from Boko Haram in 2012. Translating to the "Vanguard for the Protection of Muslims in Black Africa," the group is formally aligned with Al-Qaeda in the Islamic Maghreb (AQIM). Its explicitly stated objectives are the restoration of the historical Islamic Sokoto Caliphate across West Africa and the violently enforced defense of Muslims against Western and secular influence.

Ethno-Religious

1. **Alliance for the Re-liberation of Somalia / Union of Islamic Courts (ARS/UIC)** — Broad Somali Islamist coalition. The Islamic Courts Union rose to power in Mogadishu in 2006 by providing governance and security in the absence of a functioning state, before being ousted by the Ethiopian military intervention in December 2006. The ARS was subsequently formed in Asmara in 2007 as an opposition umbrella. The ICU imposed sharia in areas under its control, but its primary appeal was the restoration of order and Somali self-governance against foreign intervention. The coalition included moderates alongside hardliners, and the ARS later split, with the Djibouti faction under

Sheikh Sharif Sheikh Ahmed joining the transitional government and eventually assuming the presidency. Islam provided the organizational framework and legitimacy, but the movement's objectives centered on Somali sovereignty and state-building rather than transnational jihad.

2. **Islamic Salvation Army (AIS)** — Armed wing of Algeria's Islamic Salvation Front (FIS), active during the Algerian Civil War (1992–2002). The AIS was formed after the military annulled the 1991 elections that the FIS was poised to win. Its armed struggle sought to restore the political rights of Algeria's Islamist constituency rather than to impose religious governance through jihad. The AIS targeted primarily military and state installations, maintained a conventional insurgent posture, declared a unilateral ceasefire in 1997, and formally disbanded in 2000 under a government amnesty. Islam defined the movement's identity and constituency, but the armed campaign was driven by political exclusion.

3. **Harakat Sawa'id Misr (HASM)** — Militant group formed in Egypt in 2015 following the military ouster of President Mohamed Morsi. HASM has been linked by Egyptian authorities to the Muslim Brotherhood, though the Brotherhood denies the connection. The group targets Egyptian security officials and government installations, framing its struggle as nationalist resistance against an illegitimate regime. HASM's slogans and communiqués employ nationalist rather than jihadist language, and the group has not articulated goals of establishing sharia governance or a caliphate. Islam serves as an organizational identity rooted in the Brotherhood tradition, but the armed campaign is oriented toward political regime change.

4. **Jondullah (Jundallah)** — Sunni Baloch militant organization founded in 2003 by Abdolmalek Rigi in Iran's Sistan-Baluchestan province. The group's stated objectives center on securing recognition of Baloch cultural, economic, and political rights within Iran. Rigi framed the struggle as a fight for equal citizenship for Sunni Baloch, explicitly denying separatist or radical sectarian aims. However, the group recruited through Sunni religious seminaries and increasingly adopted sectarian rhetoric. Sunni Islam serves as a key component of group identity and a basis for external solidarity, but the underlying grievances are ethnic marginalization and economic underdevelopment of the Baloch population. Not to be confused with the Pakistani organization of the same name, a TTP-linked sectarian group that pledged allegiance to IS in 2014.

5. **Jaish al-Adl** — Successor organization to Jondullah, emerging in 2012 after the execution of Abdolmalek Rigi. Inherited Jondullah's organizational base and membership while adopting a more explicitly Salafi posture. Frames its insurgency as defending Sunni Baloch from Shia state oppression, blending Baloch nationalist objectives with Salafi ideology. Primarily targets Iranian Revolutionary Guards and security personnel through ambushes, assassinations, and kidnappings.

6. **Rohingya Solidarity Organisation (RSO)** — Rohingya insurgent group founded in 1982 following the Tatmadaw's mass displacement operations against the Rohingya in Arakan (Rakhine) State, Myanmar. The RSO's stated objectives center on Rohingya self-determination, restoration of political rights, and return to ancestral lands in Arakan. The group split from the more moderate Rohingya Patriotic Front as its religiously oriented faction and cultivated ties with Islamist networks in Pakistan and Bangladesh, recruiting fighters through religious channels. However, its manifesto frames the armed struggle around ethnic persecution and racial discrimination rather than religious governance, with no stated goal of establishing sharia or an Islamic state. Islam serves as a component of group identity and a basis for external solidarity rather than the primary ideological framework for armed struggle.

7. **United Tajik Opposition (UTO)** — Coalition of Islamist, democratic, nationalist, and regionalist factions formed in 1993 during the Tajikistani Civil War, led by Sayid Abdulloh Nuri. The dominant component was the Islamic Renaissance Party of Tajikistan (IRPT), which gave the coalition its Islamic identity and facilitated support from Afghan mujahideen. However, the UTO's objectives centered on political power-sharing, regional representation for marginalized Gharmi and Pamiri populations, and opposition to the post-Soviet Kulobi-Khujandi ruling elite. The coalition included secular democrats and Pamiri autonomists alongside Islamists. The IRPT never demanded an Islamic state and transitioned to legal party politics after the 1997 peace agreement, underscoring that religion served primarily as a component of group cohesion rather than the central justification for armed struggle.

8. **Hamas** — Sunni Islamist militant organization and Palestinian nationalist movement founded in 1987. While its founding charter is steeped in Islamic rhetoric and frames the liberation of Palestine as a religious duty, its primary operational objective is territorial and nationalist: the establishment of an independent Palestinian state. Religion serves as a powerful mobilizing identity and ideological framework, but the conflict is fundamentally rooted in territorial disputes rather than a transnational jihadist mandate.

9. **Hezbollah** — Lebanese Shia militant organization and political party founded in 1982 with Iranian Revolutionary Guard support in response to the Israeli invasion of Lebanon. Hezbollah's founding 1985 Open Letter declared allegiance to Ayatollah Khomeini's wilayat al-faqih and called for an Islamic republic in Lebanon, placing its origins squarely in religious-ideological territory. However, the organization underwent substantial ideological evolution, and its 2009 manifesto dropped the Islamic state objective entirely in favor of Lebanese sovereignty, resistance to Israeli occupation, and confessional power-sharing within the existing political system. Hezbollah operates an extensive social services network, holds parliamentary seats, and functions as the primary political representative of Lebanon's Shia community. Its armed struggle is framed around national resistance and defense of the Shia community rather than the imposition of religious governance, and its operational alliances with secular actors such as the Syrian Social Nationalist Party and the Free Patriotic Movement reflect a pragmatic political orientation. Shia religious identity remains central to recruitment, internal cohesion, and external patronage networks, but it operates as a marker of communal solidarity rather than the doctrinal basis for violence.

10. **Hizb-i Islami-yi Afghanistan (Hekmatyar faction)** — Afghan Islamist party founded by Gulbuddin Hekmatyar in 1977 as an offshoot of the Muslim Youth movement at Kabul University. Hizb-i Islami's founding charter explicitly calls for an Islamic state in Afghanistan, and the party received substantial Pakistani ISI and Saudi funding during the anti-Soviet jihad on the basis of its Islamist credentials. However, Hekmatyar's operational behavior consistently prioritized Pashtun factional interests over ideological objectives. During the 1992–1996 civil war, Hizb-i Islami fought other Islamist mujahideen factions with equal or greater intensity than it had fought the Soviets, allied with the secularist Dostum militia against the Islamist Rabbani government, and shelled Kabul in campaigns that followed ethnic-factional fault lines rather than ideological ones. Hekmatyar's eventual reconciliation with the Afghan government in 2016 and entry into electoral politics further suggest that Islamic governance was instrumentally deployed rather than constitutively central. The party's recruitment base drew overwhelmingly from eastern Pashtun (particularly Ghilzai) networks, and its conflicts tracked tribal and regional rivalries as much as ideological commitments.

11. **Hizb-i Islami-yi Afghanistan (Khalis faction)** — Splinter faction established by Yunus Khalis in 1979 following a break with Hekmatyar over leadership and organizational style. Khalis, a religious scholar from Nangarhar, oriented his faction more explicitly around traditional religious authority than Hekmatyar's university-educated cadre, and the faction maintained closer ties to local ulema networks in eastern Afghanistan. Nevertheless, its operational footprint was heavily concentrated among Pashtun communities in Nangarhar and Paktia provinces, and its armed struggle during the anti-Soviet jihad and subsequent civil war followed regional and tribal patterns consistent with localized power competition rather than a unified project of religious state-building. Several prominent Khalis commanders, including Jalaluddin Haqqani, later defected to the Taliban, suggesting that the faction functioned more as a vehicle for eastern Pashtun mobilization than as a cohesive ideological movement.

12. **Jam'iyyat-i Islami-yi Afghanistan** — Afghan Islamist party founded by Burhanuddin Rabbani in 1971, with Ahmad Shah Massoud as its most prominent military commander. Jam'iyyat's founding ideology drew on the Muslim Brotherhood tradition and called for Islamic governance, and Rabbani served as president of the Islamic State of Afghanistan from 1992 to 1996. However, the party's recruitment and operational base was overwhelmingly Tajik, centered in the Panjshir Valley and northeastern Afghanistan. During the civil war, Jam'iyyat's alliances and enmities tracked ethnic geography with striking consistency: cooperation with Hazara and Uzbek factions against Pashtun-dominated groups, regardless of the latter's Islamic credentials. Massoud's resistance to the Taliban was framed as much around Tajik survival and anti-Pashtun domination as around competing visions of Islamic governance. The party's ethnic character was sufficiently pronounced that its post-2001 political successor, the National Coalition of Afghanistan, operated as a de facto vehicle for Tajik political representation.

13. **Hizb-i Wahdat** — Afghan political-military organization founded in 1989 through the unification of several Hazara Shia factions, led by Abdul Ali Mazari. The party's name translates to "Party of Unity," reflecting its primary objective of consolidating Hazara political representation in a context of severe ethnic marginalization. Hazara communities had faced systematic discrimination under successive Pashtun-dominated governments, and Hizb-i Wahdat's armed struggle centered on securing Hazara political rights, territorial autonomy in the Hazarajat, and protection against ethnic violence. Iran provided patronage on the basis of Shia solidarity, and the party's religious identity was inseparable from its ethnic one, as the Hazara are Afghanistan's largest Shia community. However, Hizb-i Wahdat's conflicts were driven by ethnic survival and power-sharing demands rather than any project of imposing Shia governance, and the party fought Sunni Islamist groups (the Taliban, Hizb-i Islami) and allied with secular or Sunni actors (Dostum's Junbish-i Milli) on the basis of ethnic-strategic calculus.

14. **United Islamic Front for the Salvation of Afghanistan (UIFSA / Northern Alliance)** — Military-political coalition formed in 1996 to resist the Taliban, bringing together Jam'iyyat-i Islami, Hizb-i Wahdat, and Junbish-i Milli-yi Islami under the nominal leadership of Rabbani's government. The coalition's name invoked Islamic unity, and its component factions each carried Islamic credentials of varying degrees. However, the UIFSA's formation was driven entirely by the shared existential threat posed by the Taliban's military advance against non-Pashtun regions, and its internal cohesion derived from ethnic complementarity (Tajik, Hazara, Uzbek) rather than ideological agreement. The coalition included Dostum's Junbish-i Milli, a largely secular Uzbek militia, alongside Shia and Sunni Islamist factions whose theological positions were mutually incompatible. The UIFSA's armed struggle was framed around national sovereignty and resistance to Taliban domination rather than any shared vision of Islamic governance.

15. **Mahaz-i Milli-yi Islami-yi Afghanistan (National Islamic Front)** — Afghan mujahideen party founded by Pir Sayyid Ahmad Gailani in 1978. Gailani, a hereditary Sufi pir of the Qadiriyya order, derived his authority from traditional religious lineage rather than Islamist ideology, and his political orientation was notably moderate and pro-Western within the mujahideen spectrum. The party advocated for the restoration of the Zahir Shah monarchy and a constitutional system, positioning it closer to nationalist traditionalism than to the Islamist state-building projects of Hizb-i Islami or the Taliban. Its recruitment base drew on Gailani's Sufi network and southern Pashtun tribal affiliations. Islam functioned as a source of traditional authority and communal identity rather than as a programmatic ideology for governance, and Gailani consistently favored negotiated political settlements over maximalist religious objectives.

16. **National Resistance Front (NRF)** — An Afghan military-political alliance formed in 2021 as the de facto successor to the Northern Alliance. Operating primarily out of the Panjshir Valley and led by Ahmad Massoud, the group is structurally an ethno-nationalist resistance movement dominated by ethnic Tajiks. While entirely composed of devout Muslims, its armed struggle is fundamentally oriented toward territorial defense, anti-Taliban resistance, and the establishment of a decentralized, pluralistic Afghan republic, sharply contrasting with the Taliban's theocratic project.

17. **Supreme Council for Islamic Revolution in Iraq (SCIRI)** — An Iraqi Shia political party and militia founded in Iran in 1982. Despite its name, following the 2003 US invasion of Iraq, SCIRI integrated heavily into the newly established Iraqi political system. Its primary objectives shifted from a strict ideological revolution to securing and consolidating the political, territorial, and institutional dominance of Iraq's Shia majority against Sunni and Baathist rivals

18. **al-Mahdi Army** — An Iraqi Shia militia formed in 2003 by the populist cleric Muqtada al-Sadr. While intensely religious in its rhetoric and drawing legitimacy from the Sadr clerical lineage, the militia functioned primarily as a militant Iraqi nationalist and communal defense organization. Its armed struggle was driven by opposition to the US occupation and the defense of impoverished Shia neighborhoods in Baghdad and southern Iraq against Sunni militant groups.

19. **Syrian Insurgents** — A broad, fragmented coalition of armed factions fighting against the Assad regime during the Syrian Civil War. Similar to the United Tajik Opposition (UTO), while the most militarily dominant elements of this umbrella (such as Ahrar al-Sham or HTS) maintain strict Islamist ideologies, the overarching category includes a spectrum of actors—from secular Free Syrian Army defectors to localized defense militias. Across the coalition, Sunni Islamic identity serves as a unifying mechanism against the Alawite-dominated government, but the primary shared objective is contesting political and territorial control of the Syrian state.

20. **Kashmir Insurgents** — A broad umbrella classification encompassing the constellation of armed groups fighting the Indian state in Jammu and Kashmir since 1989. While the coalition includes hardline jihadist factions (like Lashkar-e-Taiba) alongside more secular-leaning liberation fronts (like the JKLF), the conflict is fundamentally driven by the territorial dispute over the Kashmir region. Islamic identity serves as a powerful mobilizing force and a boundary marker against the Hindu-majority Indian state, utilized to pursue either an independent Kashmiri state or integration with Pakistan.

21. **Ansarallah (Houthi movement)** — A Zaydi Shia political and militant movement operating in Yemen. Emerging in the 1990s as a theological revivalist movement to protect Zaydi cultural identity against perceived Sunni/Wahhabi encroachment, it evolved into

a powerful insurgent force. While adopting radical religious rhetoric and receiving backing from Iran, its armed struggle is deeply rooted in Yemeni ethno-political dynamics, primarily seeking territorial control, economic resources, and political dominance within the Yemeni state.

Secular Nationalist

1. **Mujahedin-e Khalq (MEK)** — An Iranian dissident militant organization founded in 1965. Originally rooted in a unique syncretism of Marxist-Leninist ideology and Islamic traditionalism, the group participated in the 1979 revolution but quickly took up arms against the newly established Islamic Republic. Its armed struggle against Tehran is fundamentally political and regime-change oriented, and the organization officially campaigns for a secular, democratic Iranian state.

2. **Popular Front for the Liberation of Palestine - General Command (PFLP-GC)** — A Palestinian nationalist militant organization founded in 1968 by Ahmed Jibril following a split from the main PFLP. The group maintains a strict secular, Marxist-Leninist ideological orientation. While operating within the heavily religiously contested geography of the Levant and receiving state backing from Syria and Iran, its foundational ideology and stated objectives are centered entirely on secular Palestinian nationalism and armed resistance against Israel, devoid of any Islamist state-building mandate.

3. **Kurdistan Workers' Party (PKK)** — A Kurdish militant and political organization founded in 1978 by Abdullah Öcalan. Originating as a strict Marxist-Leninist revolutionary movement, its initial objective was the establishment of an independent, socialist Kurdish state in southeastern Turkey. While its ideology later evolved toward "democratic confederalism," the organization remains staunchly secular. Its armed struggle is driven purely by Kurdish ethno-nationalism and the demand for political autonomy, consistently rejecting religious framing.

4. **Kurdistan Freedom Hawks (TAK)** — A radical Kurdish militant faction that emerged as a splinter group from the PKK. Sharing the secular-nationalist roots of its parent organization, TAK conducts urban terrorism and bombings across Turkey in pursuit of Kurdish ethno-nationalist retaliation against the Turkish state. It operates outside of any religious-ideological framework.

5. **Democratic Union Party (PYD)** — A Kurdish political party and militant organization (via its armed wing, the YPG) operating primarily in northern Syria. As the Syrian affiliate of the PKK's broader transnational network, it adheres to Abdullah Öcalan's secular ideology of democratic confederalism. Its armed conflict against the Syrian government and various jihadist groups (including the Islamic State) is driven by the defense of Kurdish territorial autonomy and the establishment of a secular, decentralized administrative region (Rojava).

6. **Party of Free Life of Kurdistan (PJAK)** — An Iranian Kurdish militant group founded in 2004 as part of the PKK's broader umbrella network. Following the same secular-nationalist and democratic confederalist ideology as the PKK and PYD, the group conducts an armed insurgency against the Iranian government to secure self-determination and cultural rights for the Kurdish minority in northwestern Iran, explicitly rejecting the theocratic framework of the Islamic Republic.

7. **Democratic Party of Iranian Kurdistan (KDPI)** — A Kurdish nationalist party and militant group operating in Iran, founded in 1945. As one of the oldest Kurdish political organizations, it maintains a strictly secular and left-leaning orientation. Its armed struggle

against both the Pahlavi monarchy and the subsequent Islamic Republic has been driven entirely by the goal of securing Kurdish autonomy and federalism within a democratic, secular Iran.

8. **Kurdistan Democratic Party (KDP)** — A prominent Iraqi Kurdish political party and paramilitary organization (Peshmerga) founded in 1946 by Mustafa Barzani. Functioning as a secular, traditionalist ethno-nationalist movement, its historical armed struggle against the Iraqi state was waged to achieve Kurdish self-determination, territorial autonomy, and the protection of Kurdish cultural identity, culminating in the establishment of the Kurdistan Region of Iraq.

9. **Patriotic Union of Kurdistan (PUK)** — An Iraqi Kurdish political party and paramilitary organization founded in 1975 by Jalal Talabani following a split from the KDP. The group adopted a more secular, socialist, and progressive ideological platform compared to the traditionalist KDP. Its armed insurgency against the Ba'athist Iraqi regime was fundamentally rooted in Kurdish ethno-nationalism and the pursuit of political autonomy.

10. **Communist Party of Turkey/Marxist-Leninist (TKP-ML)** — A radical Turkish insurgent organization founded in 1972. Adhering strictly to Maoist and Marxist-Leninist ideology, the group's armed struggle is aimed at the violent overthrow of the Turkish state to establish a communist regime. Its motivations and operational framework are entirely secular and rooted in revolutionary class struggle.

11. **Maoist Communist Party (MKP)** — A Turkish revolutionary insurgent group that emerged as a splinter faction from the TKP-ML. The group pursues a hardline Maoist ideological agenda, conducting armed attacks against Turkish state targets with the objective of sparking a communist revolution. It operates completely outside of any religious or ethno-religious parameters.

12. **Peace at Home Council (Yurtta Sulh Konseyi)** — A clandestine faction within the Turkish Armed Forces that orchestrated the violent, failed coup d'état attempt on July 15, 2016. In their official broadcast declaration, the plotters framed their armed takeover as a necessary intervention to restore the secular, democratic, and constitutional order of the Turkish Republic, making their stated ideological justification explicitly secular-nationalist.

13. **Revolutionary People's Liberation Party/Front (DHKP-C)** — A radical Marxist-Leninist militant organization operating in Turkey, founded in 1978 (originally as Revolutionary Left). The group pursues the violent overthrow of the Turkish government to establish a socialist state. It conducts assassinations and bombings motivated purely by secular, anti-imperialist, and communist ideology.

14. **Popular Front for the Liberation of Palestine (PFLP)** — A Palestinian Marxist-Leninist and revolutionary socialist organization founded in 1967 by George Habash. The group's ideology is fundamentally secular, viewing the Israeli-Palestinian conflict through the lens of anti-imperialism and class struggle rather than religious duty. It pursues the establishment of a secular, democratic Palestinian state.

15. **Balochistan Liberation Army (BLA)** — A secular ethno-nationalist militant organization based in the Balochistan region of Pakistan. The group wages an armed insurgency against the Pakistani state to achieve independence for the Baloch people. Its grievances are rooted in the economic exploitation of regional resources and political marginalization, and its ideology explicitly rejects religious framing in favor of secular Baloch nationalism.

16. **Balochistan Liberation Front (BLF)** — A Baloch ethno-nationalist militant group founded in 1964, operating primarily in Pakistan's Balochistan province. With historical roots in Marxist ideology and left-wing nationalism, the organization's armed struggle against the Pakistani state is strictly secular. It seeks an independent Balochistan based on ethnic identity rather than religious affiliation.

17. **Baloch Republican Army (BRA)** — A secular, ethno-nationalist insurgent group in Pakistan's Balochistan province, largely serving as the militant wing of the Baloch Republican Party. Like other Baloch separatist organizations, its armed struggle is driven by demands for ethnic sovereignty, political rights, and control over natural resources, entirely separate from the religious militant networks operating in the same region.

18. **Baloch Raaji Aajoi Sangar (BRAS)** — An operational umbrella organization formed in 2018 to unify the efforts of several secular Baloch ethno-nationalist insurgent groups, including the BLA, BLF, and BRA. The coalition coordinates armed attacks against Pakistani security forces and Chinese infrastructure projects. Its foundational mandate is strictly secular and separatist, consolidating the armed struggle for an independent Baloch state.

19. **Muttahida Qaumi Movement (MQM)** — A secular political party in Pakistan representing the Muhajir community (Urdu-speaking migrants from India). Its armed factions engaged in extensive urban conflict in Karachi to secure political and economic power for Muhajirs, operating strictly along ethnic and secular lines

20. **Purbo Banglar Communist Party (PBCP)** — A clandestine Marxist-Leninist and Maoist insurgent group operating in Bangladesh. The group's armed struggle is driven by revolutionary communist ideology, seeking to overthrow the state through agrarian revolution and class warfare, firmly placing it in the secular-ideological category.

21. **Purbo Banglar Communist Party - Janajuddha (PBCP-J)** — A hardline splinter faction of the PBCP in Bangladesh. It maintains the same secular, Maoist revolutionary ideology as its parent organization, conducting assassinations and extortion to fund its localized class-based armed struggle.

22. **Parbatya Chattagram Jana Samhati Samiti (PCJSS)** — A secular political organization representing the indigenous Jumma people of the Chittagong Hill Tracts in Bangladesh. Its armed wing, the Shanti Bahini, waged a decades-long ethno-nationalist insurgency to secure autonomy, land rights, and cultural preservation against Bengali settlement and state marginalization.

23. **Sudan People's Liberation Movement/Army (SPLM/A)** — The primary secular insurgent movement that fought the Sudanese government during the Second Sudanese Civil War. Founded by John Garang, the group initially pursued a Marxist-oriented vision for a unified, secular, and democratic "New Sudan," vigorously opposing the Khartoum government's imposition of Islamic law (sharia) on the diverse and largely non-Muslim southern population.

24. **Sudan People's Liberation Movement-North (SPLM/A-North)** — The northern branch of the SPLM/A that continued fighting the Sudanese state after the secession of South Sudan in 2011. Operating primarily in the Nuba Mountains and Blue Nile states, it fights against regional marginalization and advocates for a secular, democratic, and decentralized Sudan.

25. **Sudan Liberation Movement/Army (SLM/A)** — A secular, ethno-nationalist rebel group founded in 2002 to fight for the rights of the marginalized non-Arab communities in the

Darfur region of Sudan. It explicitly rejects the Islamist and Arab-nationalist policies of the central government, demanding political representation and economic development.

26. **Sudan Liberation Movement/Army - Minni Minnawi (SLM/A - MM)** — A major splinter faction of the SLM/A, drawing its support predominantly from the Zaghawa ethnic group. While it has occasionally signed peace agreements with the government, its ideological foundation remains rooted in secular Darfuri ethno-nationalism and regional autonomy.

27. **Justice and Equality Movement (JEM)** — A powerful Darfuri rebel group. While its early leadership had historical ties to the Islamist politician Hassan al-Turabi, the organization's armed struggle is overwhelmingly centered on regional equality, Darfuri ethno-nationalist rights, and the overthrow of the Bashir regime. It allied with strictly secular rebel groups, placing its operational focus squarely on regional political objectives rather than a religious mandate.

28. **Sudan Revolutionary Front (SRF)** — A broad alliance formed in 2011 comprising major Sudanese rebel groups, including the SPLM-N, JEM, and various SLM factions. The coalition's unified mandate is entirely secular and political, aiming to overthrow the Islamist regime in Khartoum and establish a democratic, federal state that respects the rights of Sudan's marginalized peripheries.

29. **Darfur Joint Resistance Forces** — An umbrella coalition of Darfuri rebel factions formed to coordinate military operations against Sudanese government forces and allied militias. Its objectives are firmly rooted in the defense of Darfuri communities and secular regional autonomy.

30. **National Democratic Alliance (NDA)** — An umbrella coalition of Sudanese opposition groups formed in 1989 to oppose Omar al-Bashir's Islamist military regime. It brought together the secular SPLM/A with northern political parties, unified by a secular, democratic, and anti-regime political platform.

31. **South Sudan Democratic Movement/Army (SSDM/A)** — A militant group formed in South Sudan shortly before its independence. Driven by ethnic grievances (particularly among the Murle and Shilluk) and opposition to the dominance of the SPLM government, its insurgency was entirely secular and based on localized political and ethnic power struggles.

32. **South Sudan Liberation Movement/Army (SSLM/A)** — A predominantly Nuer rebel militia in South Sudan. Formed in opposition to the Dinka-dominated SPLM/A state apparatus, its armed campaign was driven by secular ethno-nationalist grievances, seeking greater political power and resource distribution for its constituency.

33. **Free Aceh Movement (GAM)** — An ethno-nationalist separatist organization that waged an insurgency against the Indonesian government from 1976 to 2005. Despite Aceh's strong Islamic identity and the eventual implementation of regional sharia law, GAM's foundational ideology under founder Hasan di Tiro was explicitly secular, anti-colonial, and ethno-nationalist. The group's stated objective was independence from perceived Javanese neo-colonialism and the restoration of the historic Acehnese state, consciously rejecting the religious-state objectives of earlier rebellions like Darul Islam.

34. **Free Papua Movement (OPM)** — An umbrella term for the decentralized ethno-nationalist separatist movement in West Papua, Indonesia. Founded in 1965, the movement's armed

factions wage a low-level guerrilla war against the Indonesian state. Their armed struggle is strictly secular and driven by demands for indigenous self-determination, independence, and the cessation of economic exploitation and demographic marginalization by the Indonesian government.

35. **National Transitional Council (NTC)** — The de facto political body and armed coalition formed by anti-Gaddafi forces during the 2011 Libyan Civil War. While functioning as a broad umbrella encompassing various local and Islamist militias, the central leadership of the NTC officially pursued the overthrow of the Gaddafi regime to establish a pluralistic, democratic, and secular civil state, prioritizing national political transition over ideological mandates.

36. **Forces of the House of Representatives (LNA)** — The military coalition, predominantly known as the Libyan National Army (LNA) led by Khalifa Haftar, aligned with the Tobruk-based parliament during the Second Libyan Civil War. The faction's stated objective was to defeat rival Islamist and UN-backed coalitions in Tripoli to unify Libya under a centralized military-backed government. Haftar explicitly framed his military campaigns, such as Operation Dignity, as a secular-nationalist war to eradicate Islamist militias and political Islam from Libya.

37. **Movement of Democratic Forces of Casamance (MFDC)** — A separatist movement operating in the Casamance region of Senegal since 1982. The group's armed insurgency is driven by the ethno-nationalist grievances of the predominantly Diola population, citing economic neglect and cultural marginalization by the northern Wolof-dominated government in Dakar. Its demands for independence are secular and rooted entirely in regional and ethnic identity.

38. **Polisario Front** — A Sahrawi nationalist liberation movement fighting for the independence of Western Sahara from Moroccan control. Originating with strong Marxist-Leninist and anti-colonial influences, the organization is strictly secular. Its armed struggle and diplomatic efforts are dedicated entirely to securing territorial sovereignty and self-determination through the establishment of the Sahrawi Arab Democratic Republic (SADR).

39. **Lord's Resistance Army (LRA)** — A radical syncretic militant group operating across Uganda, South Sudan, the Central African Republic, and the DRC. Founded by Joseph Kony in 1987, the group's foundational objective was the violent overthrow of the Ugandan government to establish a state governed by the biblical Ten Commandments and Acholi mysticism. While lacking a cohesive political manifesto and driven by a rigid, prophetic religious-ideological mandate rather than secular political grievances, the group operates entirely outside the Islamic ideological frameworks that characterize other religiously motivated insurgencies in this dataset.

40. **Junbish-i Milli-yi Islami** — Afghan political and paramilitary organization founded by Abdul Rashid Dostum in 1992. Despite the word "Islami" in its name, it is a staunchly secular, predominantly ethnic Uzbek militia. It fought for secular governance and Uzbek factional power during the Afghan civil wars, allying at different times with both Soviet-backed and Western forces against Islamist factions.

41. **Afghanistan Freedom Front (AFF)** — An anti-Taliban militant group formed in 2022 following the Taliban takeover. Composed largely of former Afghan National Defense and Security Forces (ANDSF) personnel, the group's objectives are secular and nationalist. It aims to overthrow the Taliban's theocratic regime and restore a democratic, constitutional republic.

42. **Republic of Artsakh** — The unrecognized breakaway state (formerly the Nagorno-Karabakh Republic) that fought for independence and integration with Armenia against Azerbaijan. Its armed forces functioned as a conventional military fighting a strictly ethno-nationalist and territorial war, entirely devoid of religious-ideological goals, despite the broader religious differences between the warring populations.

43. **Somali National Movement (SNM)** — A Somali insurgent group founded in 1981, drawing its support almost exclusively from the Isaaq clan. The group waged a secular, ethno-nationalist guerrilla war against the Siad Barre regime, ultimately declaring the independent (though internationally unrecognized) state of Somaliland in 1991.

44. **United Somali Congress / Somali Salvation Alliance (USC/SSA)** — A major military faction in the Somali Civil War, originally founded in 1989 and rooted primarily in the Hawiye clan. Its armed struggle to overthrow Siad Barre and its subsequent factional warfare in Mogadishu were driven strictly by clan-based political and territorial competition.

45. **Somali Reconciliation and Restoration Council (SRRC)** — A coalition of secular, clan-based Somali warlords formed in 2001 (with significant backing from Ethiopia) to oppose the Transitional National Government and emerging Islamist groups. Its objectives were entirely political and factional, seeking power-sharing and territorial control within a decentralized Somali state.

46. **Forces of Khudoberdiyev** — A rebel military faction led by mutinous Tajik warlord Colonel Mahmud Khudoberdiyev. He led secular military uprisings in 1997 and 1998 against the Tajik government, driven by regional factionalism and strict opposition to the integration of Islamist forces (from the United Tajik Opposition) into the national army following the civil war peace accords.

47. **Mouvement Populaire de l'Azawad (MPA)** — A Tuareg rebel organization founded in 1990 in Mali. While its founder, Iyad Ag Ghaly, much later embraced jihadism, the MPA itself operated during the 1990s Tuareg rebellion as a strictly secular, ethno-nationalist movement seeking autonomy and political representation for the marginalized Tuareg people.

48. **Anjouan People's Movement (MPA / Republic of Anjouan)** — A separatist movement that declared the independence of the island of Anjouan from the Comoros in 1997. The armed struggle was driven entirely by localized political and economic grievances against the central government in Moroni, operating without any religious-ideological framework.

49. **Niger Movement for Justice (MNJ)** — A predominantly Tuareg rebel group that waged an insurgency against the government of Niger from 2007 to 2009. The group's objectives were secular and ethno-nationalist, demanding a greater share of the region's uranium wealth, regional development, and political integration for northern nomadic populations.

50. **Coordination of Azawad Movements (CMA)** — A coalition of Tuareg independence and Arab nationalist movements in northern Mali formed in 2014. Serving as the primary secular and ethno-nationalist counterweight to jihadist coalitions like JNIM, the CMA seeks political autonomy and self-determination for the Azawad region, consistently rejecting the imposition of transnational sharia governance.

51. **National Council of Timorese Resistance (CNRT)** — An umbrella organization formed in 1998 representing the East Timorese struggle for independence from Indonesia. Integrating the armed wing FALINTIL with diplomatic and political fronts, the movement was strictly secular, nationalistic, and focused on achieving sovereign statehood and self-determination for East Timor.

52. **Front for the Restoration of Unity and Democracy (FRUD)** — A rebel group in Djibouti that launched an insurgency in 1991. Drawing its support primarily from the Afar ethnic group, the organization fought against the perceived political and economic marginalization by the Issa-dominated central government. Its objectives were strictly secular and focused on ethno-political power-sharing.

53. **Rally of Democratic Forces of Guinea (RFDG)** — A Guinean rebel coalition that launched cross-border attacks from Liberia and Sierra Leone into Guinea in 2000 and 2001. Backed by Liberian President Charles Taylor, the group's armed campaign was a secular, political proxy conflict aimed at destabilizing the government of Lansana Conté, entirely devoid of religious framing.

54. **Syrian Democratic Forces (SDF)** — A multi-ethnic and multi-religious military alliance formed in 2015, dominated by the Kurdish YPG but including Arab, Assyrian, and Turkmen militias. The SDF was the primary ground force fighting the Islamic State in Syria. Its ideology is rooted in secular democratic confederalism, aiming to establish a decentralized, secular, and pluralistic governance model in northeastern Syria.

55. **Patriotic Salvation Movement (MPS)** — A Chadian rebel group formed in 1990 by Idriss Déby. Operating from bases in Sudan, the group launched a successful insurgency to overthrow the dictatorial regime of Hissène Habré. Its foundational objectives and subsequent governance as Chad's ruling party were rooted in secular political control and regional factionalism rather than any religious mandate.

56. **Popular Front for the Liberation of Azawad (FPLA)** — A Tuareg rebel faction active during the 1990s Tuareg rebellion in Mali. Splintering from other groups over strategic differences, the FPLA pursued a strictly secular, ethno-nationalist agenda, fighting for the political and economic rights of the Tuareg population in the northern Azawad region.

57. **Sudan Liberation Forces Alliance (SLFA)** — A rebel coalition operating in the Darfur region of Sudan. Formed by merging several secular Darfuri factions, its armed struggle is driven by ethno-nationalist grievances, seeking regional autonomy, political representation, and an end to the marginalization of Darfur's non-Arab populations by the central government.

58. **Rwandan Secular Militant Factions (RJF)** — An armed faction operating within the context of the Rwandan conflict system. Its armed struggle is driven by secular political objectives, regional grievances, and factional competition for state resources, operating entirely outside of a religious-ideological mandate.

59. **Popular Resistance Committees (PRC)** — A Palestinian militant coalition operating primarily in the Gaza Strip. Formed in 2000 during the Second Intifada, it functions as a cross-ideological umbrella group. While operating alongside Islamist groups, its foundational mandate is rooted in secular Palestinian nationalism and armed resistance against Israel, lacking a specific Islamist state-building agenda.

60. **Popular Front of Tajikistan (PFT)** — A secular, pro-government paramilitary organization active during the Tajikistani Civil War. Composed primarily of Kulyabi and Uzbek

militias, the group fought on behalf of the post-Soviet secular government against the Islamist and regionalist forces of the United Tajik Opposition (UTO), motivated by regional factionalism and the preservation of the secular state apparatus.

61. **United Baloch Army (UBA)** — A secular ethno-nationalist militant group in Pakistan's Balochistan province, formed as a splinter faction from the Balochistan Liberation Army (BLA). The group wages a separatist armed struggle against the Pakistani state, driven entirely by demands for Baloch sovereignty and control over regional resources, distinct from religious militant networks.

62. **Alliance of Tuareg of Niger and Mali for Change (ATNMC)** — A cross-border Tuareg rebel alliance formed to coordinate insurgent activities in both Mali and Niger. The group's objectives were strictly secular and ethno-nationalist, demanding economic development, political integration, and cultural rights for the marginalized nomadic Tuareg populations across the Sahel.

63. **Tigray People's Liberation Front (TPLF)** — An ethno-nationalist paramilitary group and political party in Ethiopia. Rooted in Marxist-Leninist ideology at its founding, the organization pursues secular, ethno-nationalist objectives. Following its ouster from the federal government, its recent armed struggle was driven by demands for Tigrayan regional autonomy and resistance against the central government, entirely outside any religious framework.

64. **OPON Forces** — The Special Purpose Police Unit (OPON) of Azerbaijan. In 1995, this heavily armed state security faction, led by Colonel Rovshan Javadov, engaged in a mutinous armed rebellion against the government of Heydar Aliyev. The conflict was a strictly secular, factional power struggle for control of the Azerbaijani state.

65. **Military Faction (Red Berets)** — An elite airborne commando unit of the Malian army. Following the 2012 military coup by the rival "Green Berets," this faction (loyal to ousted President Amadou Toumani Touré) launched an armed counter-coup attempt in Bamako. The fighting was a secular, internal military factional dispute over state control.

66. **Military Faction (Forces of Shah Nawaz Tanai)** — A mutinous faction of the Afghan communist military led by Defense Minister General Shah Nawaz Tanai, who launched an attempted coup against President Mohammad Najibullah in 1990. The armed uprising was a secular, factional power struggle within the Soviet-backed regime.

67. **Military Faction (Forces of Suret Huseynov)** — A rogue military faction in Azerbaijan led by warlord and army commander Suret Huseynov, who launched a military rebellion in 1993 that ousted President Abulfaz Elchibey. The uprising was an entirely secular, politically motivated military coup.

68. **Presidential Guard (Comoros)** — A mercenary-led military unit, commanded by Bob Denard, that launched an armed takeover of the Comorian state in 1989. Despite its origins as a state security force, it operated during the conflict as an independent, secular military faction driven by political control rather than ideological motives.

69. **Sudan Allied Revolutionary Forces (SARC)** — A rebel coalition operating in the Darfur region. Like other Darfuri groups in this category, its armed struggle against the Sudanese government is driven by secular, ethno-nationalist demands for regional autonomy and an end to systemic marginalization.

70. **Baloch Ittehad** — An ethnic separatist alliance operating in Pakistan's Balochistan province. Its armed struggle is firmly secular and ethno-nationalist, coordinating efforts to achieve an independent Baloch state and control over regional resources.

71. **Council for the Restoration of Arakan (CRA)** — An ethnic Rakhine insurgent group in Myanmar. Operating in Rakhine State, its militant campaign against the central government is driven by secular ethno-nationalist objectives, seeking self-determination and political rights for the Rakhine people.

72. **Front pour la Défense de la République (FDR)** — An armed rebel faction in Burundi. Operating within the context of the Burundian Civil War, the group's mobilization and armed struggle were driven by secular ethno-political grievances and factional competition for state power.

73. **Front de Libération de l'Aïr et de l'Azawad (FLAA)** — A prominent Tuareg rebel organization active during the early 1990s insurgency in Niger. Its armed struggle was strictly secular and ethno-nationalist, demanding political integration, economic development, and autonomy for the marginalized Tuareg populations of the Aïr region.

74. **Front Patriotique de Résistance du Niger (FPRN)** — A rebel faction active in Niger. The group's insurgency was driven by secular political opposition and regional grievances against the central government, entirely devoid of religious-ideological framing.

75. **FRUD-Combattant (FRUD-C)** — A hardline splinter faction of the Front for the Restoration of Unity and Democracy (FRUD) in Djibouti. After the mainstream FRUD signed a peace agreement with the government, this faction continued a secular, ethno-nationalist armed struggle to secure greater political representation for the Afar people.

76. **Forces of Khalifa al-Ghawil** — The armed factions and militias loyal to Khalifa al-Ghawil's National Salvation Government in Tripoli during the Libyan civil war. The conflict with rival governments (like the UN-backed GNA or Haftar's LNA) was a secular political struggle for institutional legitimacy and territorial control over the Libyan state.

77. **Ogaden National Liberation Front (ONLF)** — An ethno-nationalist rebel group operating in the Somali Region of Ethiopia. The group's armed struggle against the Ethiopian state is strictly secular, driven by demands for self-determination, regional autonomy, and the protection of ethnic Somali rights against perceived marginalization by the central government.

78. **Popular Front for the Liberation of Libya (PFLL)** — A Libyan political and militant organization comprised of loyalists to the ousted Muammar Gaddafi regime. Formed in 2016, the group seeks to restore the secular, nationalist Jamahiriya system, operating outside of the religious and tribal factionalism that characterizes much of the ongoing Libyan conflict.

79. **Revolutionary Forces of 1 April** — A Chadian rebel faction that opposed the dictatorial regime of Hissène Habré. Formed in the late 1980s, the group later merged with other secular rebel factions to form the Patriotic Salvation Movement (MPS), which successfully overthrew Habré in 1990. The group's armed struggle was driven by secular political opposition and ethnic factionalism rather than a religious ideology.

80. **Somali Patriotic Movement (SPM)** — A Somali rebel organization founded in 1989, drawing its support predominantly from the Ogaden clan. The group waged an armed insurgency during the Somali Civil War, driven strictly by clan-based factionalism, territorial control, and political power-sharing following the collapse of the Siad Barre regime.

81. **Union des Forces de la Résistance de l'Azawad (UFRA)** — A Tuareg rebel group operating in Mali. Active during the ethno-nationalist insurgencies in the Sahel, the group's

armed struggle is strictly secular, seeking regional autonomy, political representation, and the defense of the marginalized Tuareg populations in the Azawad region.

82. **United Liberation Army (ULA)** — An ethno-nationalist militant group operating in Pakistan's Balochistan province. Like other secular Baloch separatist organizations, its armed struggle against the Pakistani state is driven by demands for ethnic sovereignty, political rights, and control over natural resources, explicitly rejecting the religious militant networks operating in the same region.

83. **United Somali Congress / Somali National Alliance (USC/SNA)** — A powerful Somali political and military alliance led by General Mohamed Farah Aidid. Drawing its primary strength from the Hawiye clan, the faction waged a secular, clan-based armed struggle for control of Mogadishu and the broader Somali state during the civil war, notably clashing with rival warlords and UN peacekeeping forces.

84. **Zintan Military Council** — A powerful coalition of militias based in the Libyan city of Zintan. Emerging during the 2011 revolution, the group later aligned with secular and nationalist forces (such as Khalifa Haftar's LNA) against Islamist and UN-backed coalitions in Tripoli. Its operations are driven by local tribal interests, anti-Islamist sentiment, and the pursuit of political power within the fragmented Libyan state.